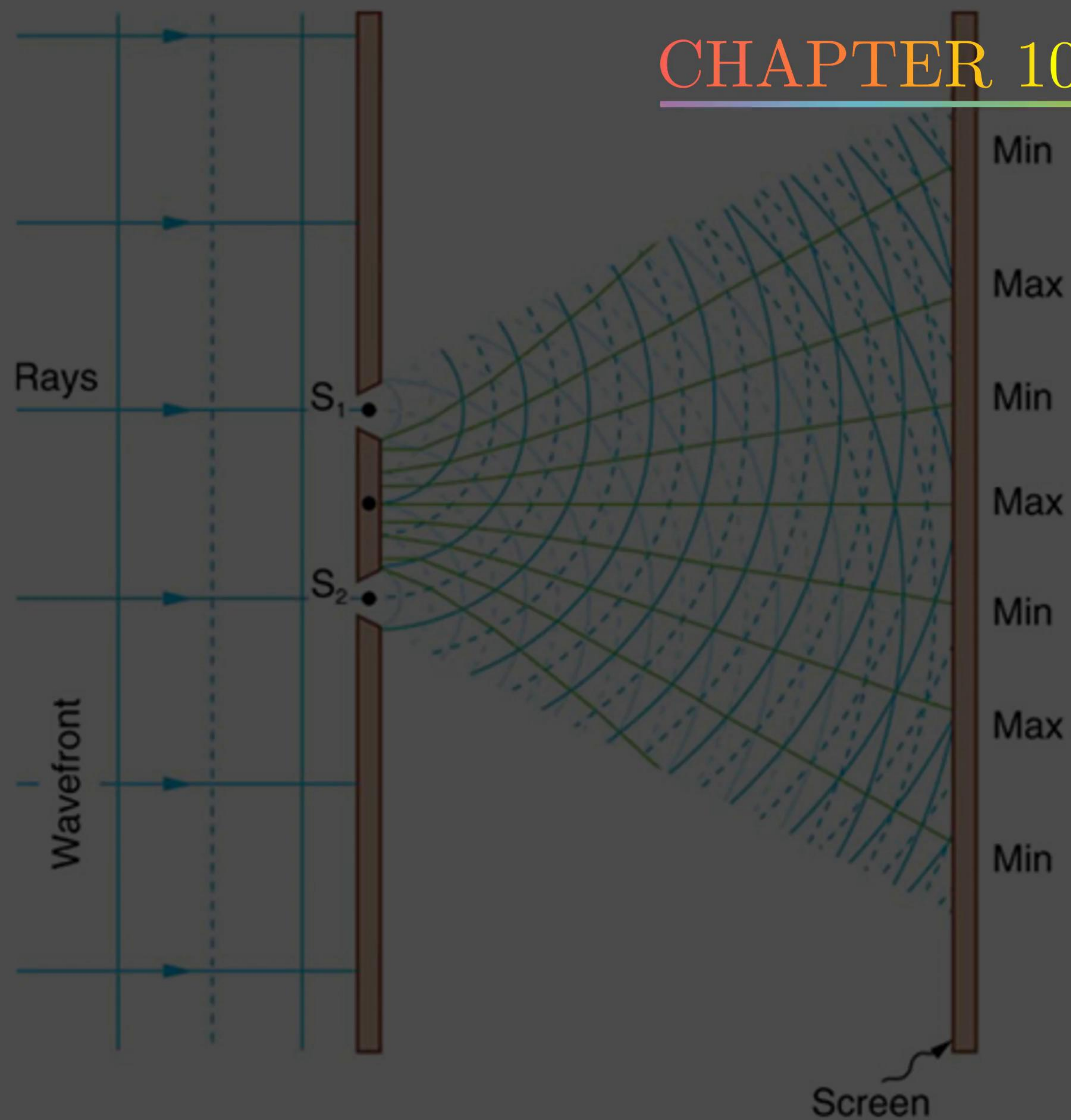
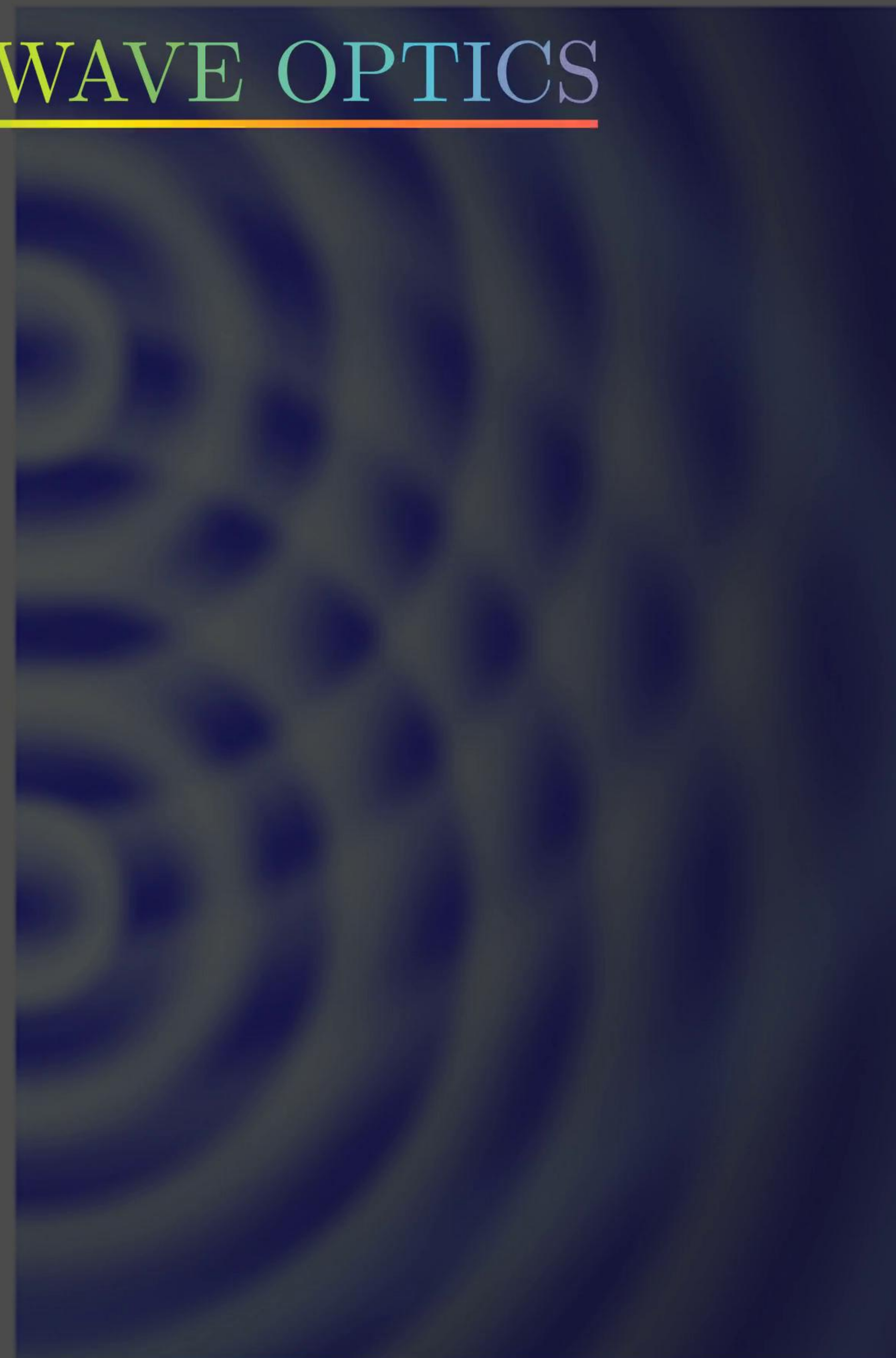


CHAPTER 10 : WAVE OPTICS



(a)

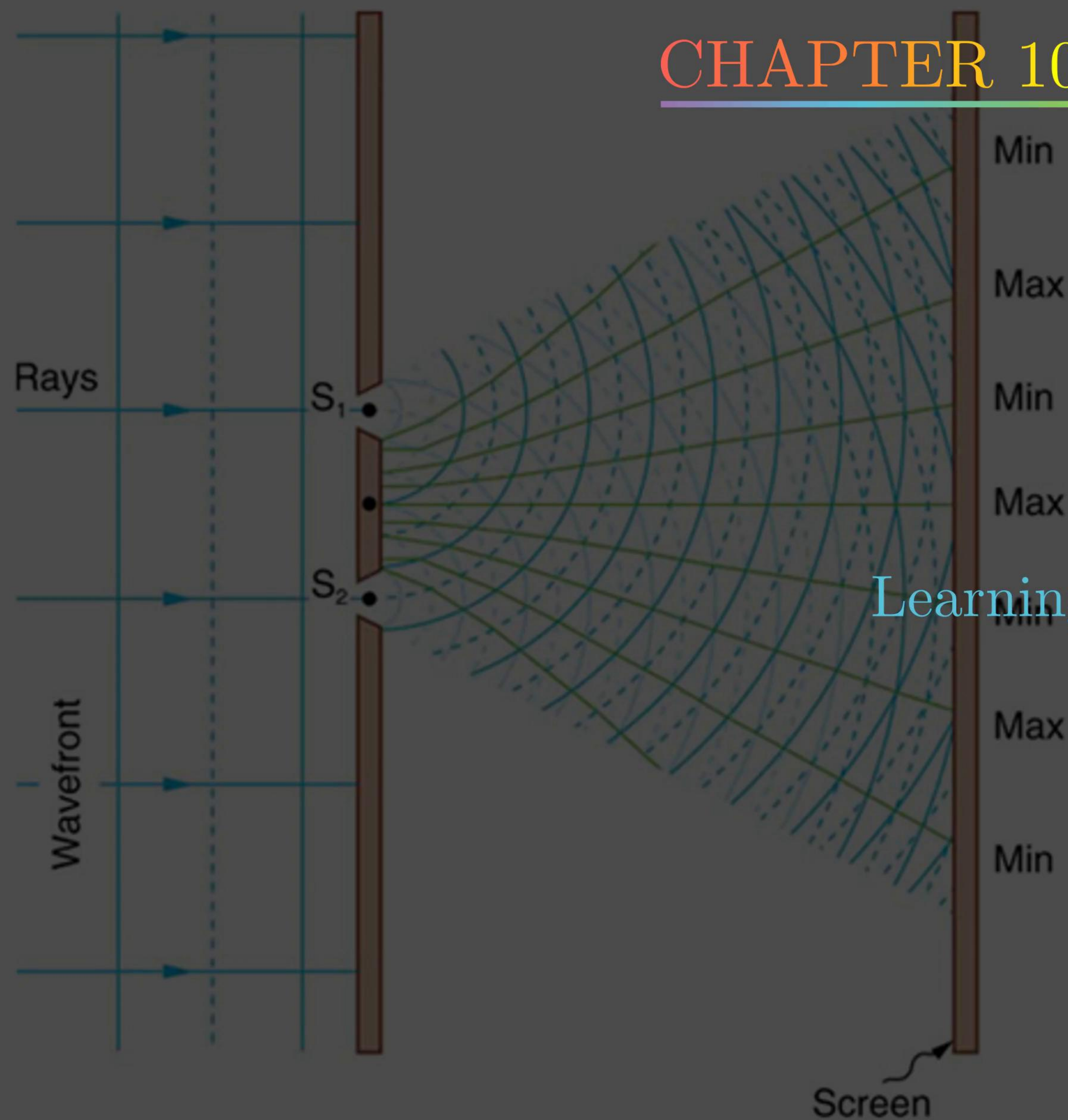


(b)



(c)

CHAPTER 10 : WAVE OPTICS

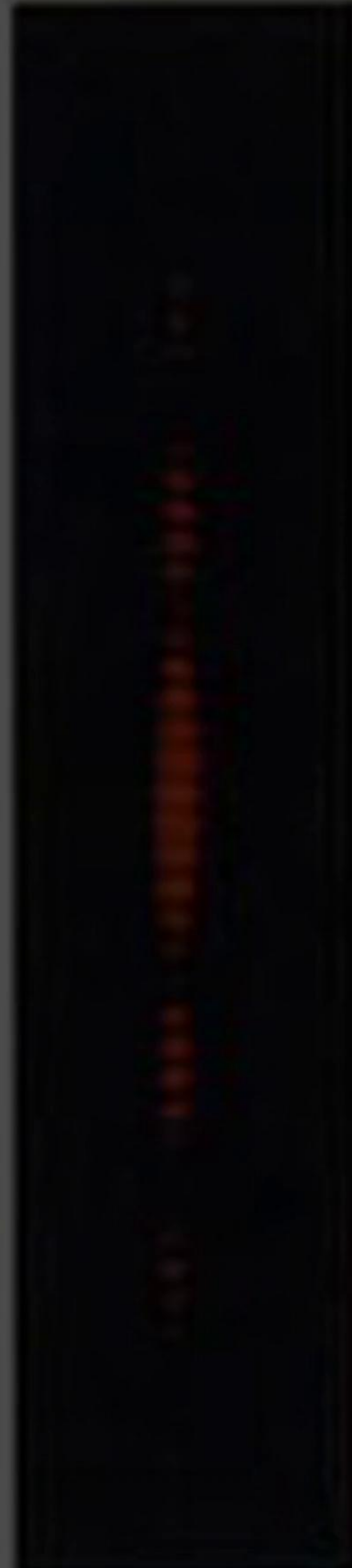


(a)

Learning Objectives :



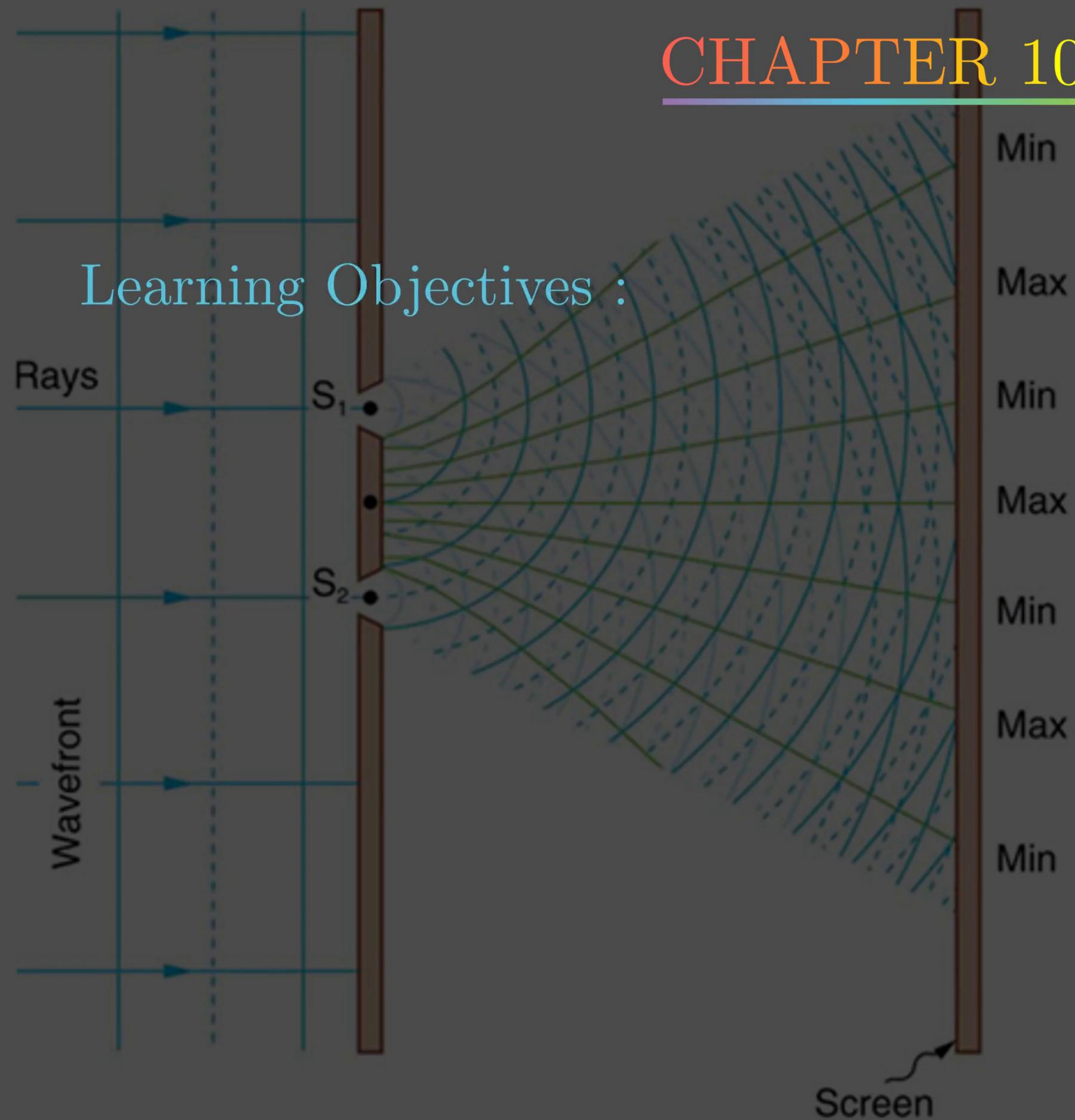
(b)



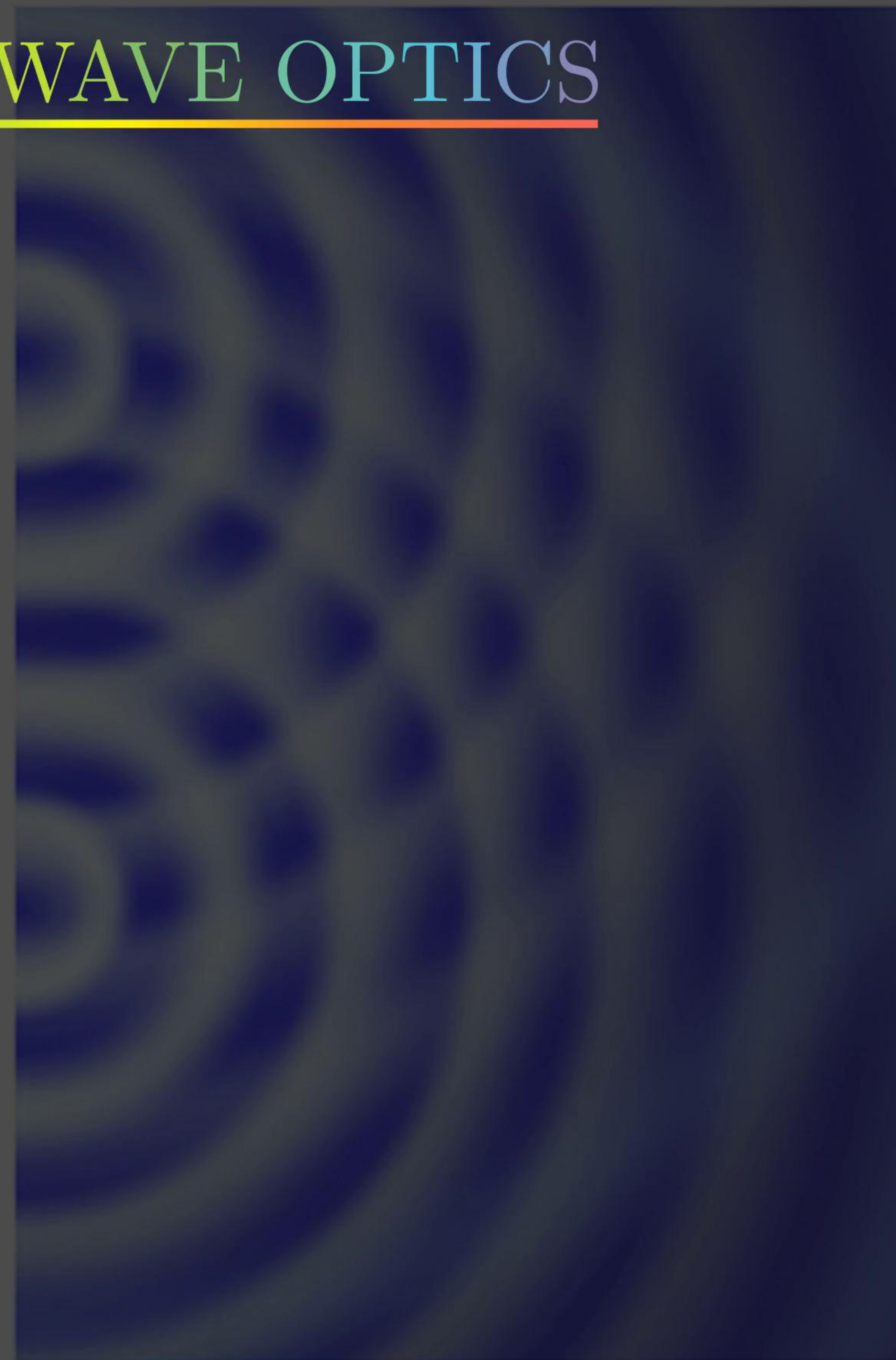
(c)

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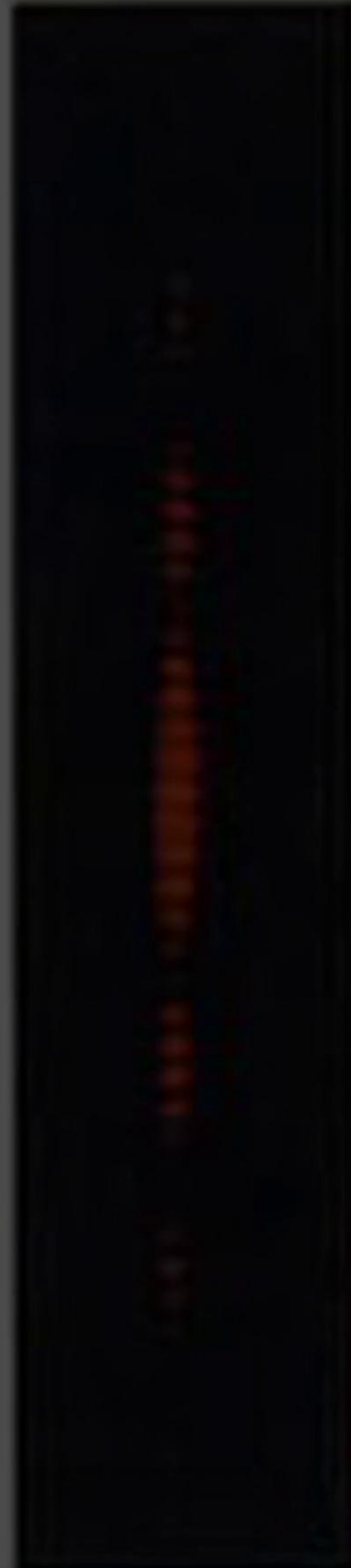
Learning Objectives :



(a)



(b)

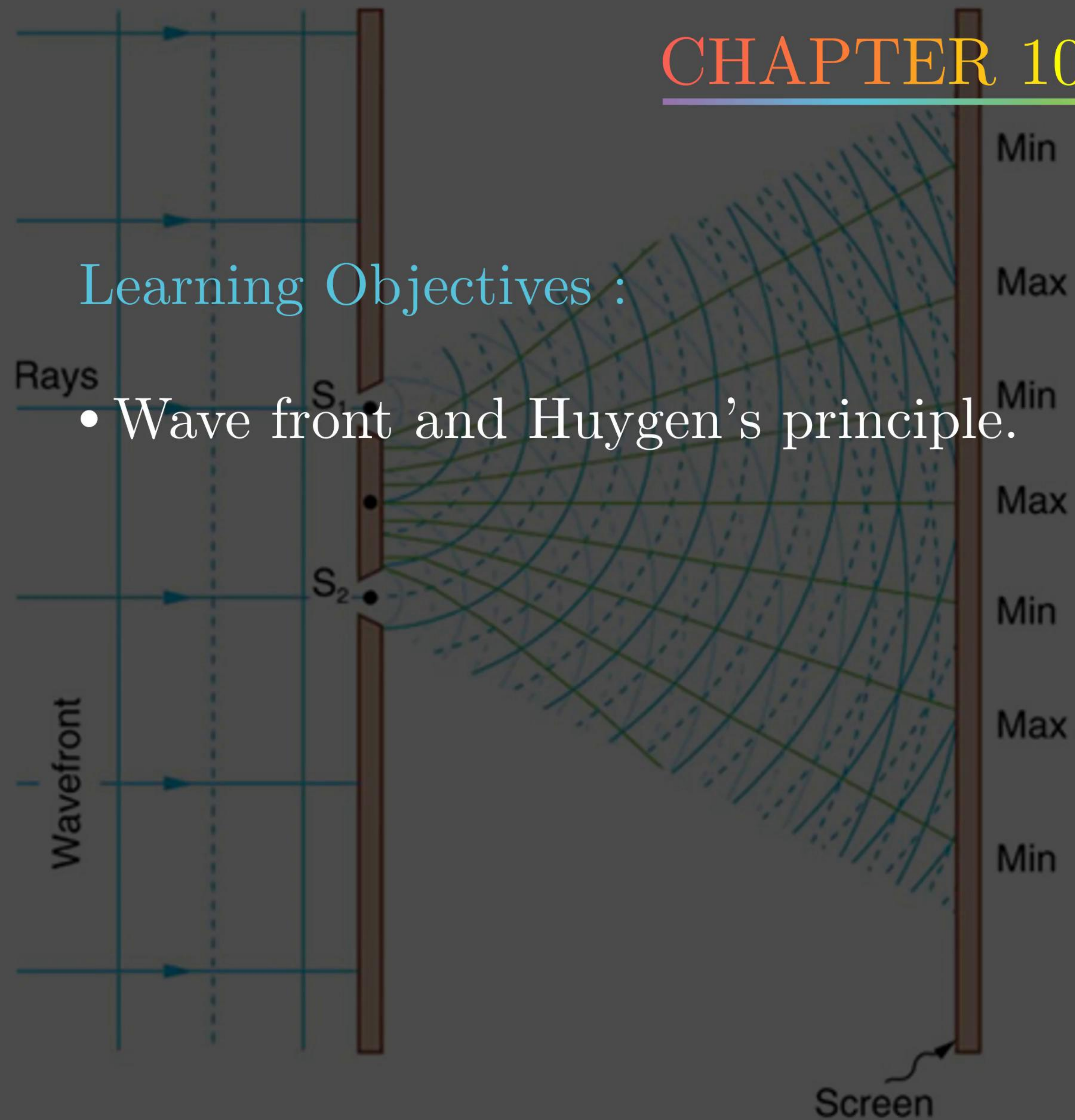


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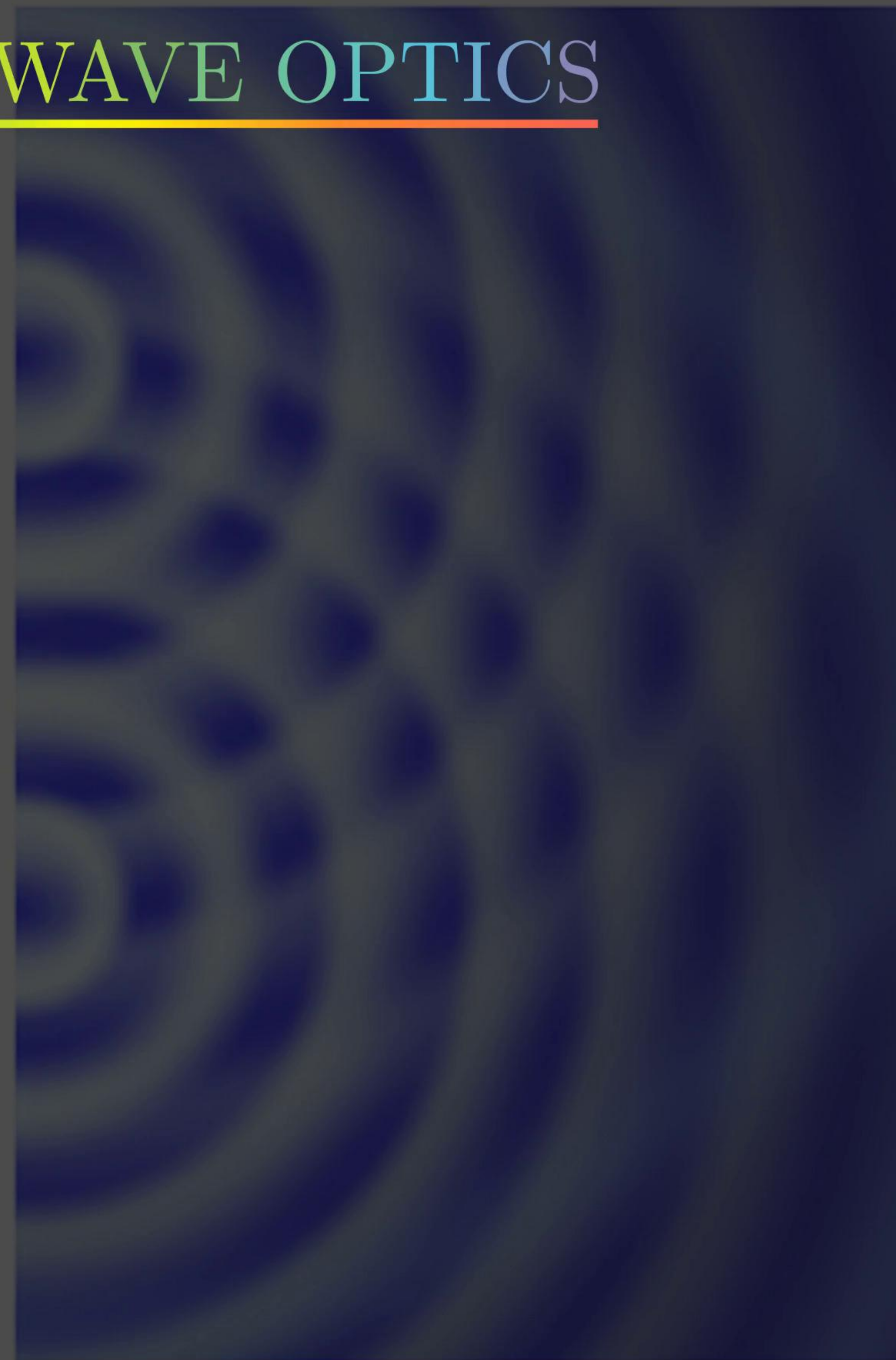
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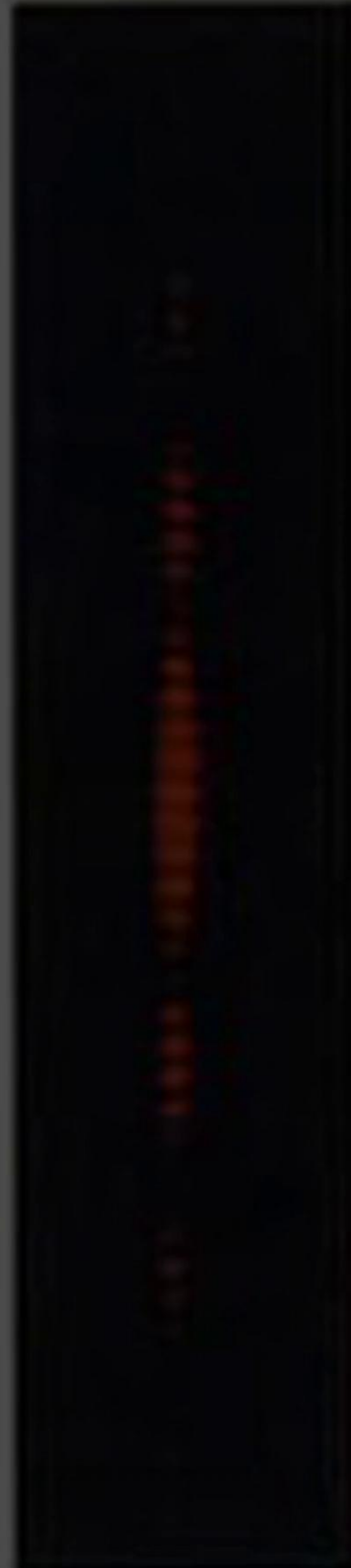
- Wave front and Huygen's principle.



(a)



(b)

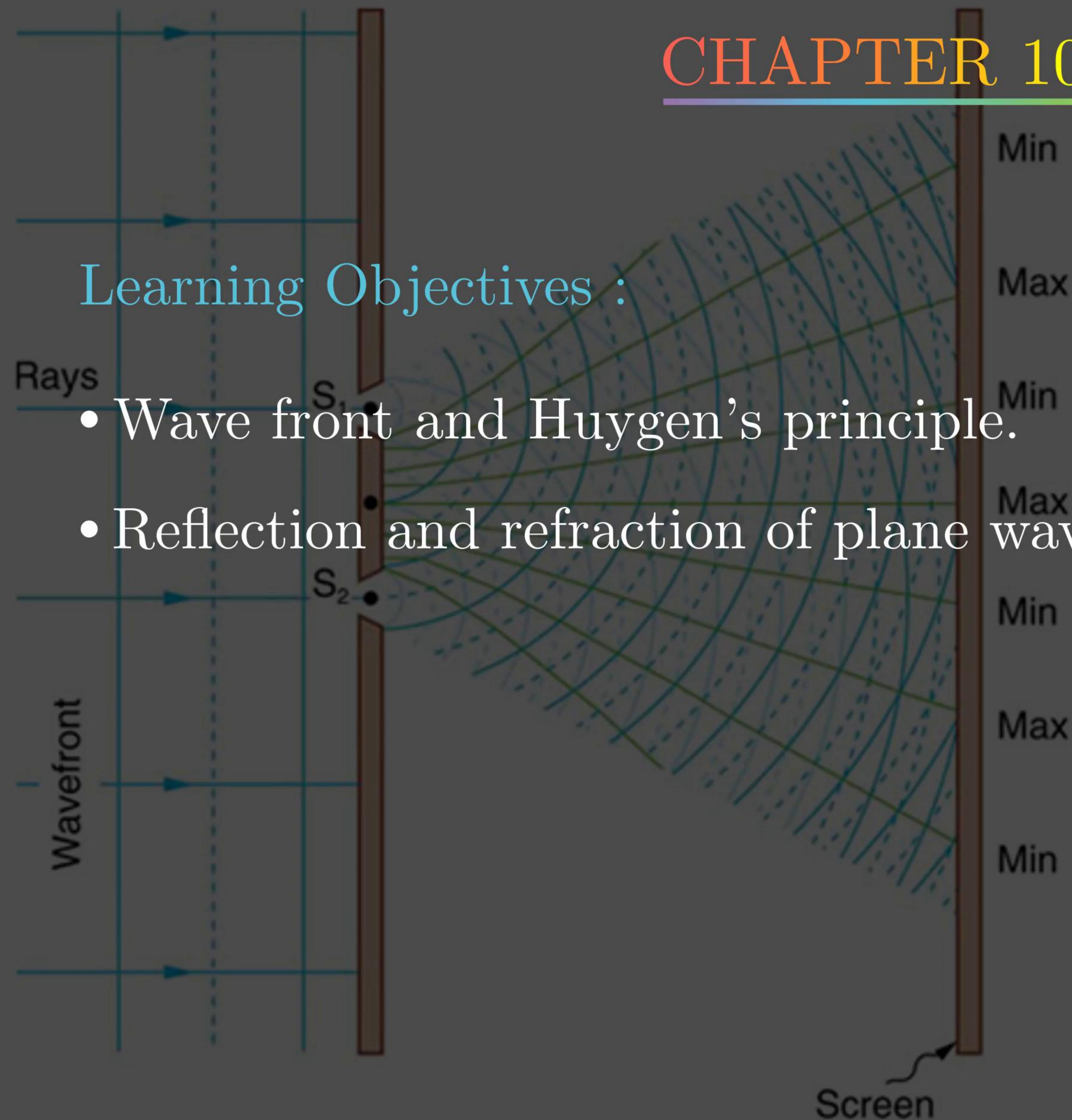


(c)

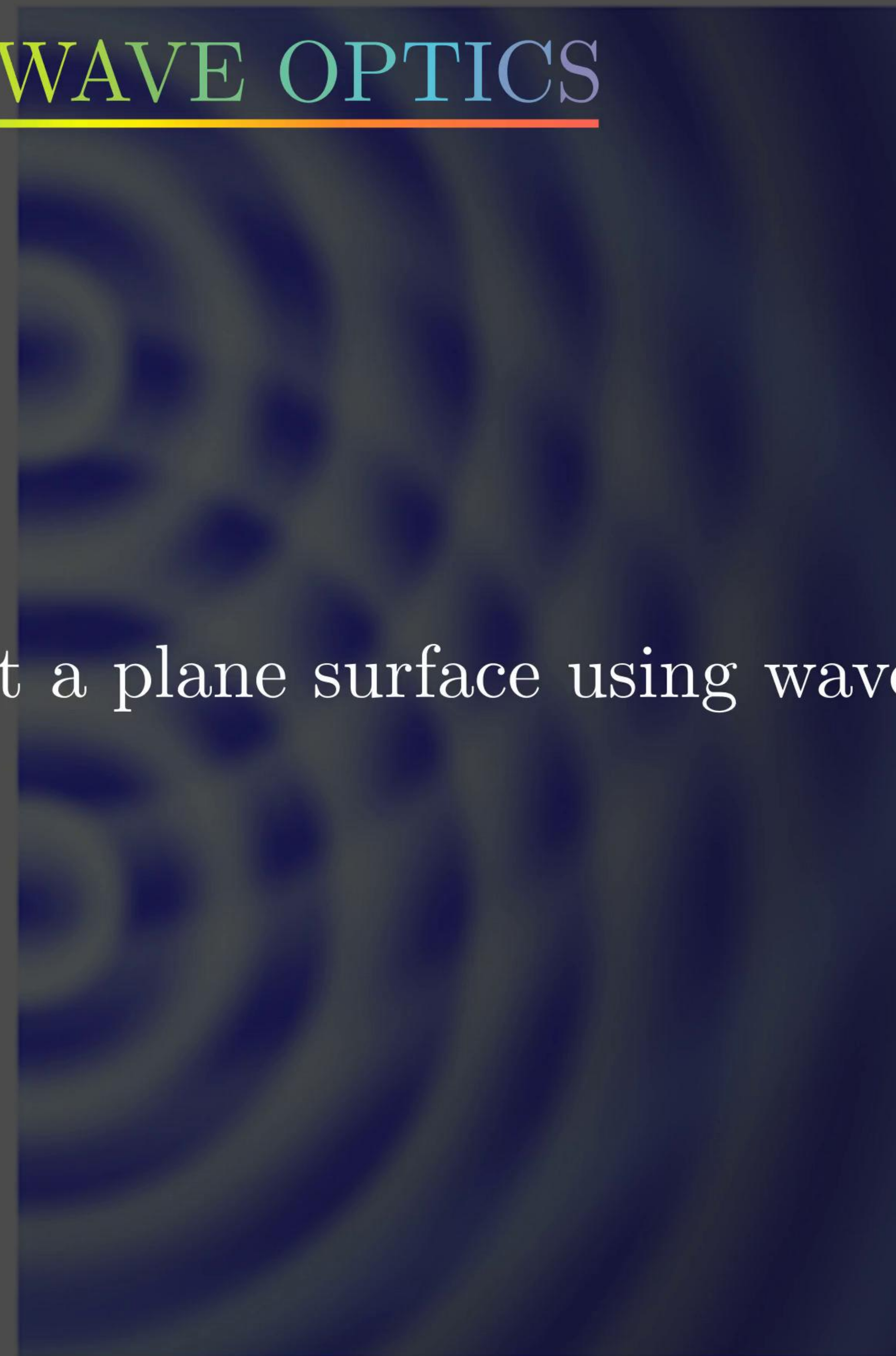
CHAPTER 10 : WAVE OPTICS

Learning Objectives :

- Wave front and Huygen's principle.
- Reflection and refraction of plane wave at a plane surface using wave fronts.



(a)



(b)

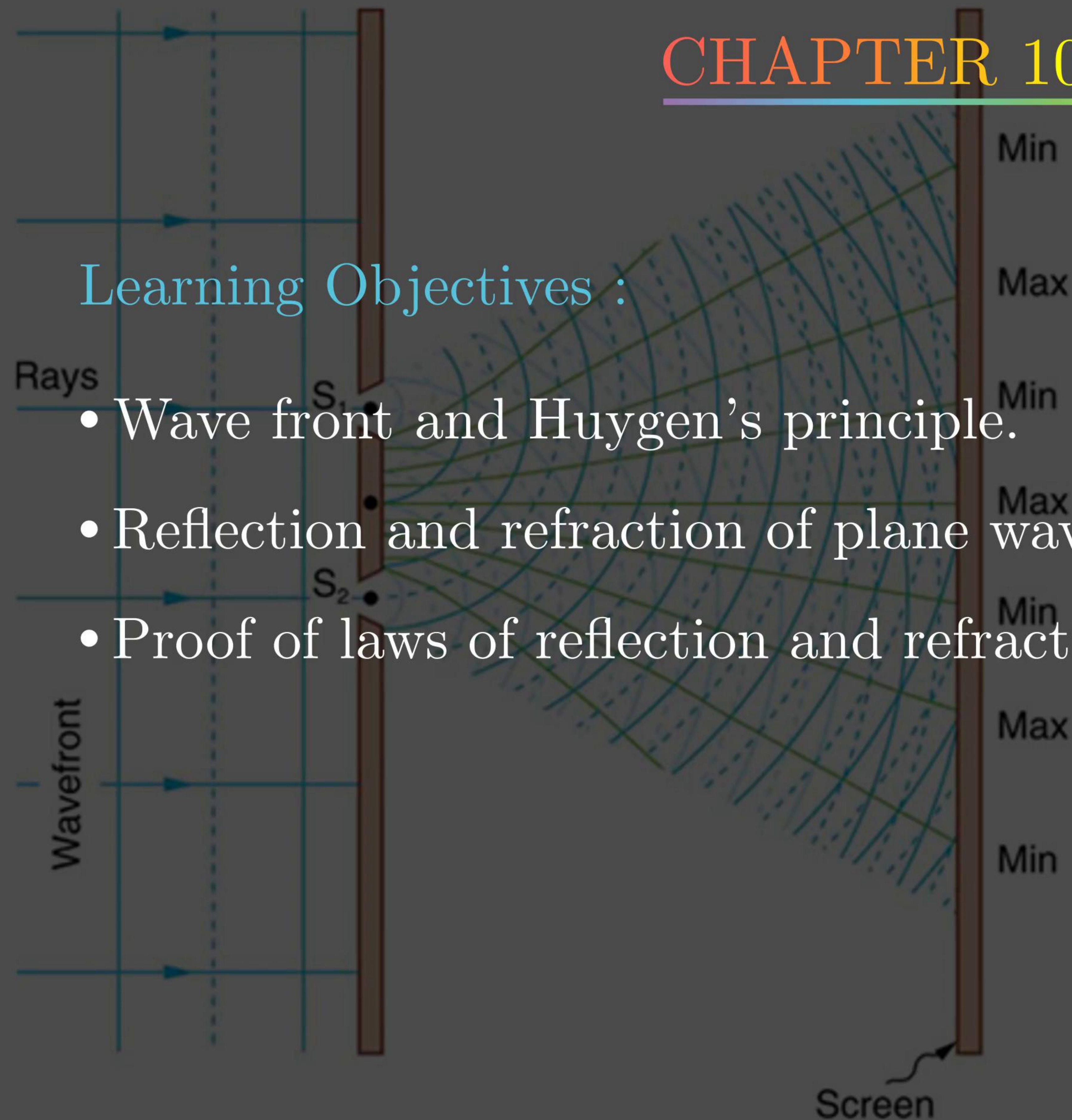


(c)

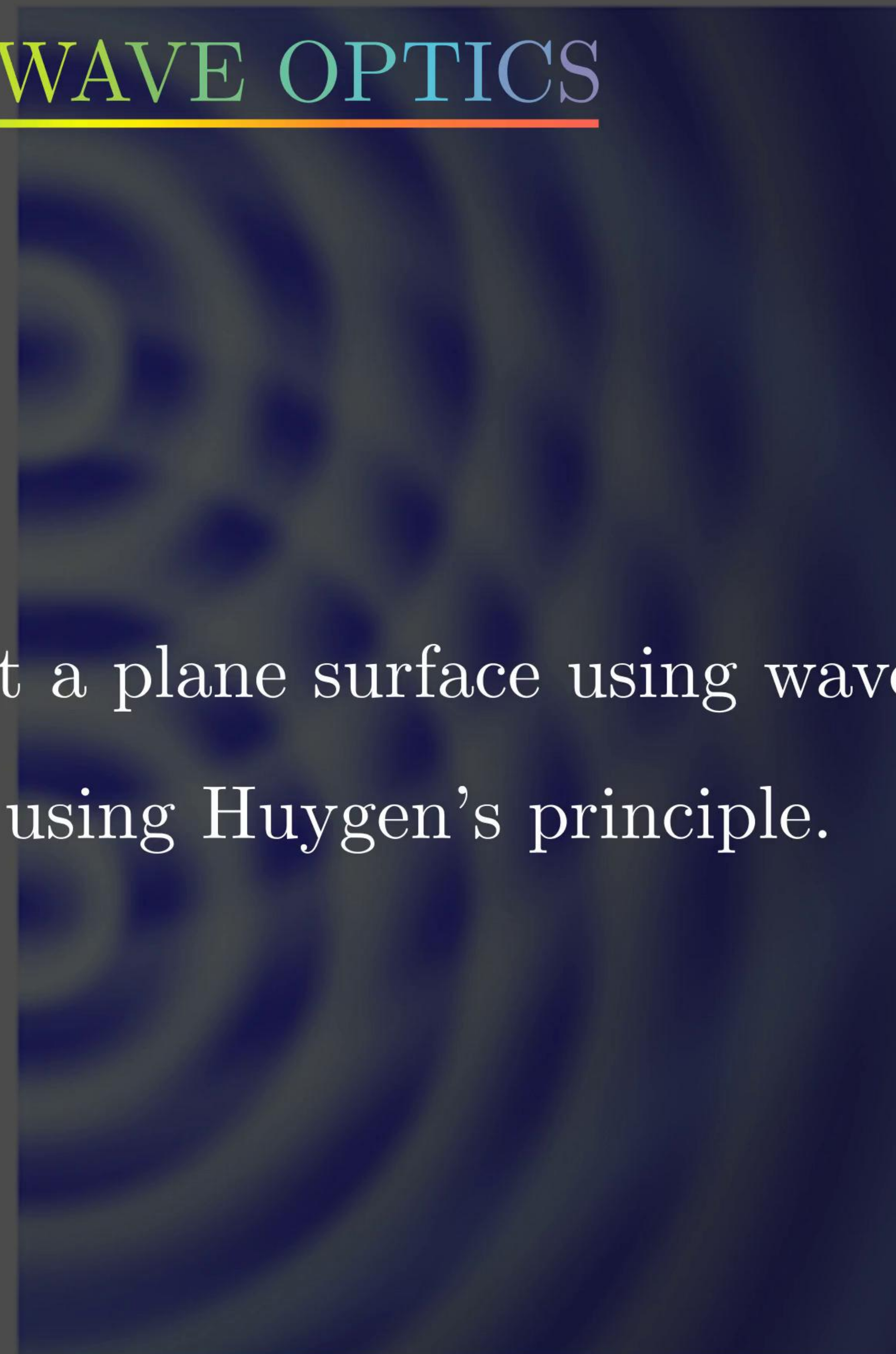
CHAPTER 10 : WAVE OPTICS

Learning Objectives :

- Wave front and Huygen's principle.
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- Proof of laws of reflection and refraction using Huygen's principle.



(a)



(b)

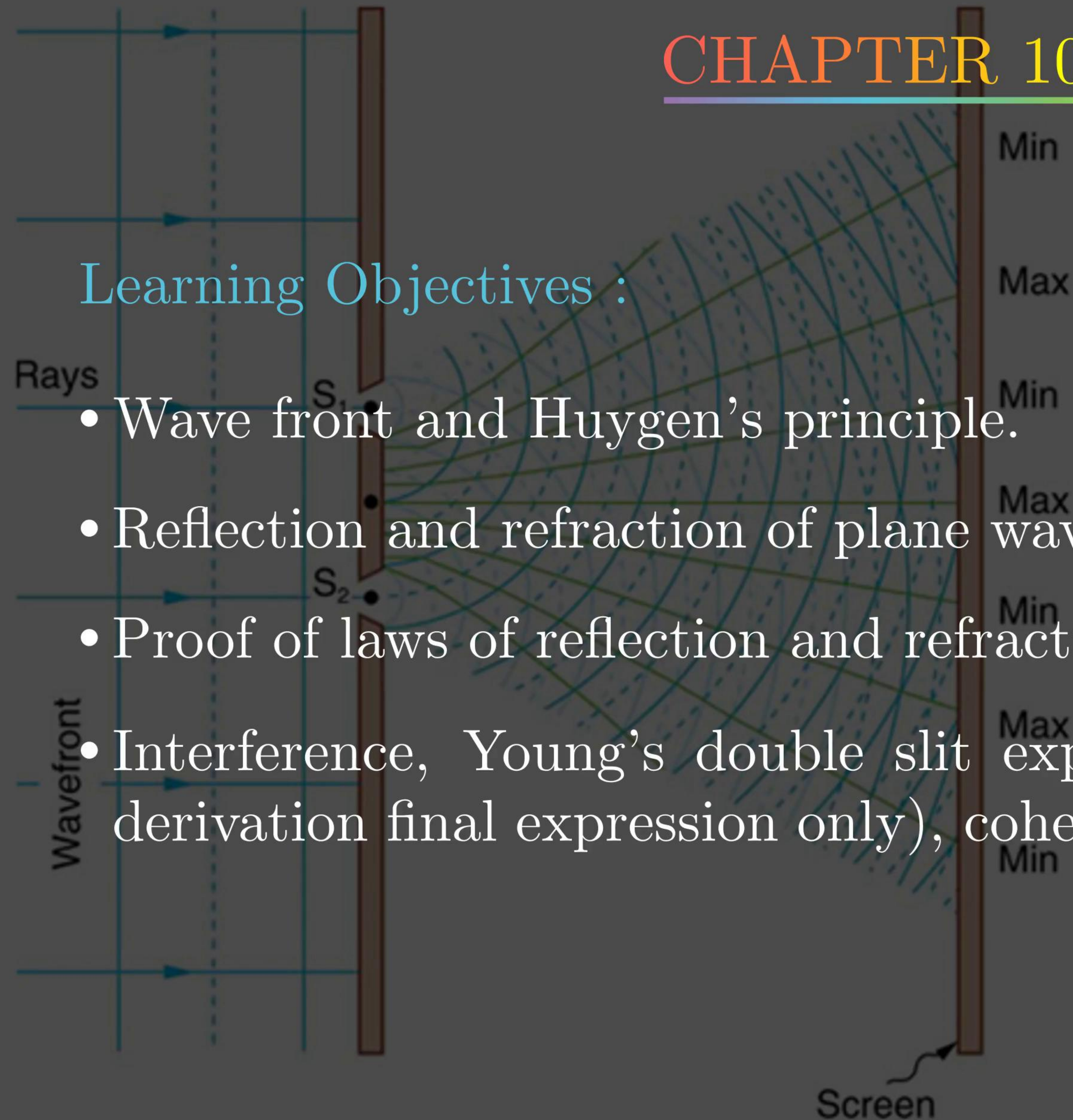


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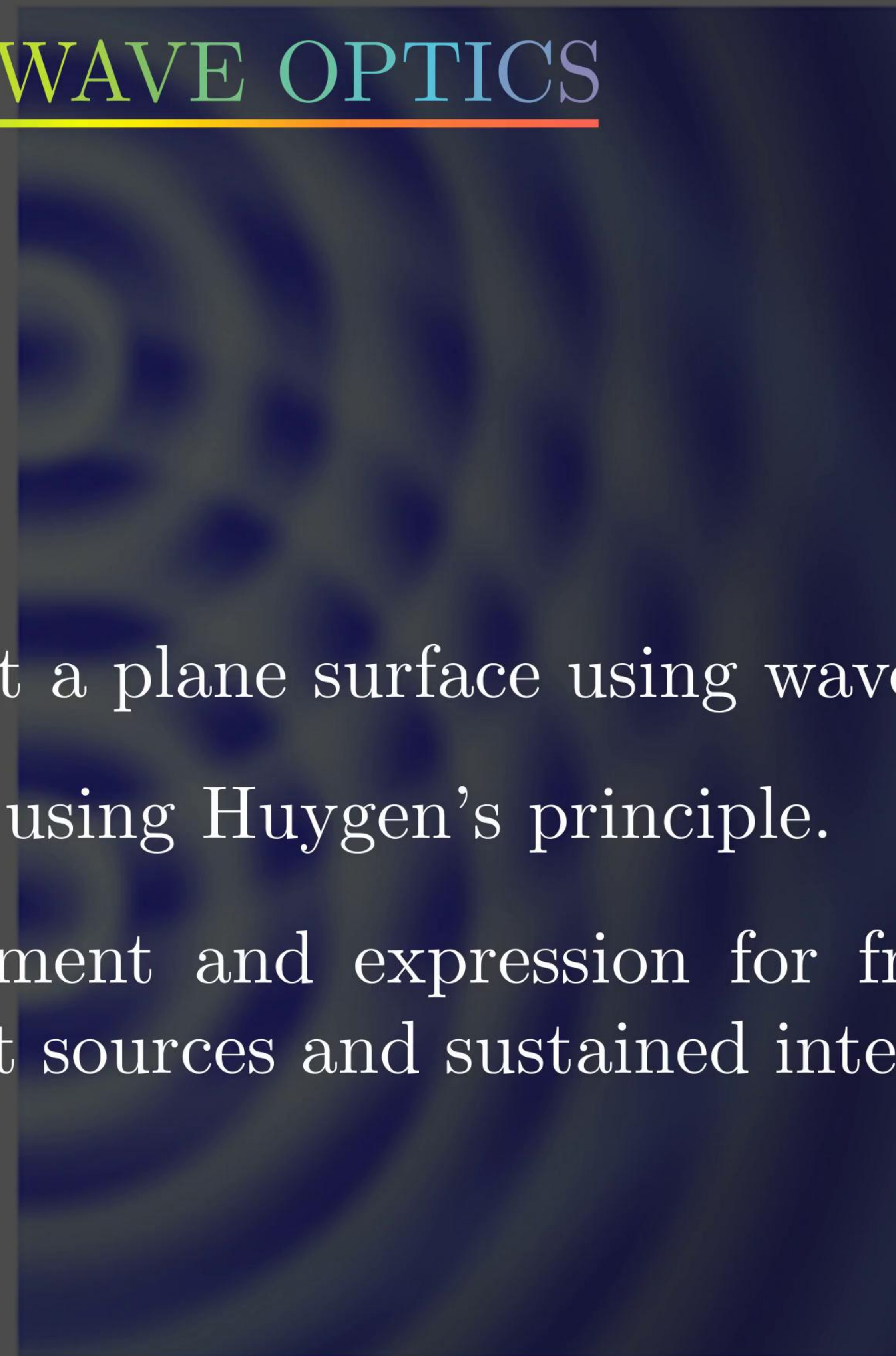
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Learning Objectives :

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- Interference, Young's double slit experiment and expression for fringe width (No derivation final expression only), coherent sources and sustained interference of light.



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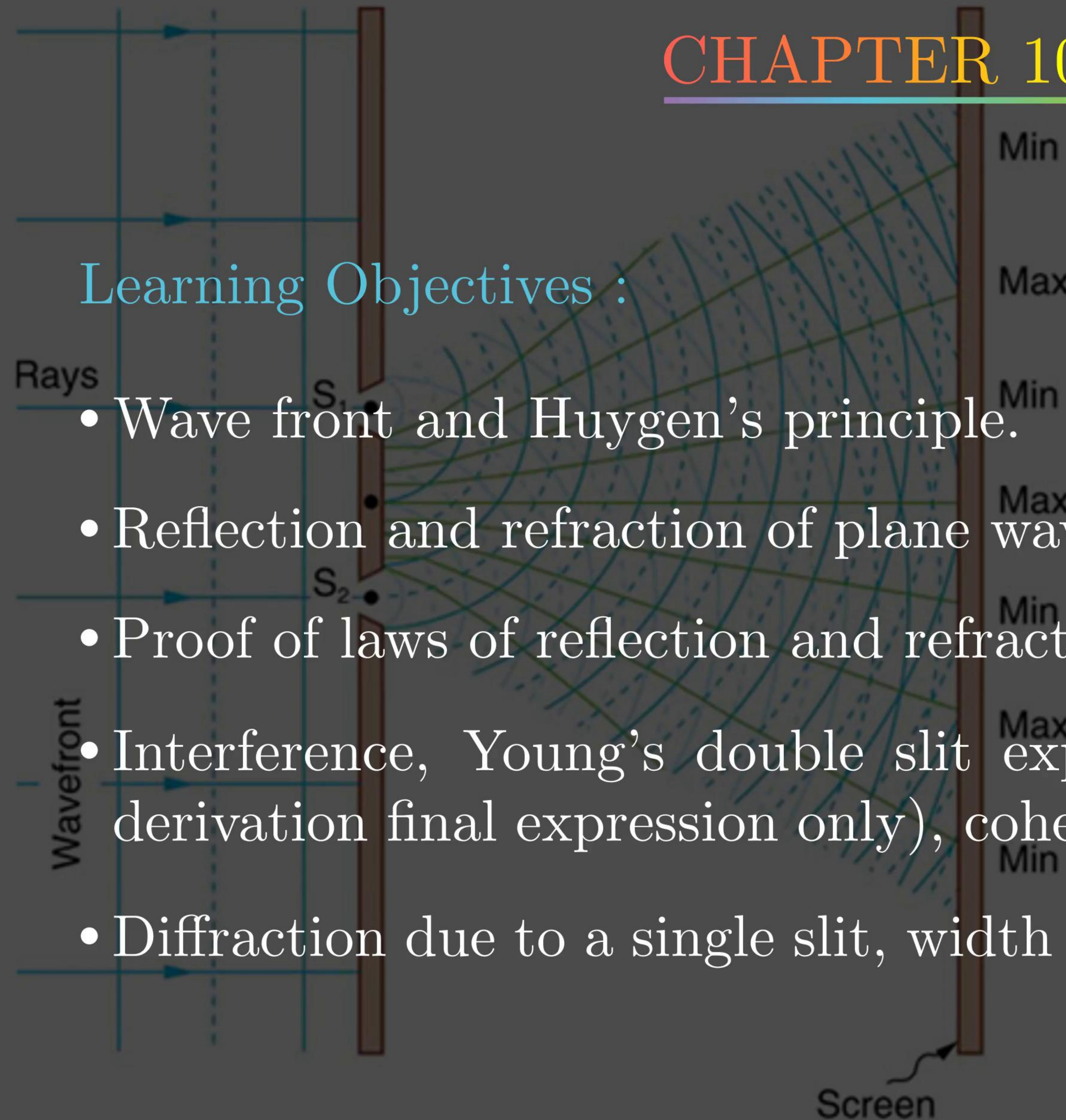


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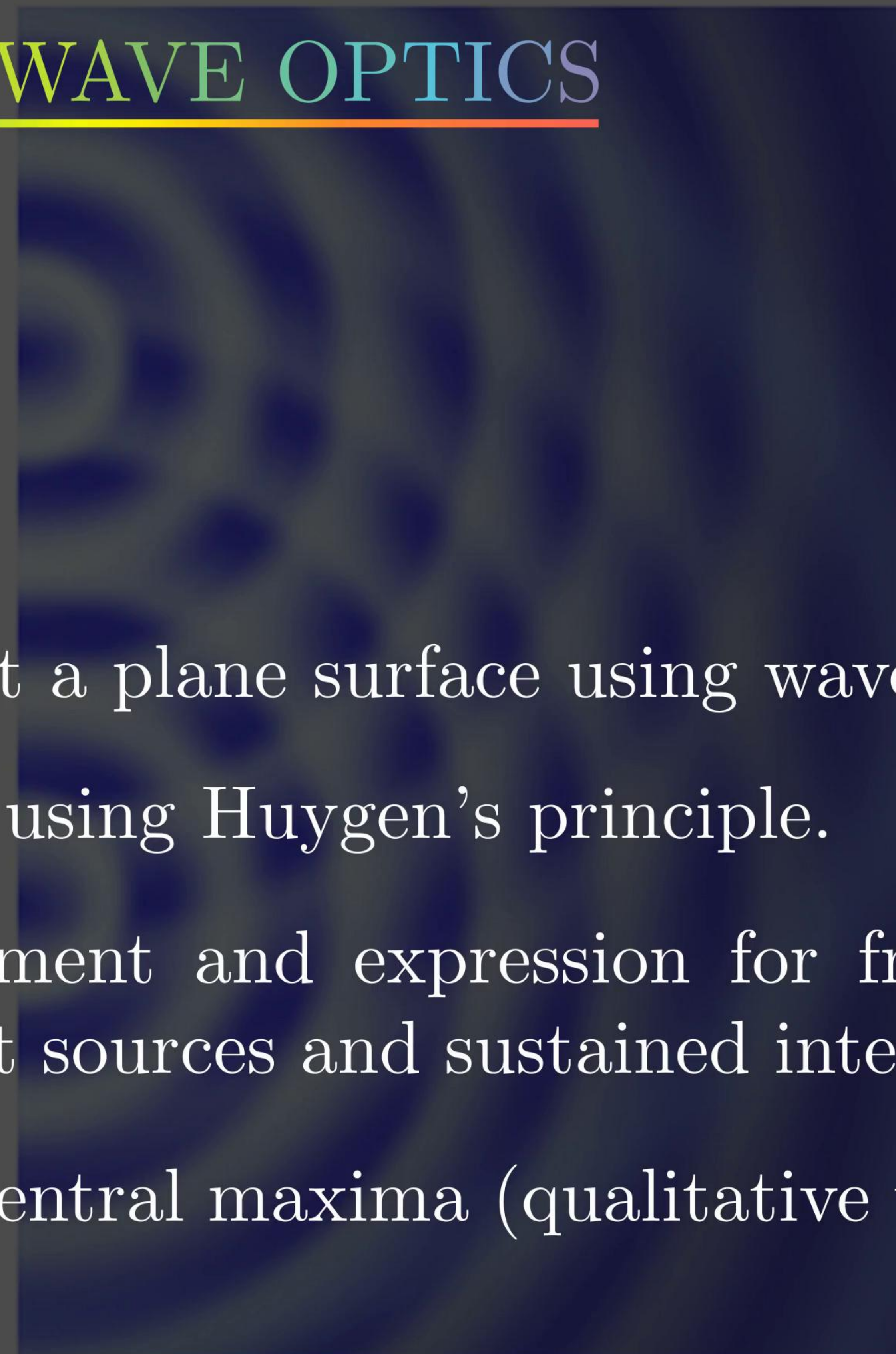
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- Diffraction due to a single slit, width of central maxima (qualitative treatment only).



(a)



(b)



(c)

WAVE OPTICS

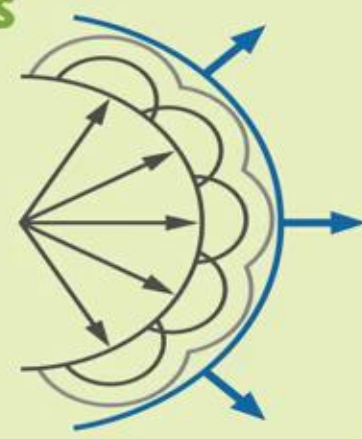
HUYGEN'S wave Theory

According to HUYGEN'S PRINCIPLE a body emits light in the form of waves. Each Point Source of light is a center of disturbance from which waves propagate in all direction.

Wave front

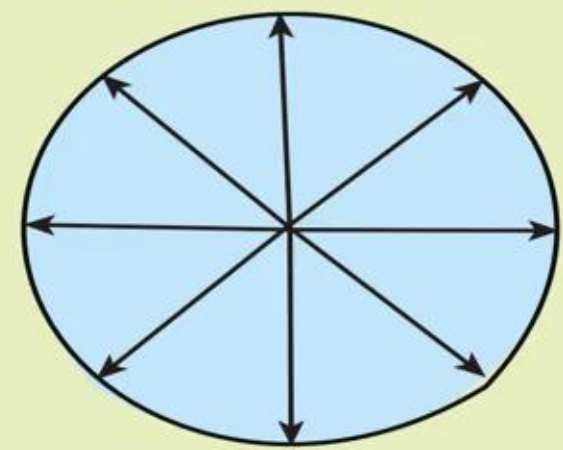
A wavefront is a surface along which the waves phase remains constant.

- (i) The energy of wave travels in a direction perpendicular to wavefront.
- (ii) Rays are perpendicular to wavefront.
- (iii) The time taken by light to travel from one wavefront to another is the same along anyray.



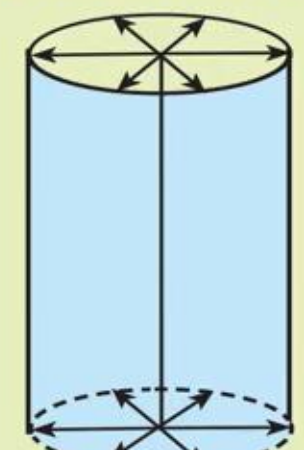
Types of wavefronts

Spherical wavefront



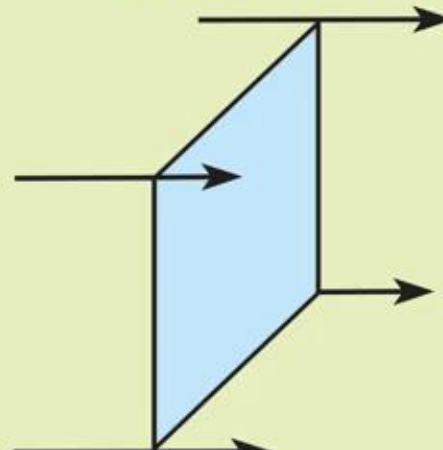
(Due to point source of light).

Cylindrical wavefront



(Due to line source of light).

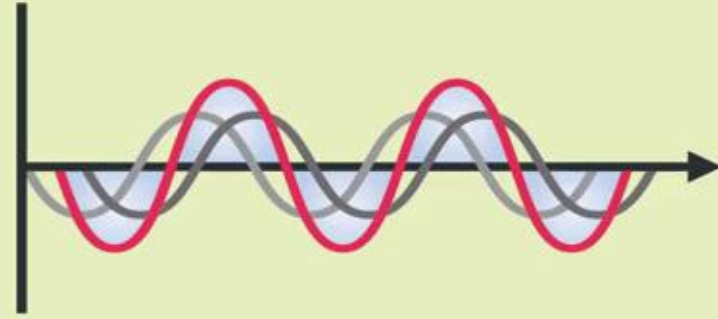
Plane wavefront



(Due to line source of light).

INTERFERENCE

Interference is a phenomenon of superposition of two coherent waves through which they transfer energy and momentum.



MATHEMATICAL INTERPRETATION OF INTERFERENCE OF TWO WAVES

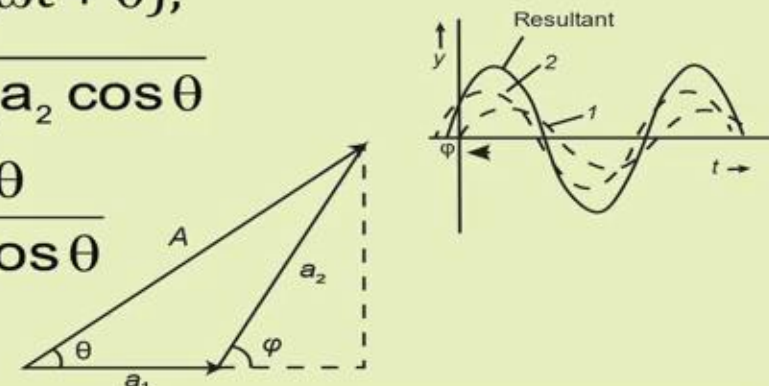
Let a_1 and a_2 be amplitudes of the waves and ϕ the phase difference between them.

$$y_1 = a_1 \sin \omega t; y_2 = a_2 \sin(\omega t + \phi);$$

$$Y = y_1 + y_2 = A \sin(\omega t + \theta);$$

$$A = \sqrt{a_1^2 + a_2^2 + 2a_1a_2 \cos \theta}$$

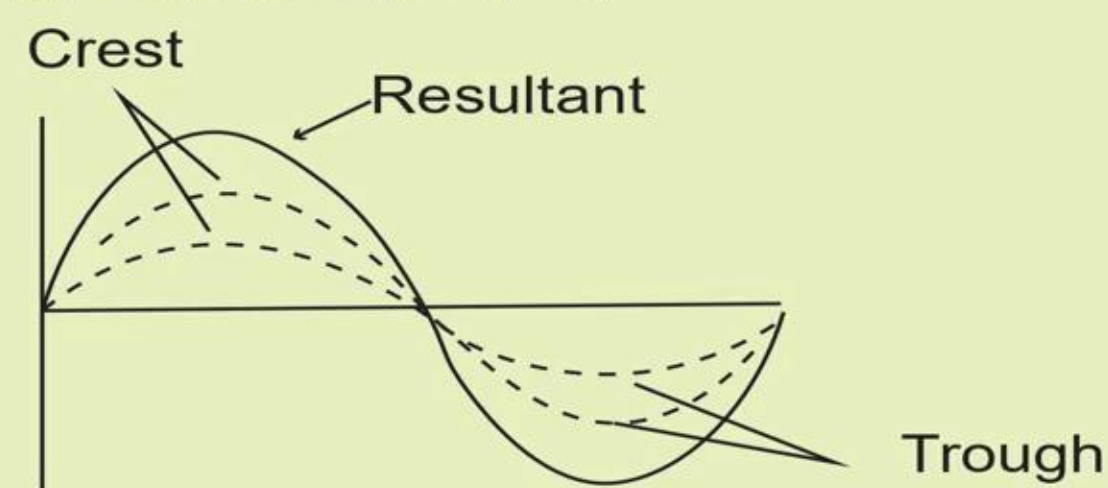
$$\tan \theta = \frac{a_2 \sin \theta}{a_1 + a_2 \cos \theta}$$



TYPES OF INTERFERENCE

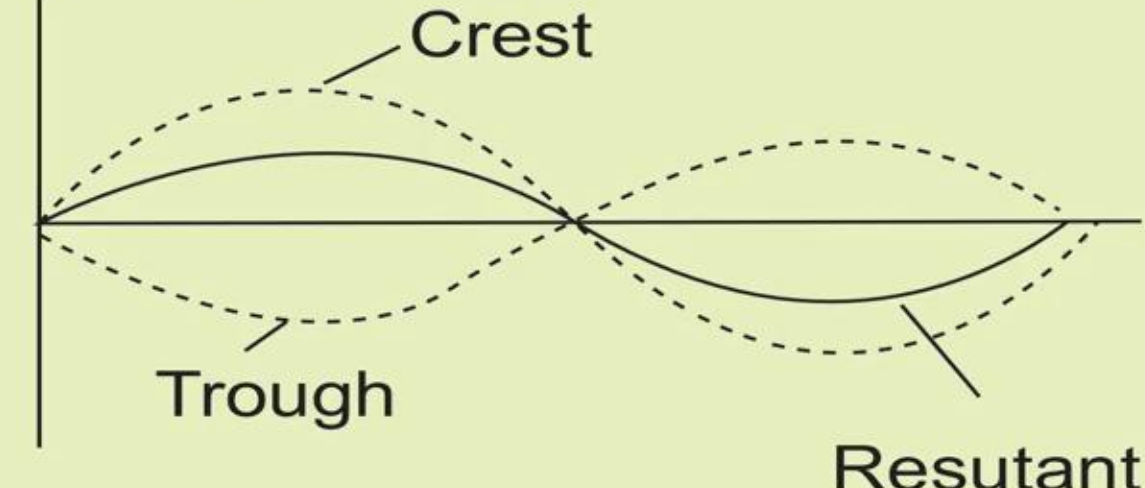
Constructive Interference

- Phase difference $\rightarrow (\Delta\phi) = 2n\pi; n = 0, 1, 2, \dots$
- Path difference $\rightarrow \Delta x = 2n \left(\frac{\lambda}{2} \right)$
- Time Interval $\rightarrow \Delta T = 2n \left(\frac{T}{2} \right)$
- Resultant Amplitude $\rightarrow A = a_1 + a_2$; if $\phi = 0, 2\pi, 4\pi, \dots, 2n\pi$
- Resultant Intensity $\rightarrow I_{\max} = I_1 + I_2 + 2\sqrt{I_1 I_2} = (\sqrt{I_1} + \sqrt{I_2})^2$
- $I_{\max} = 4I$ where $(I_1 = I_2 = I)$



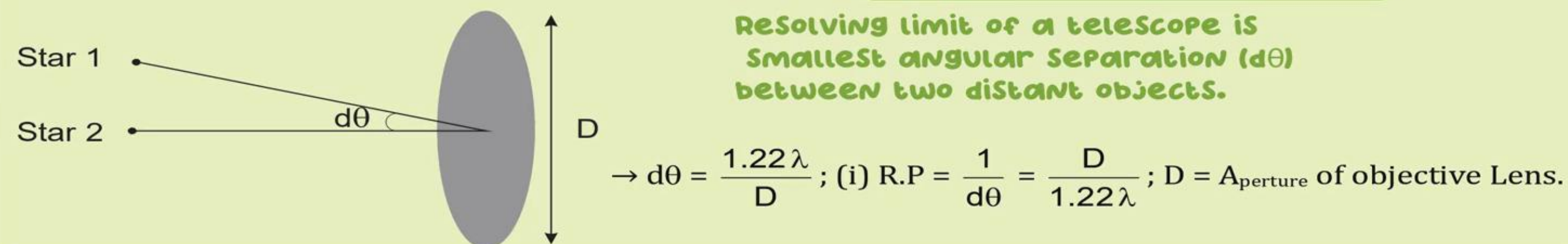
Destructive Interference

- Phase difference $\rightarrow (\Delta\phi) = (2n - 1)\pi$, Where $n = 1, 2, 3, \dots$
- Path difference $\rightarrow \Delta x = (2n - 1) \frac{\lambda}{2}$
- Time Interval $\rightarrow \Delta T = (2n - 1) \frac{T}{2}$
- Resultant Amplitude $\rightarrow A = a_1 - a_2$; If $\phi = \pi, 3\pi, 5\pi, \dots, (2n - 1)\pi$
- Resultant Intensity $\rightarrow I_{\max} = I_1 + I_2 - 2\sqrt{I_1 I_2} = (\sqrt{I_1} - \sqrt{I_2})^2$
- $I_{\min} = 0$ (When $I_1 = I_2 = I$)



R.P for Telescope

Resolving limit of a telescope is smallest angular separation ($d\theta$) between two distant objects.



YOUNG'S DOUBLE SLIT EXPERIMENT

(i) For Bright Fringes

$$S_2P - S_1P = \frac{X_n d}{D} = n\lambda; d = \text{slit width}$$

$$X_n = \frac{n\lambda D}{d}; X_n =$$

Distance between Central fringe and Nth Bright fringe : λ = wavelength

= Bright fringes are also called maxima's.

(ii) For Dark Fringes $S_2P - S_1P = \frac{X_n d}{D} = \frac{(2n - 1)\lambda}{2}$

$$\Rightarrow X_n = \frac{(2n - 1)D}{d}$$

X_n = Distance between central fringe and Nth dark fringe

= Dark fringes are also called minima's

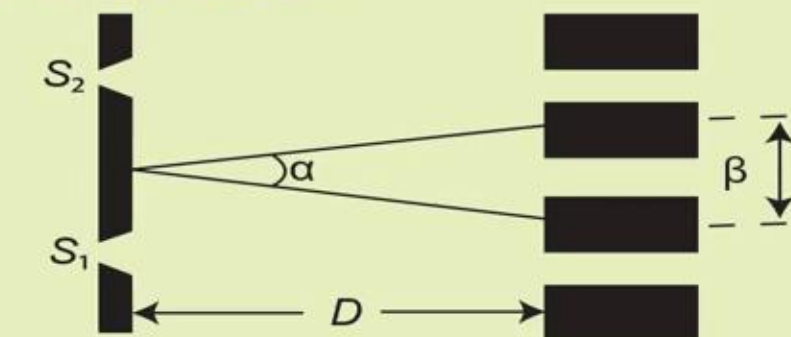
Fringe width:-

Fringe width of dark & bright fringes are same and given by

$$\beta = X_n - X_{n-1} = \frac{\lambda D}{d}$$

Angular width of fringe:-

$$\alpha = \frac{\beta}{d} = \frac{\lambda}{d}$$



Resolving Power (R.P)

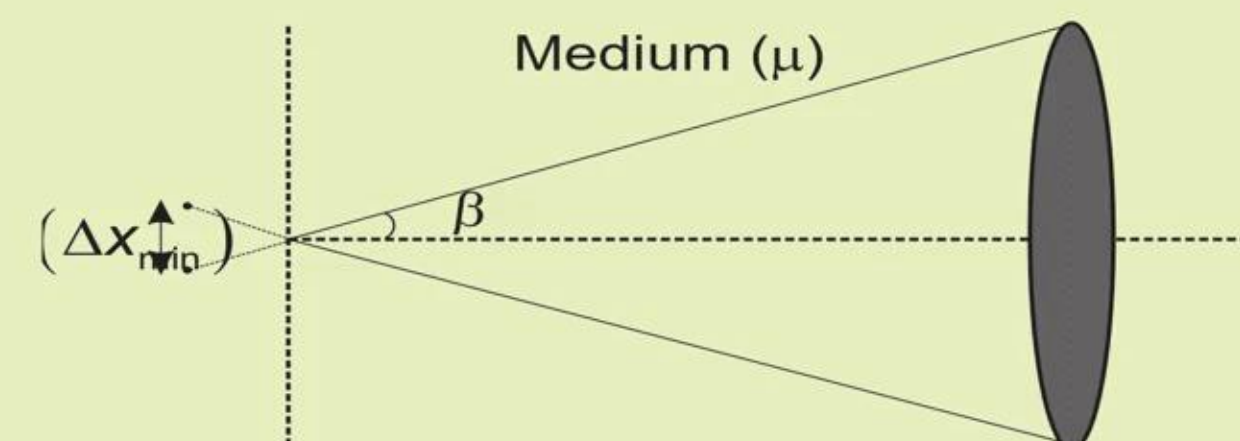
Resolving Power an optical instrument is its ability to distinguish two closely placed point.

R.P for Microscope

(i) The minimum distance to form separate images of two objects.

$$\Delta X_{\min} = \frac{1.22 \lambda}{2\mu \sin \beta}$$

$$R.P = \frac{1}{\Delta X_{\min}} = \frac{2\mu \sin \beta}{1.22 \lambda}$$



DIFFRACTION

Bending of light waves around the sharp edges of opaque obstacles or aperture and their encroachment in the geometrical shadow of obstacles or aperture.

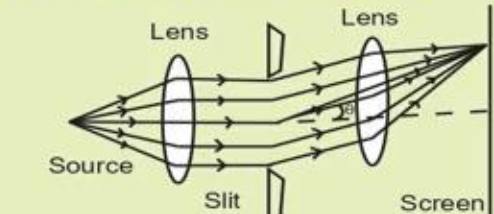
(i) Necessary Condition:- Size of obstacle

(a) must be the order of wavelength (λ) . i.e $\frac{a}{\lambda} > 1$

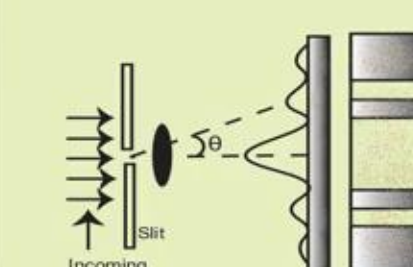


Types of Diffraction

Fresnel Diffraction:- Fresnel Diffraction involves spherical wavefronts. So that source 'S' and point 'P' are at finite distance.



Fraunhofer Diffraction:- It deals with plane wavefronts and an effective viewing distance of infinity.



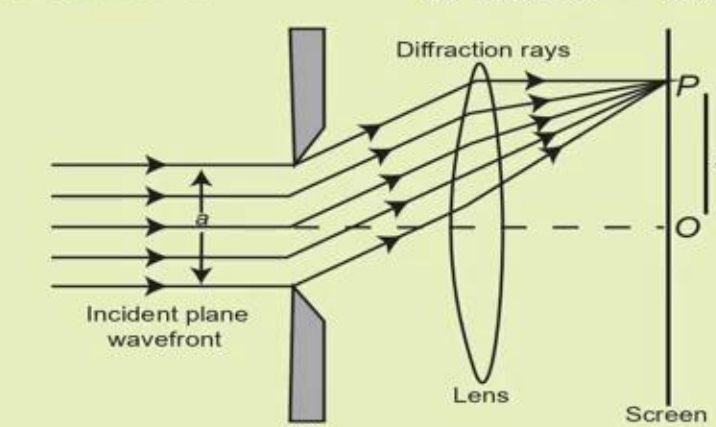
FRAUNHOFER DIFFRACTION FOR SINGLE SLIT

In this diffraction pattern central maxima is bright on the both side of it. maxima & minima occurs symmetrically.

(i) Position of Secondary Maxima in diffraction $\alpha \sin \theta = (2n - 1) \frac{\lambda}{2}$

$$\Rightarrow X_n = \frac{(2n - 1)D}{2a} \lambda$$

(ii) Position of Secondary Minima in diffraction:- $\alpha \sin \theta = n\lambda \Rightarrow X_n = \frac{n\lambda D}{a}$

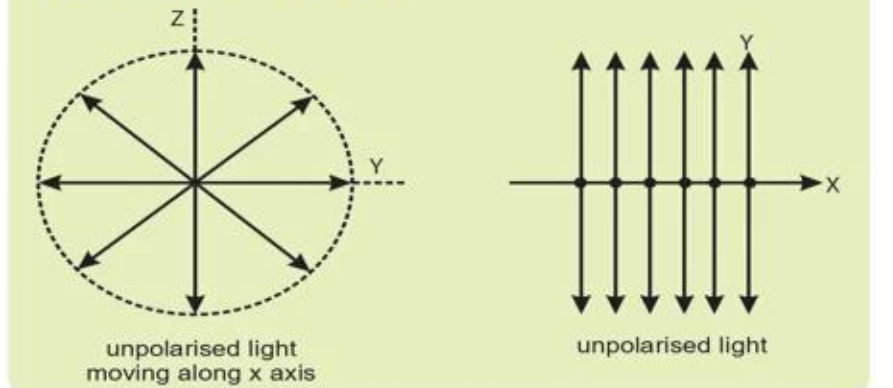


POLARISATION

The process of confining the vibrations of unpolarised light in one single plane using polariser is called polarisation.

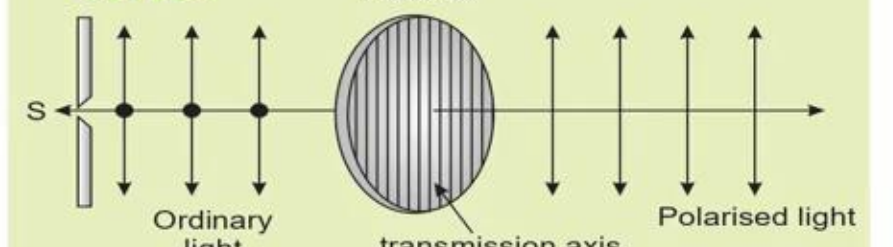
Unpolarised light

An ordinary beam of light consists of a large number of waves emitted by the atoms or molecules of the light source.



Plane Polarised light

Beam of light in which vibration of electric field vector are perpendicular to wave motion and confined to single plane only.

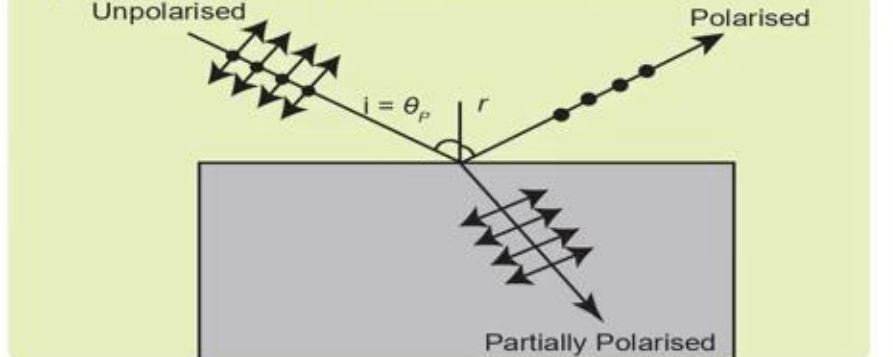


BREWSTER'S LAW

This law states that when light is incident on a transparent substance at polarising angle θ_p , the reflected light is completely plane polarised.

$\mu = \tan \theta_p$; θ_p = Polarising angle.

μ = Refractive Index of medium

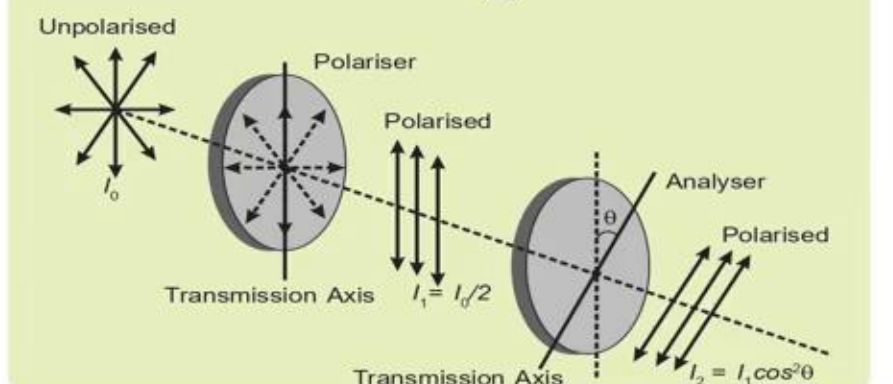


MALUS' LAW

$$I_2 = K (A \cos \theta)^2 = K A^2 \cos^2 \theta$$

$$I_2 = I_1 \cos^2 \theta; I = \text{intensity}$$

$$(i) I_{\text{polarised}} = \frac{I_{\text{unpolarised}}}{2}$$



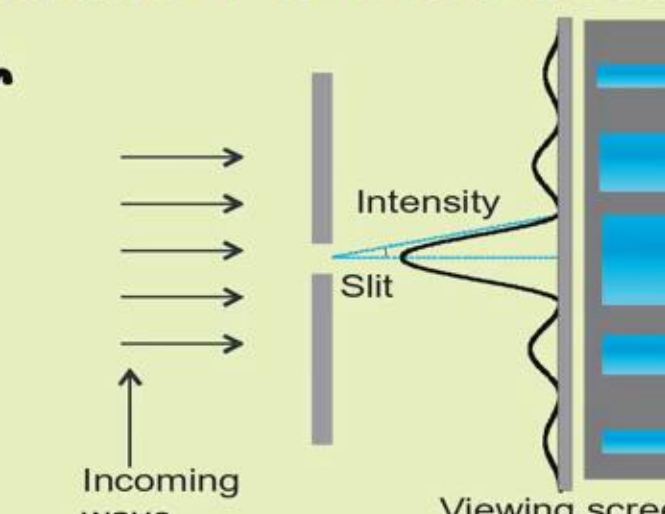
WIDTH OF CENTRAL MAXIMUM

The distance between two secondary minima formed on two sides of central maximum is known as width of central maximum.

$$W = \frac{2f\lambda}{a}$$

f = focal length of convex lenses

a = Slit width



Introduction

Historical Background of Theories of Light

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* Main Ideas

1. Light is made up of tiny material particles (corpuscles) emitted by luminous bodies.
2. These corpuscles travel in straight lines with very high speed.
3. It explained the laws of reflection and refraction of light at an interface.
4. The corpuscular model predicted that if the ray of light (on refraction) bends towards the normal then the speed of light would be greater in the second medium.

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* Isaac Newton: Developed Descartes' corpuscular theory further in his famous book "Opticks".
Due to his influence, it was widely accepted.

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Confirmed that light travels slower in water than in air, validating the wave theory.

- * Acceptance of Wave Theory

Initially rejected because:

- # Newton's authority.

- # Belief that waves require a medium, yet light travels through a vacuum.

1801 – Thomas Young:

Conducted the double-slit interference experiment, establishing the wave nature of light.

Introduction

* Geometrical Optics vs. Wave Optics

Wavelength of visible light: $\sim 0.6 \mu\text{m}$ (very small).

For large-scale objects (mirrors, lenses), light appears to travel in straight lines $\rightarrow$ Geometrical Optics.

When wavelength effects are considered (interference, diffraction, etc.) $\rightarrow$ Wave Optics.

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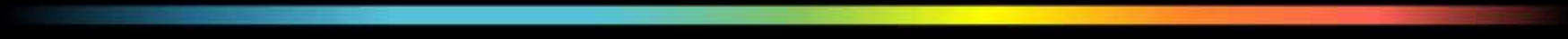
* Maxwell's Electromagnetic Theory (1860s)

Unified electricity and magnetism; predicted electromagnetic waves.

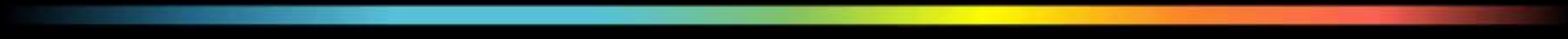
Calculated the speed of EM waves $\approx$ speed of light $\rightarrow$ light is an electromagnetic wave.

Light consists of oscillating electric and magnetic fields that propagate through space and vacuum.

Huygen's Principle



Huygen's Principle

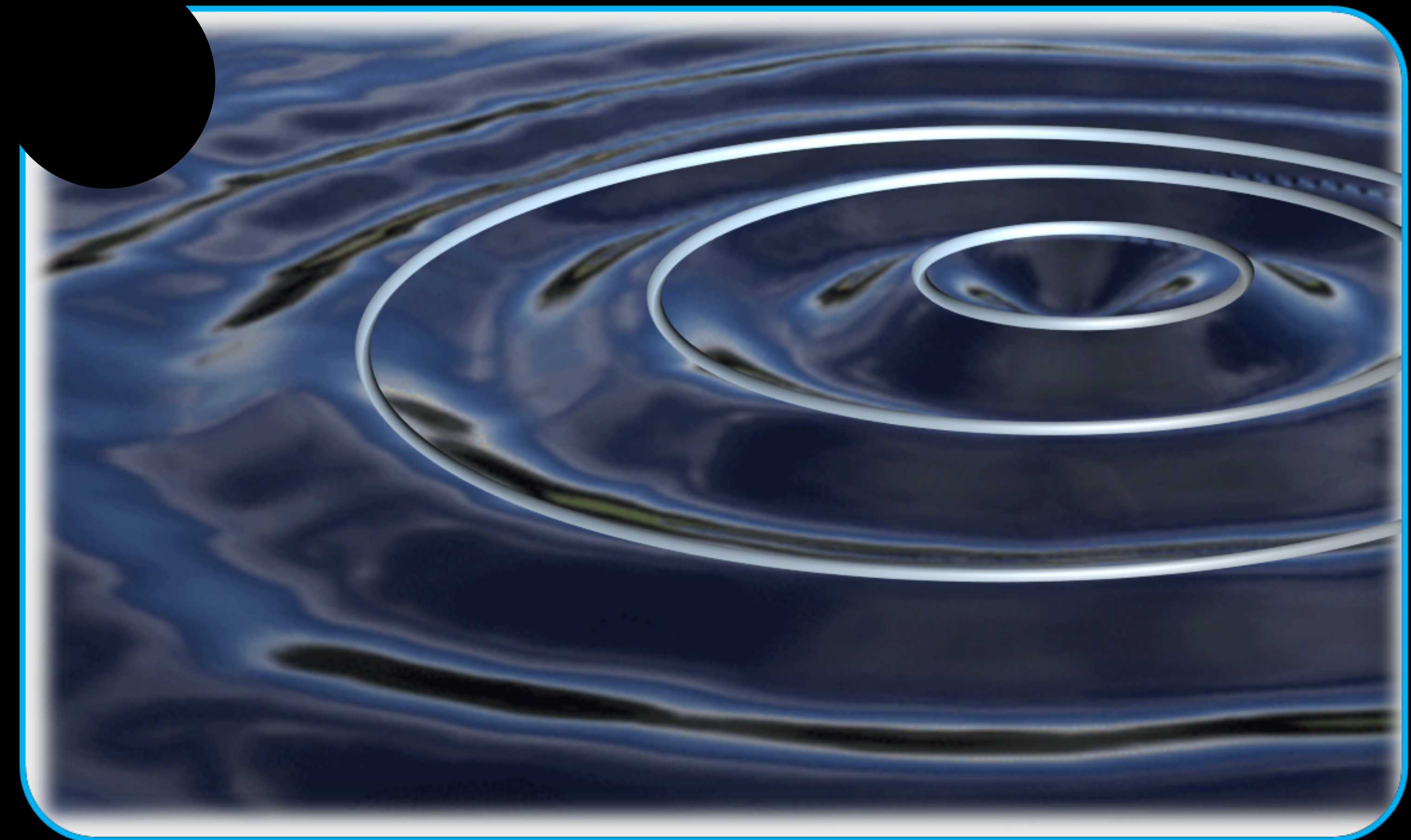


Wavefront :

Huygen's Principle

Wavefront :

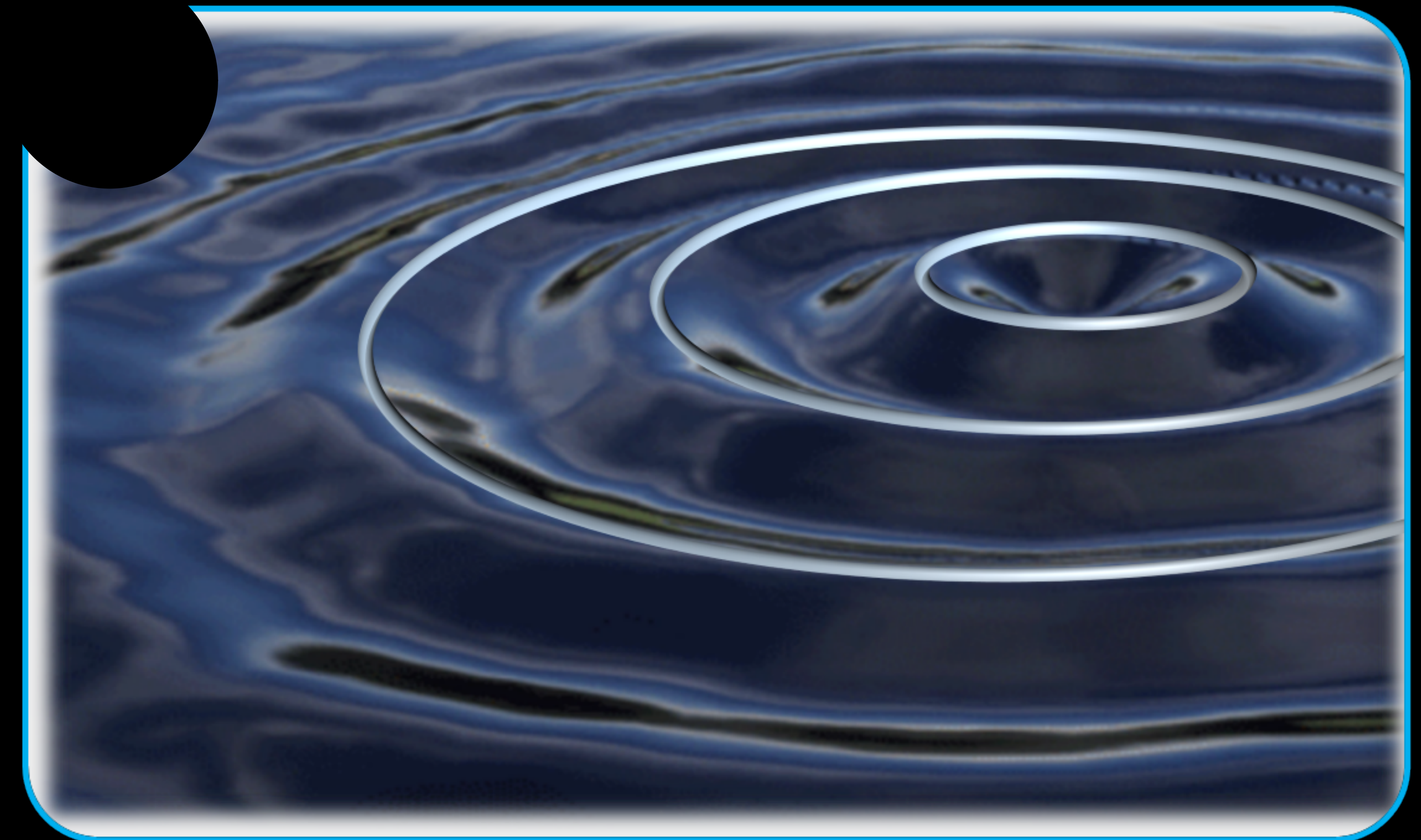
- * When a small stone is dropped in calm water, waves spread out as concentric circles — each circle represents a wavefront.
- * Definition: A wavefront is the locus of all points vibrating in the same phase.



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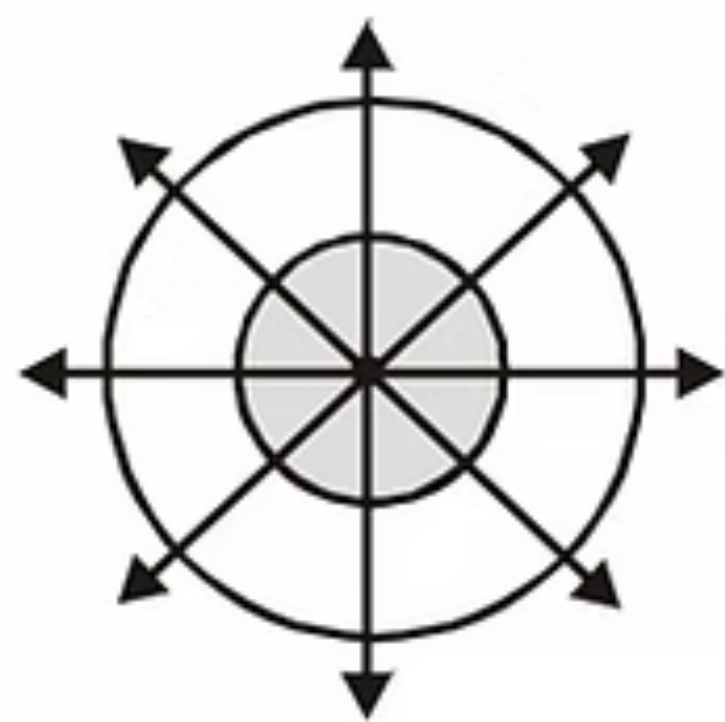


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- * Energy propagates perpendicular to the wavefront. The lines normal to wavefronts are called rays.

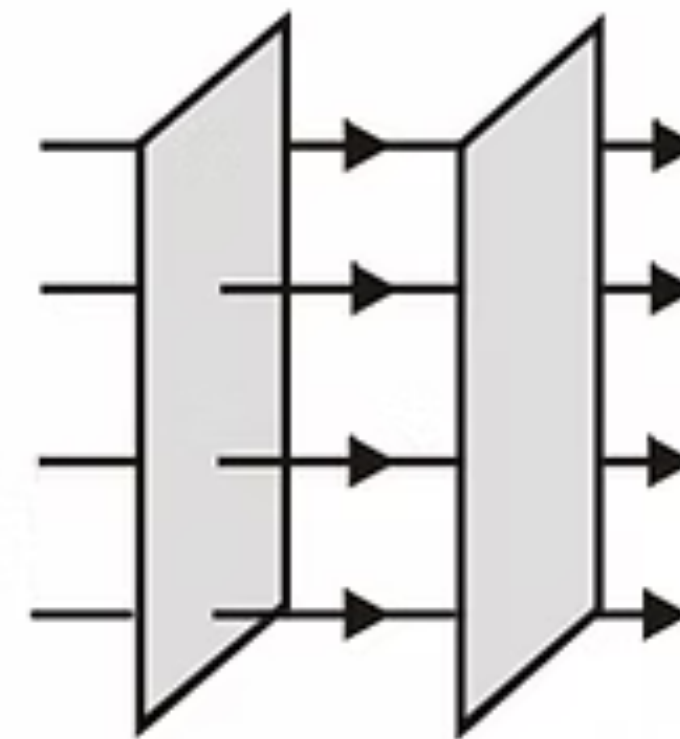
TYPES OF WAVEFRONTS



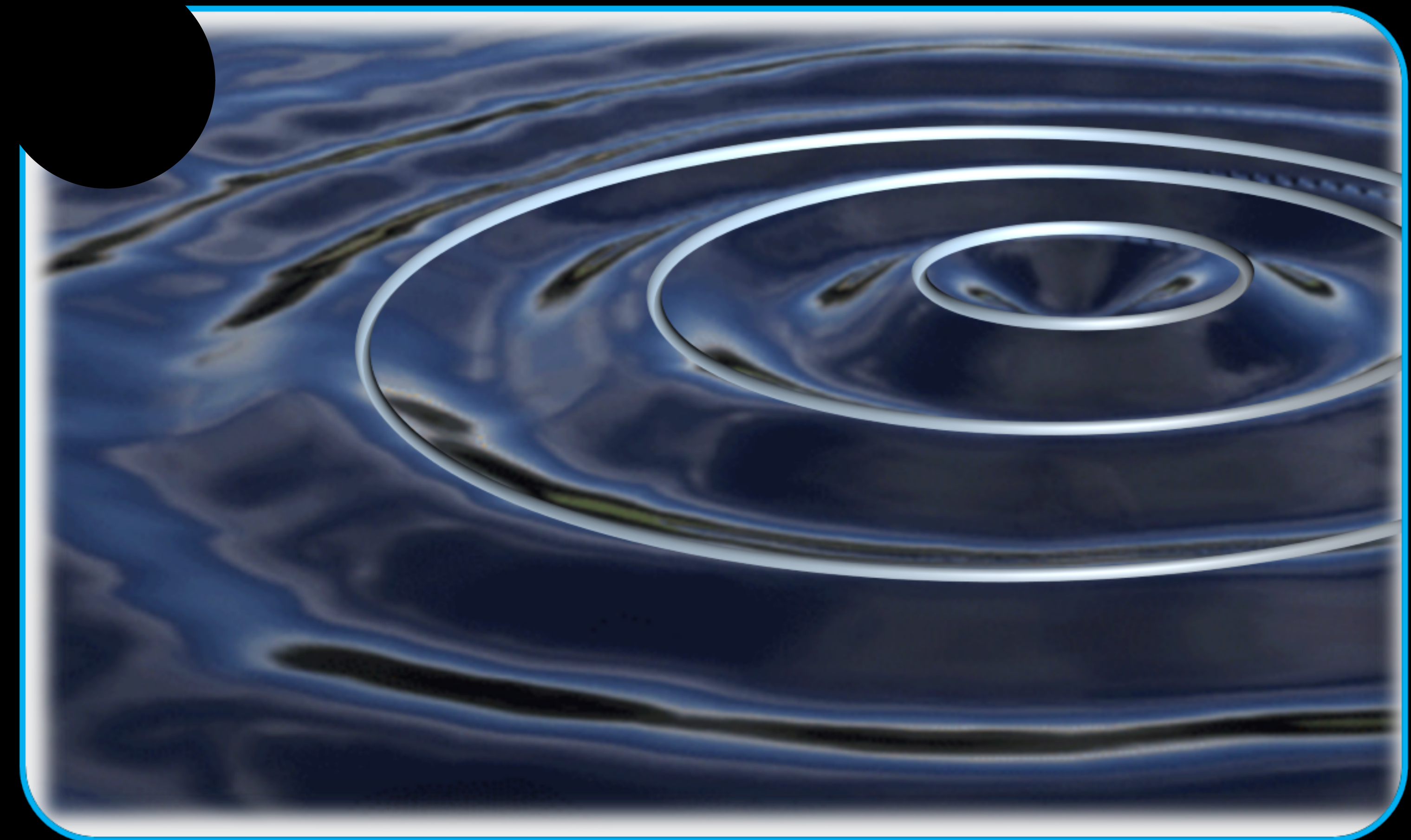
SPHERICAL



CYLINDRICAL



PLANAR



Huygen's Principle



Statement :

Huygen's Principle

Statement :

- * Every point on a given wavefront acts as a source of new secondary spherical wavelets spreading out in all directions with the speed plane wave. of the wave.
- * The new wavefront at any later time is the common tangent (envelope) to all the secondary wavelets.

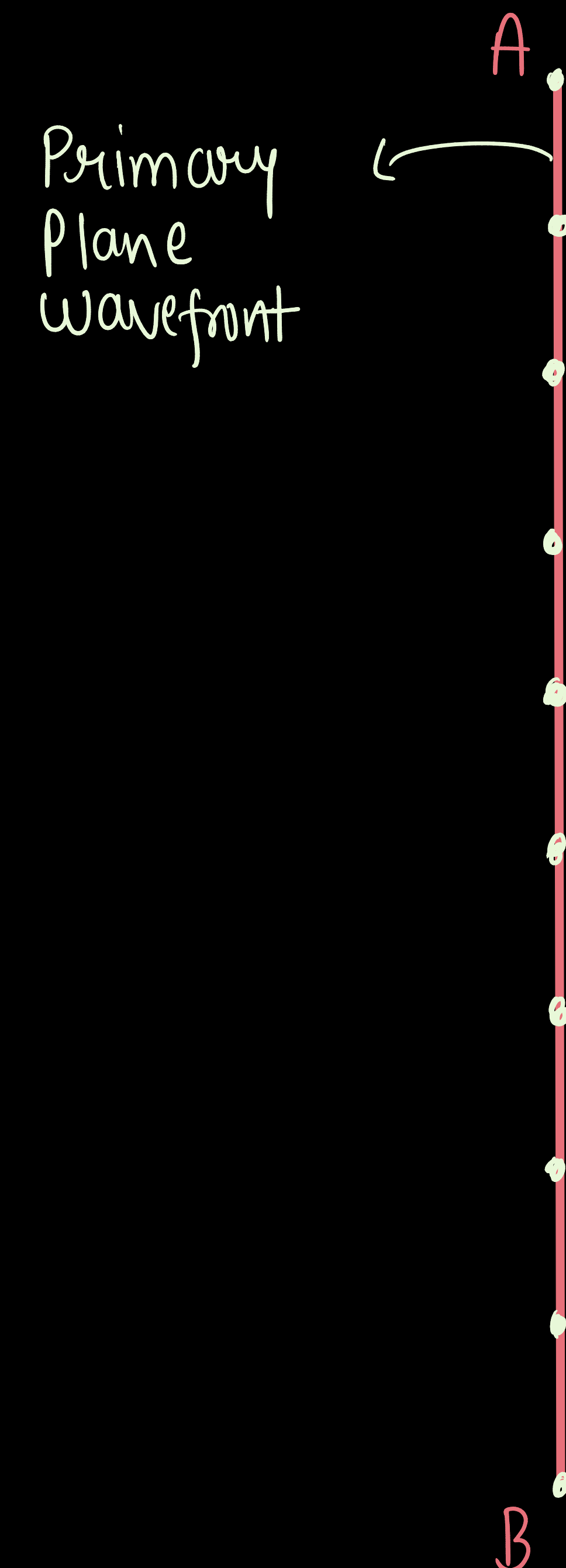
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Let a wavefront AB represent the position of a wave at time $t = 0$.



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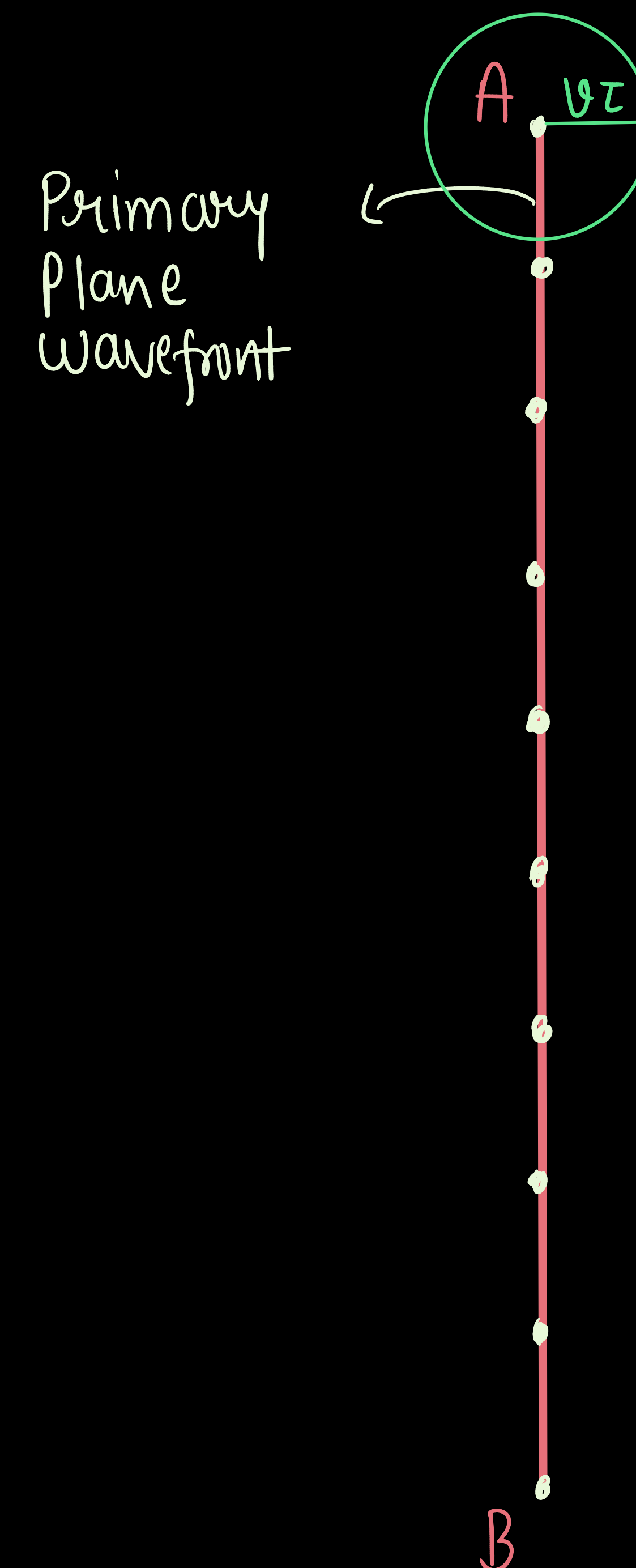
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After a short time τ :

Each point on AB emits secondary wavelets.

Each wavelet travels a distance $v\tau$,

where v = speed of wave in the medium.



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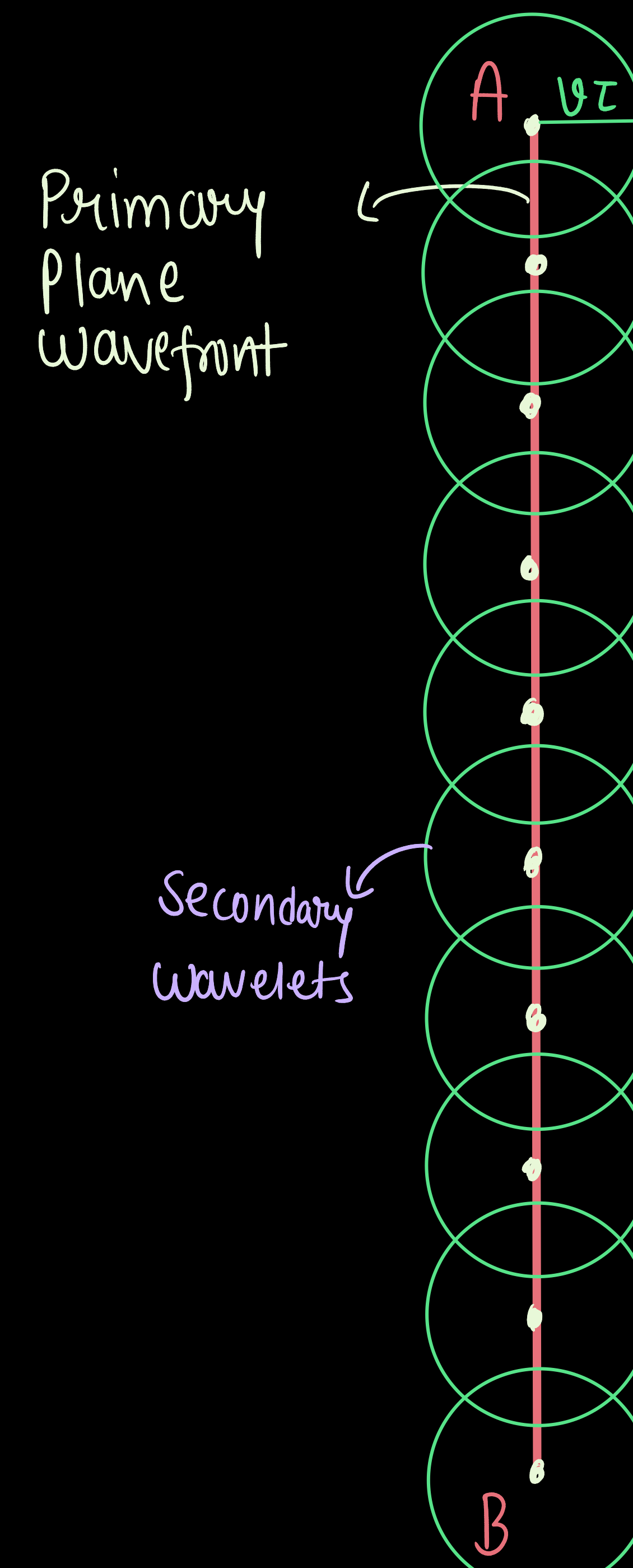
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Let a wavefront AB represent the position of a wave at time $t = 0$.

After a short time τ :

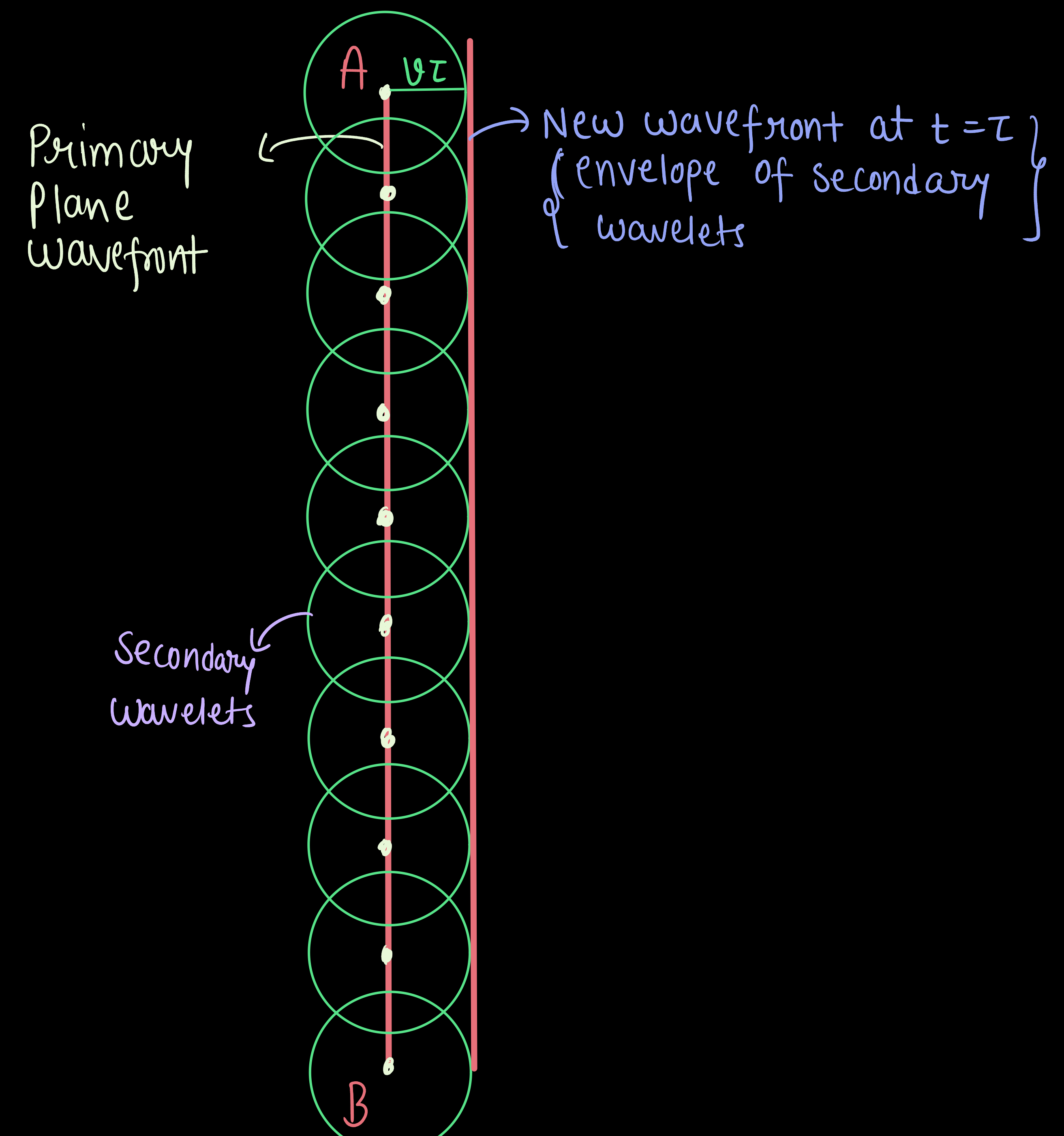
Each point on AB emits secondary wavelets.

Each wavelet travels a distance $v\tau$,

where v = speed of wave in the medium.

The envelope CD of all these secondary wavelets gives the new wavefront at time $t = \tau$.

New Wavefront = Envelope of Secondary Wavelets



Huygen's Principle

Statement :

- * Every point on a given wavefront acts as a source of new secondary spherical wavelets spreading out in all directions with the speed plane wave. of the wave.
- * The new wavefront at any later time is the common tangent (envelope) to all the secondary wavelets.

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Let a wavefront AB represent the position of a wave at time $t = 0$.

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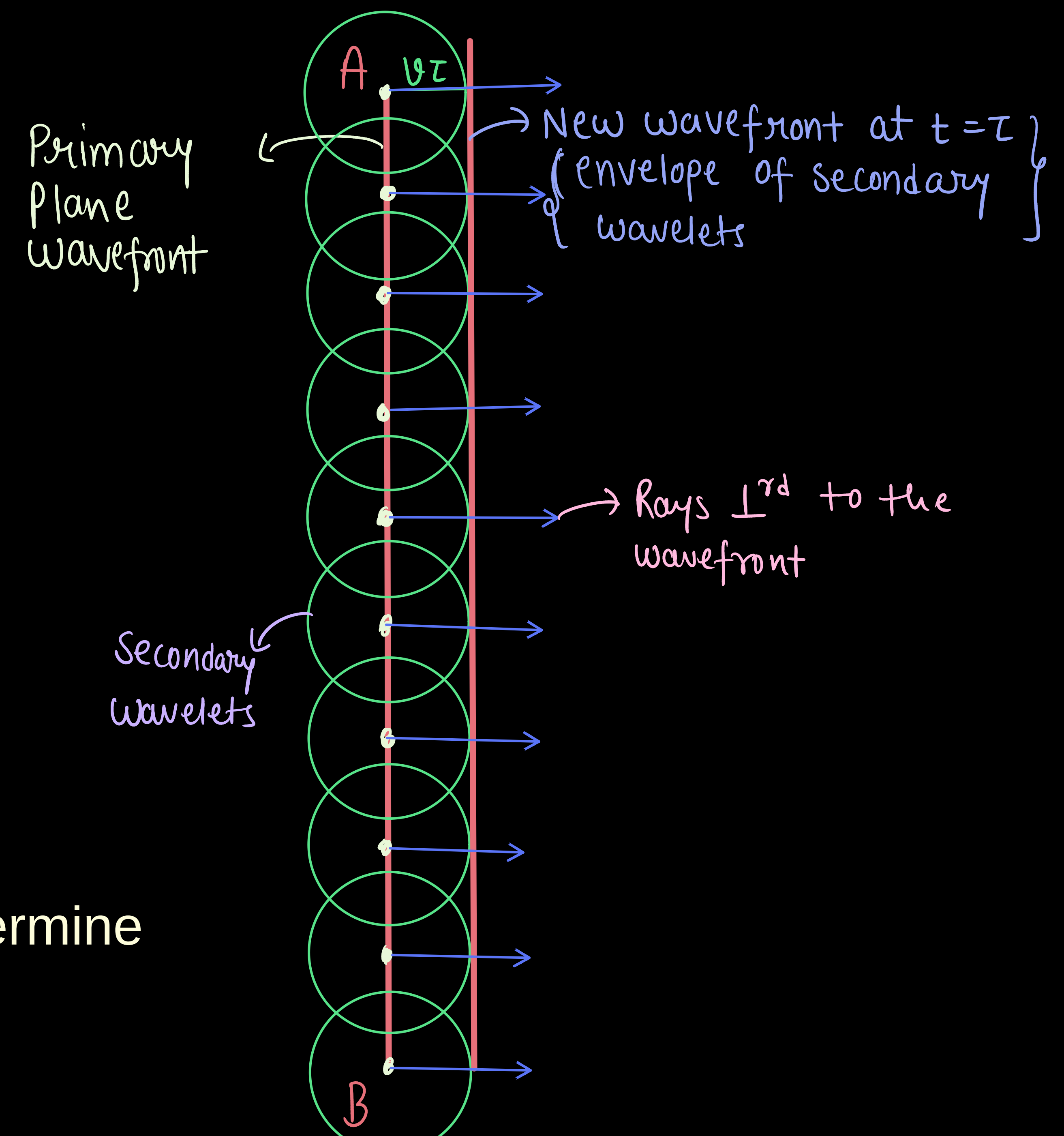
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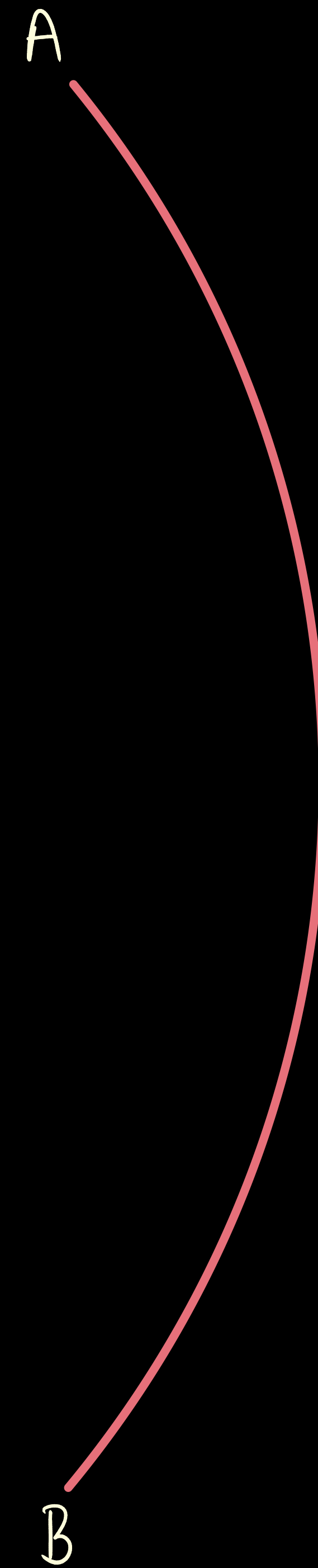
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New Wavefront = Envelope of Secondary Wavelets

- * Thus, Huygens principle is essentially a geometrical construction, which given the shape of the wavefront at any time allows us to determine the shape of the wavefront at a later time.



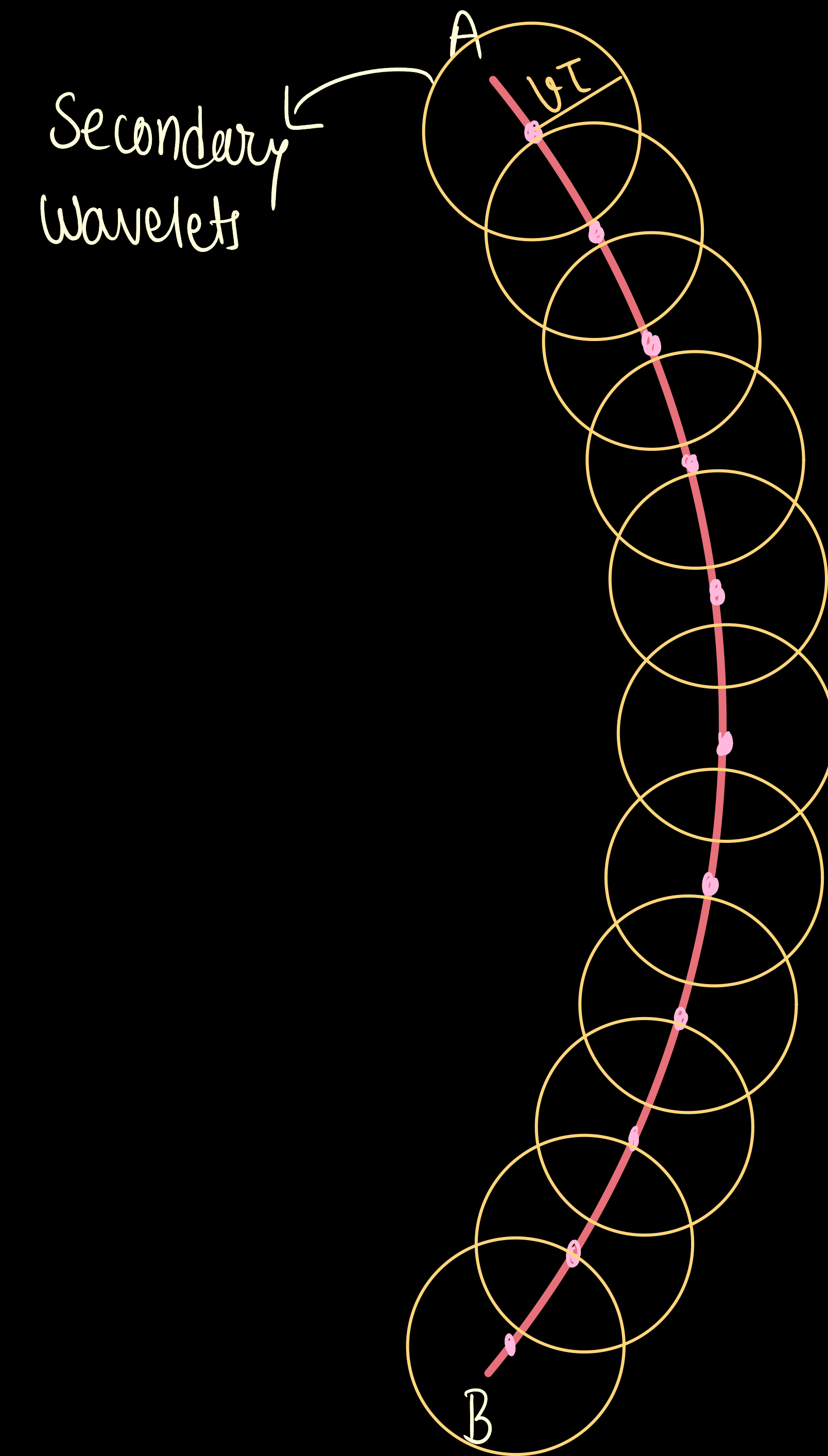
Huygen's Principle



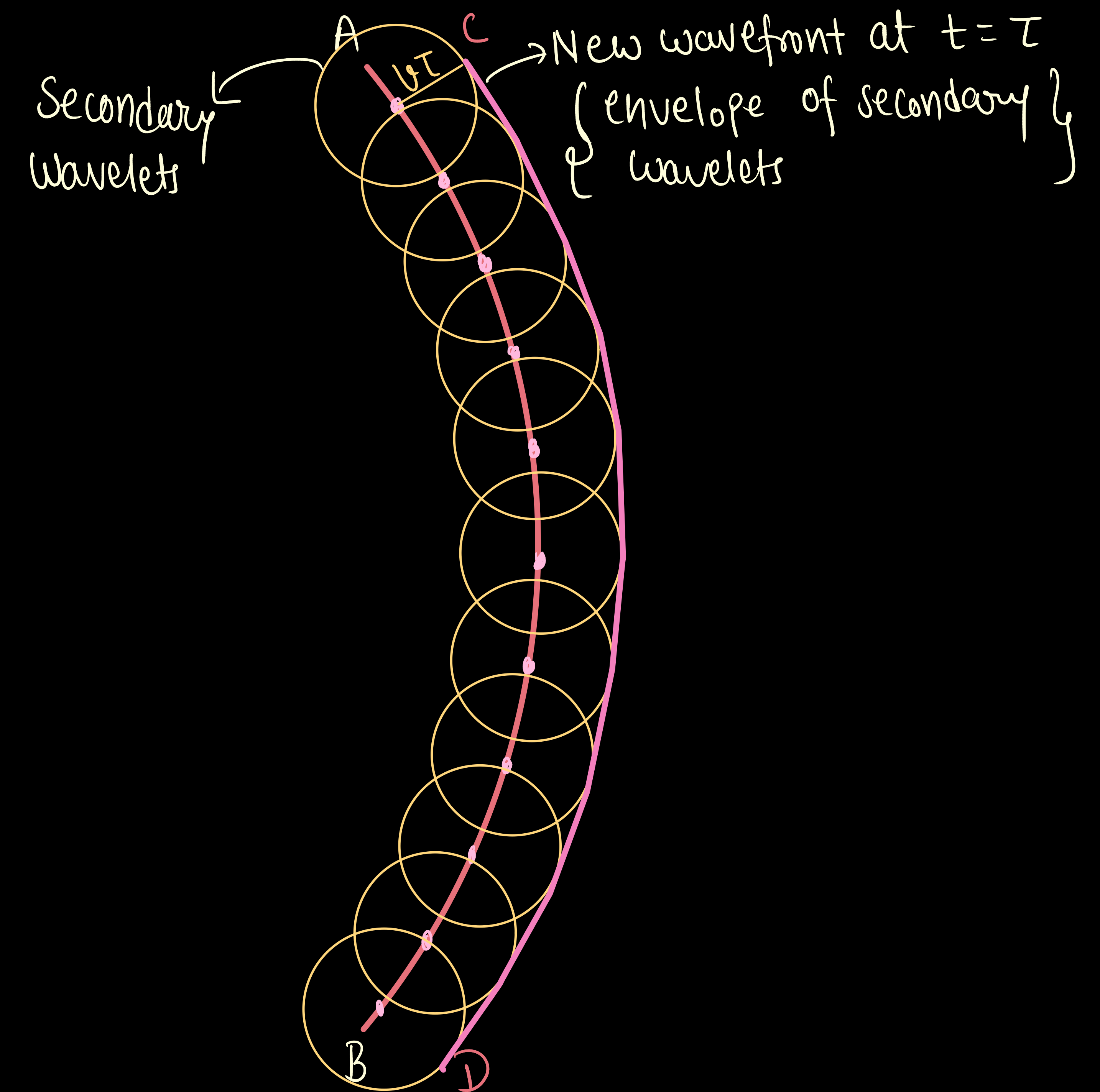
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Huygen's Principle



Huygen's Principle



Huygen's Principle

* Forward and Backward Wavelets

The construction gives two possible envelopes:

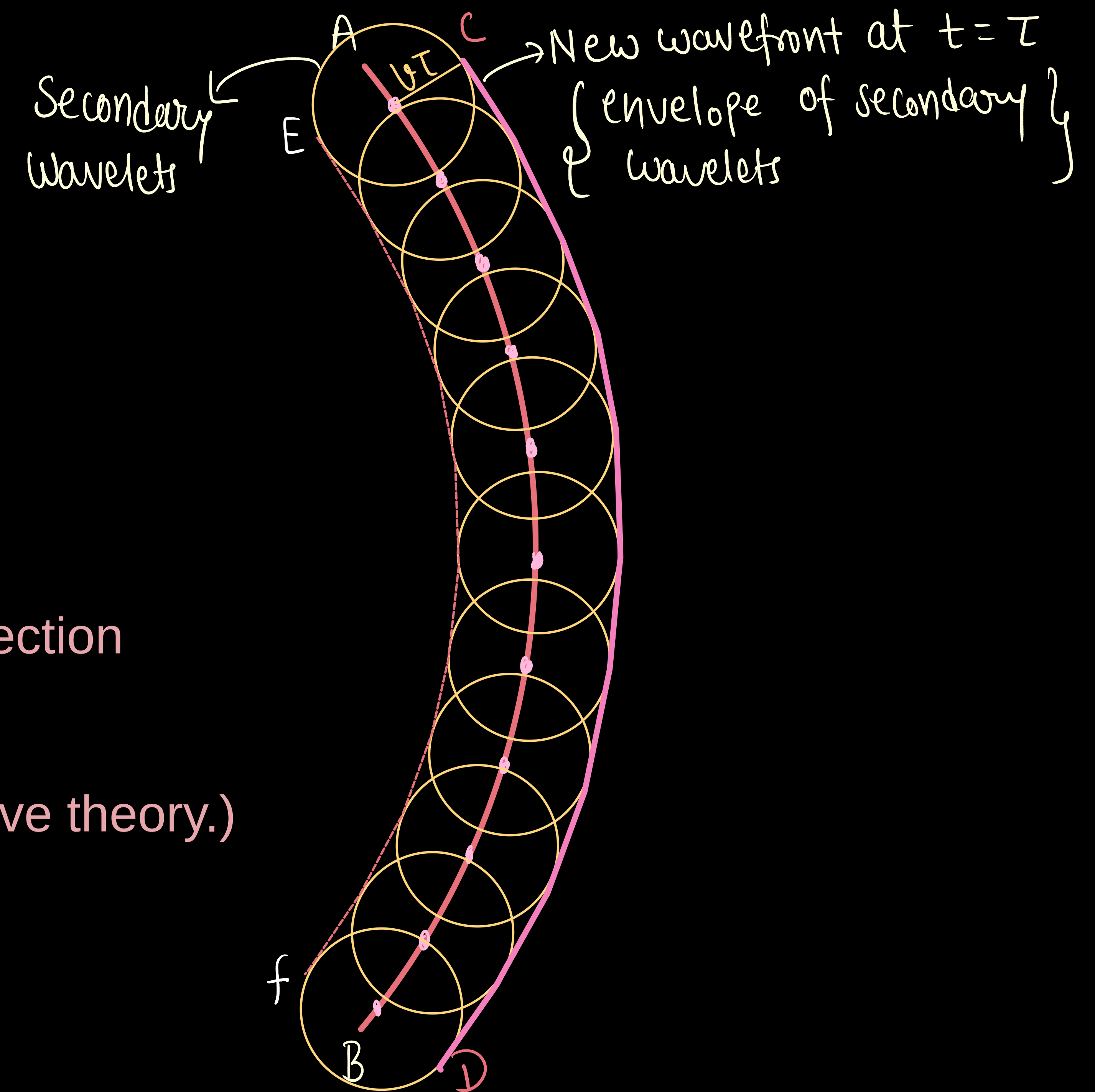
Forward wavefront (CD): represents actual propagation of light.

Backward wavefront (EF): does not exist physically.

* Huygens explained the absence of the backwave by assuming:

Amplitude of secondary wavelets is maximum in the forward direction and zero in the backward direction.

(This assumption was later justified rigorously by electromagnetic wave theory.)



Huygen's Principle

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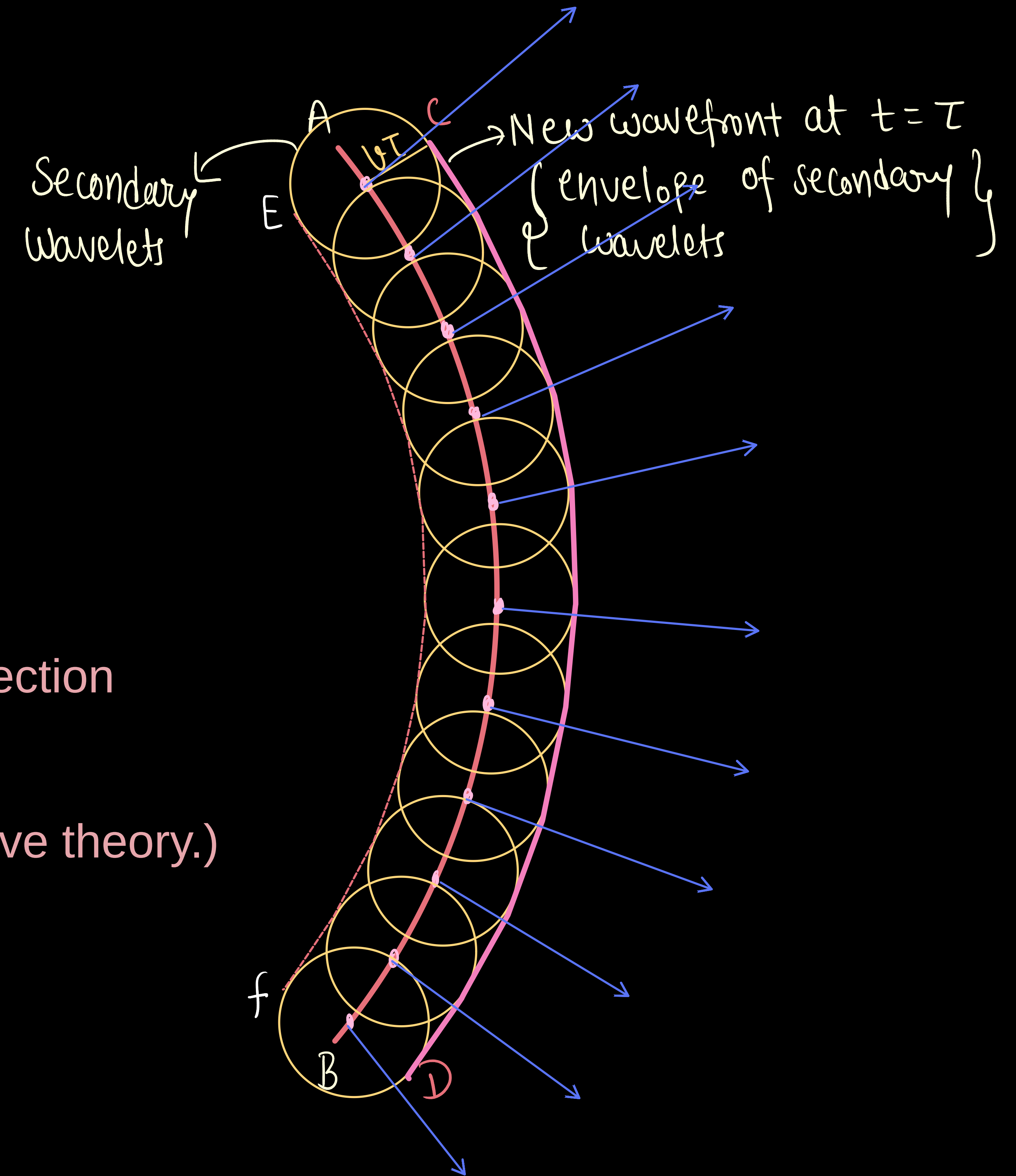
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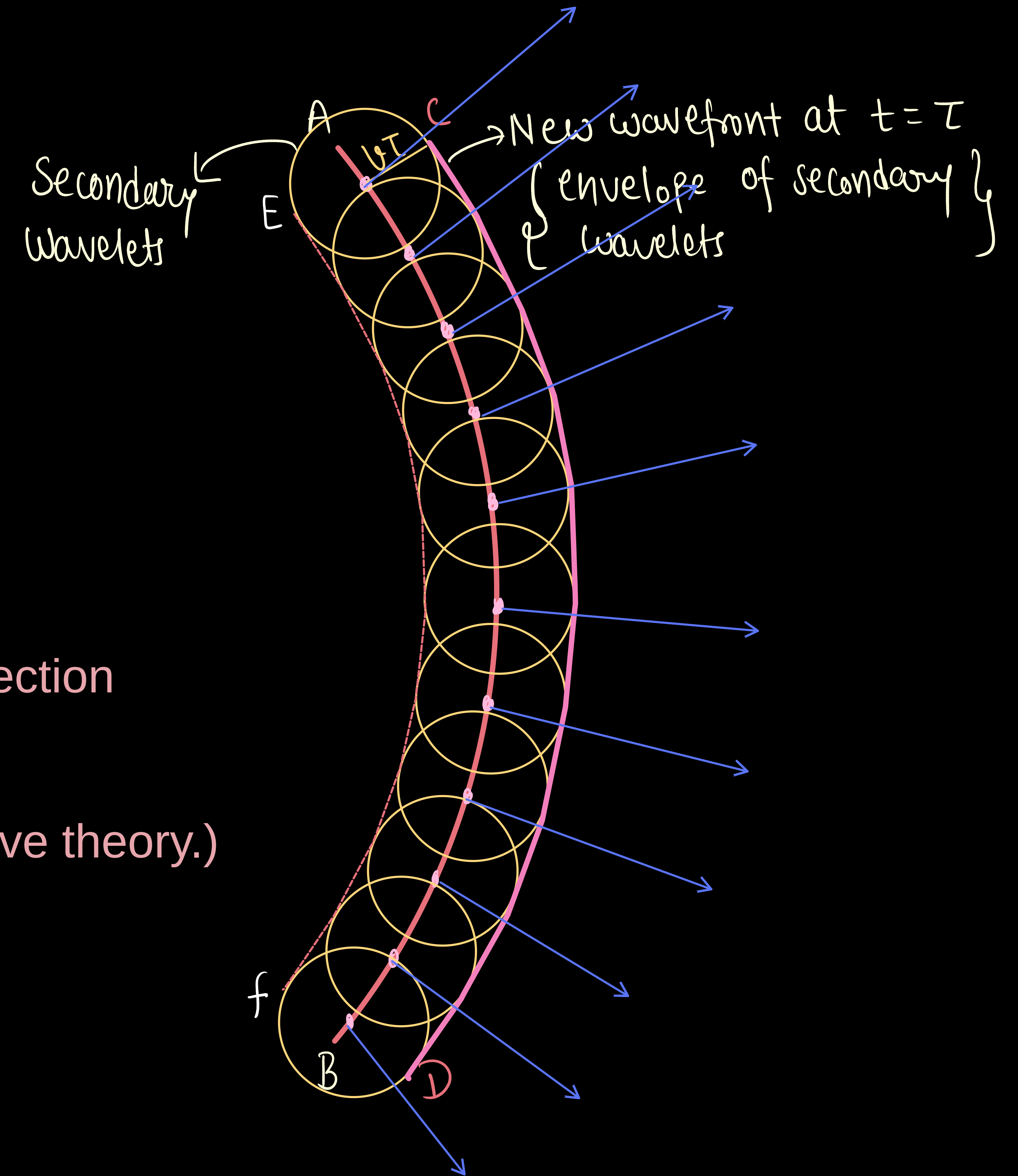
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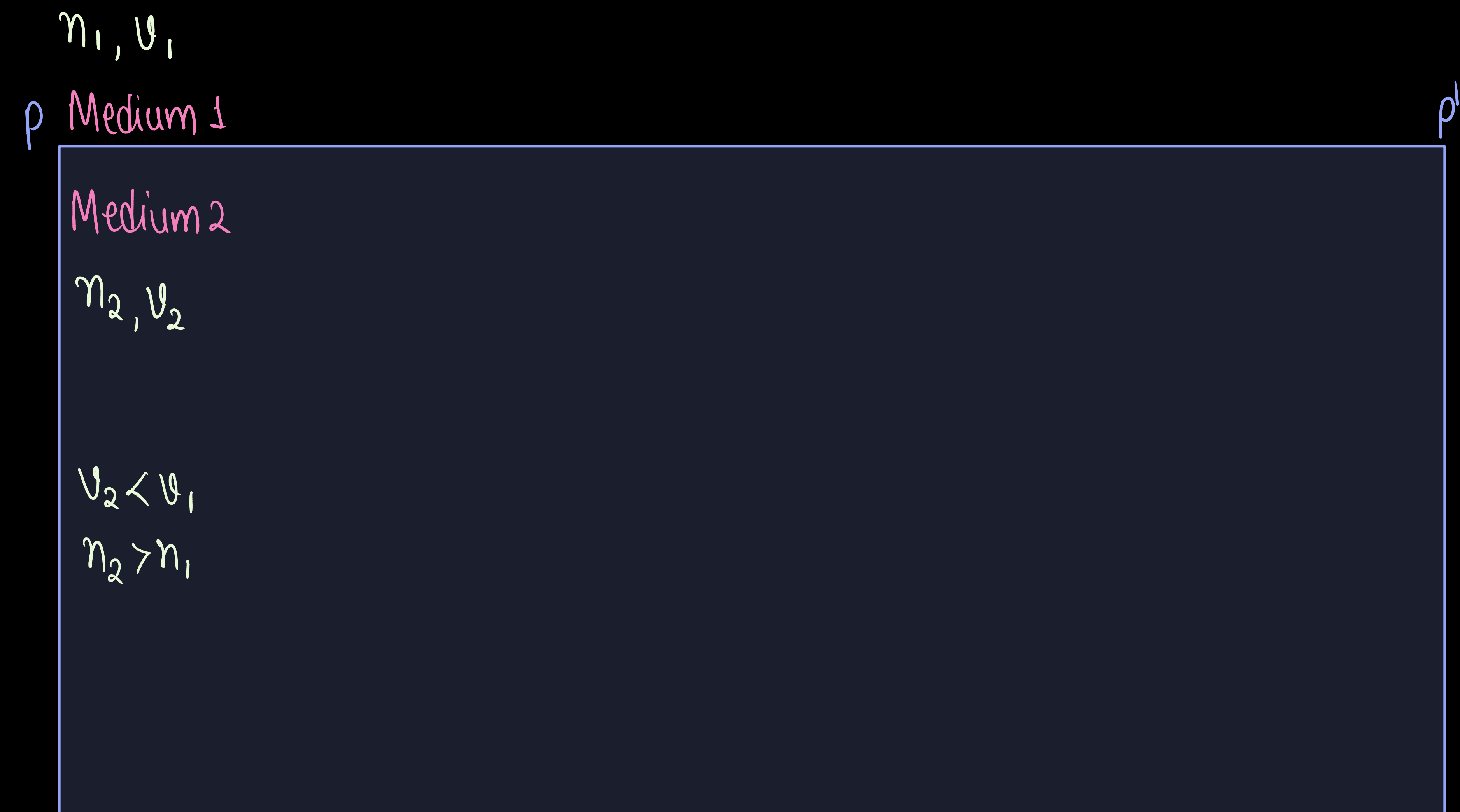


Reflection and Refraction of Plane Wave at a Plane Surface

(1) Refraction of a Plane Wave (Using Huygens' Principle)

Reflection and Refraction of Plane Wave at a Plane Surface

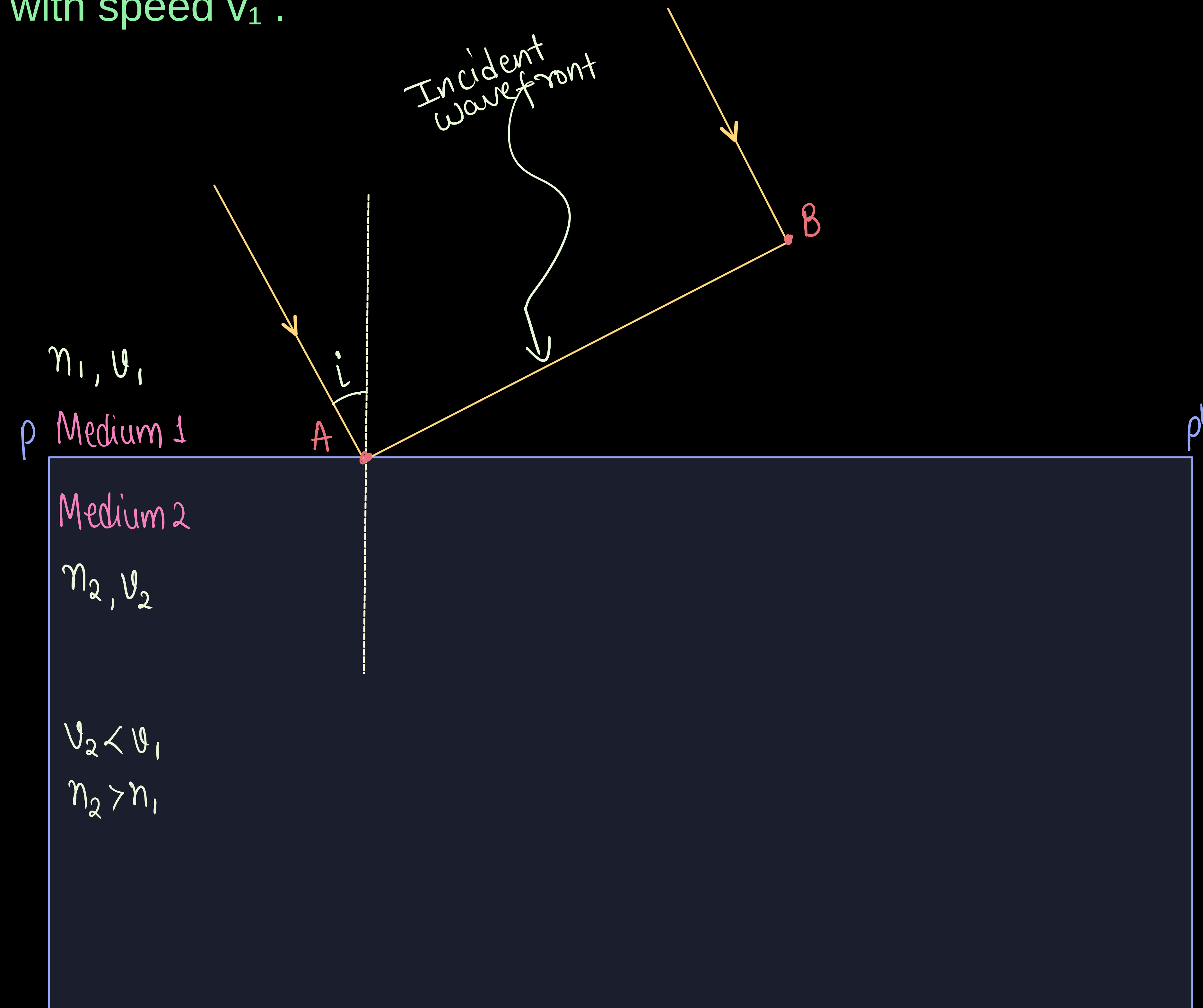
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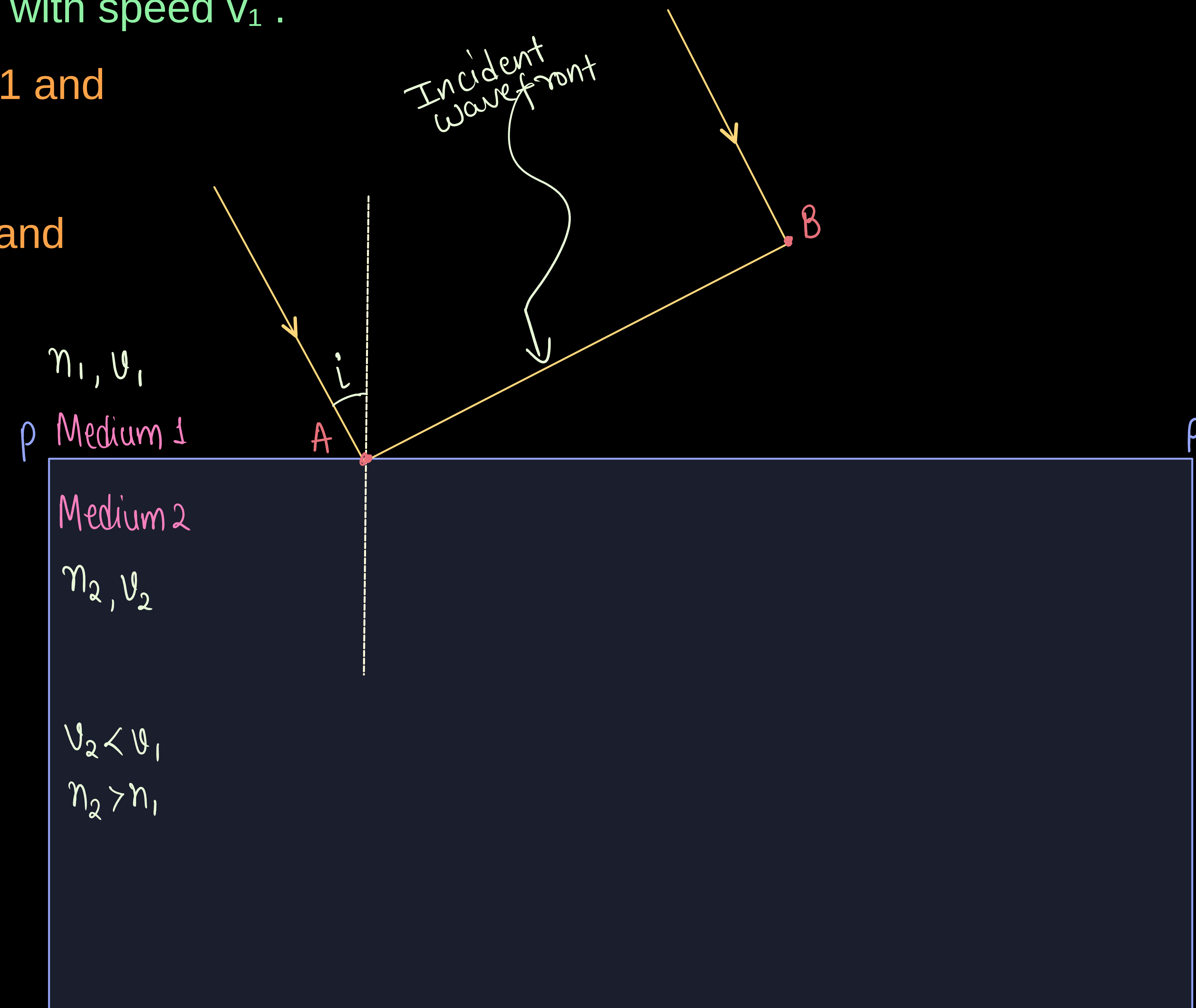
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Reflection and Refraction of Plane Wave at a Plane Surface

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- * Consider a plane wavefront AB travelling in medium 1 with speed v_1 .
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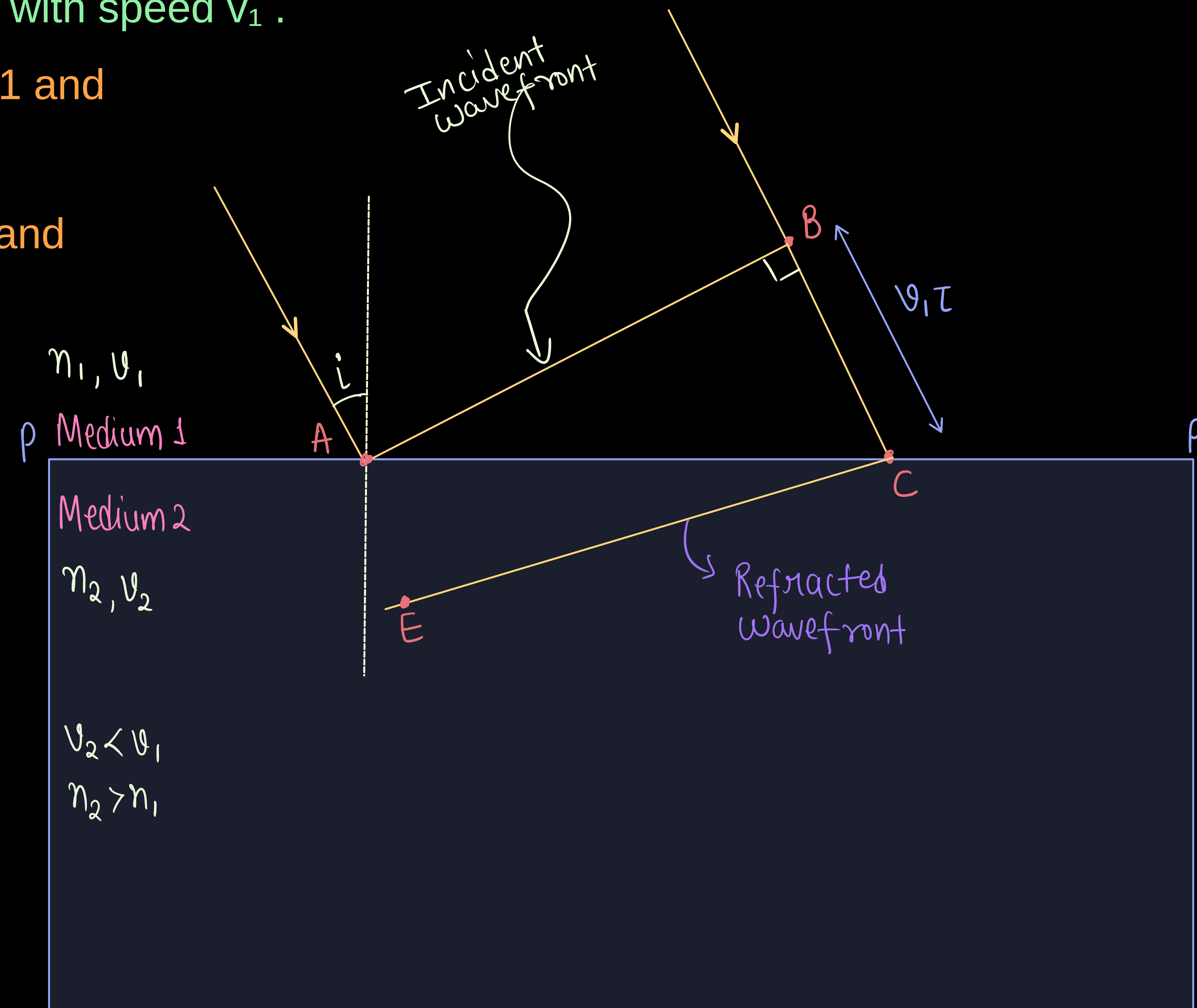
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Huygens' Construction

- (1) Let the wavefront AB strike the boundary at point A first.
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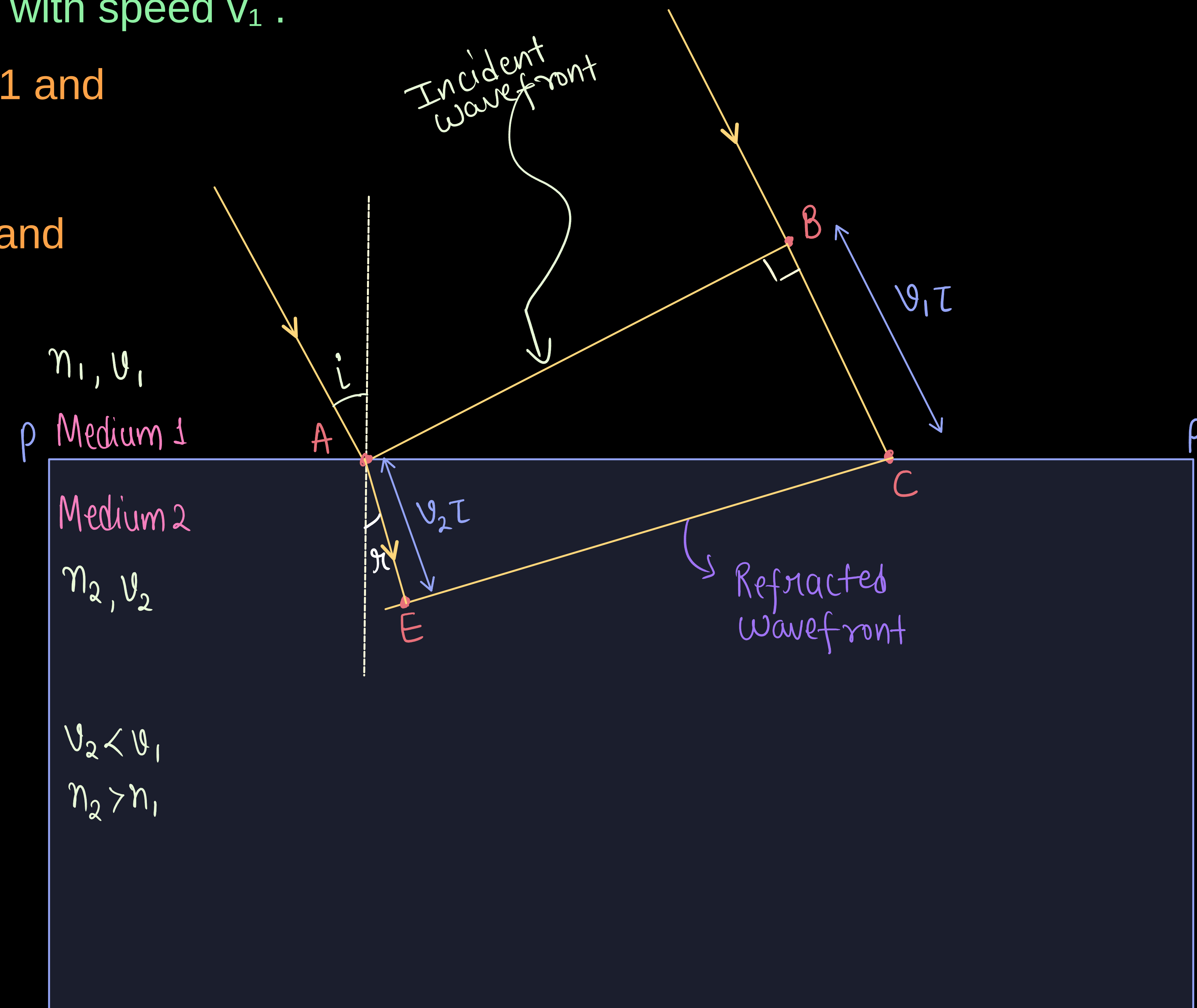
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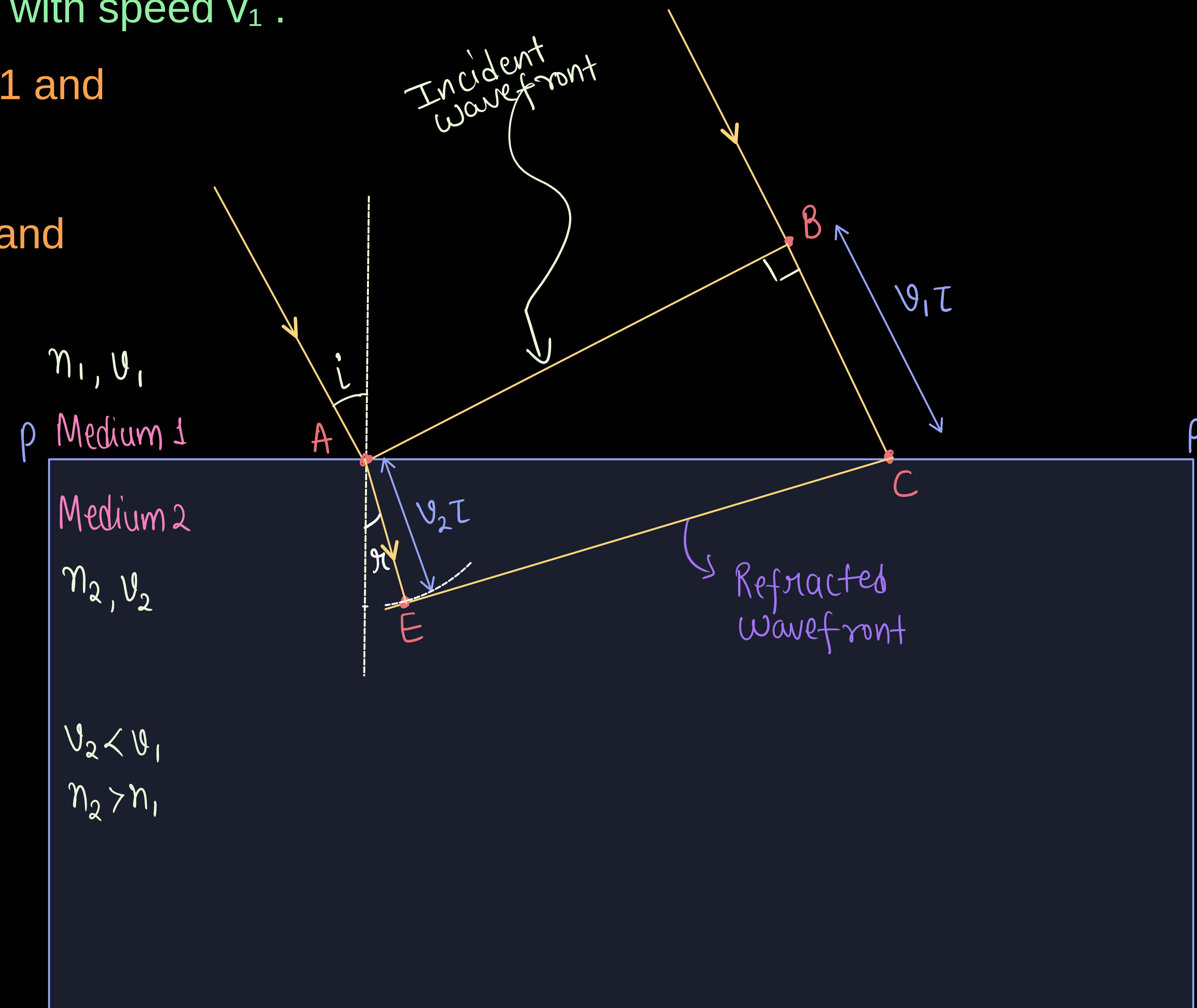
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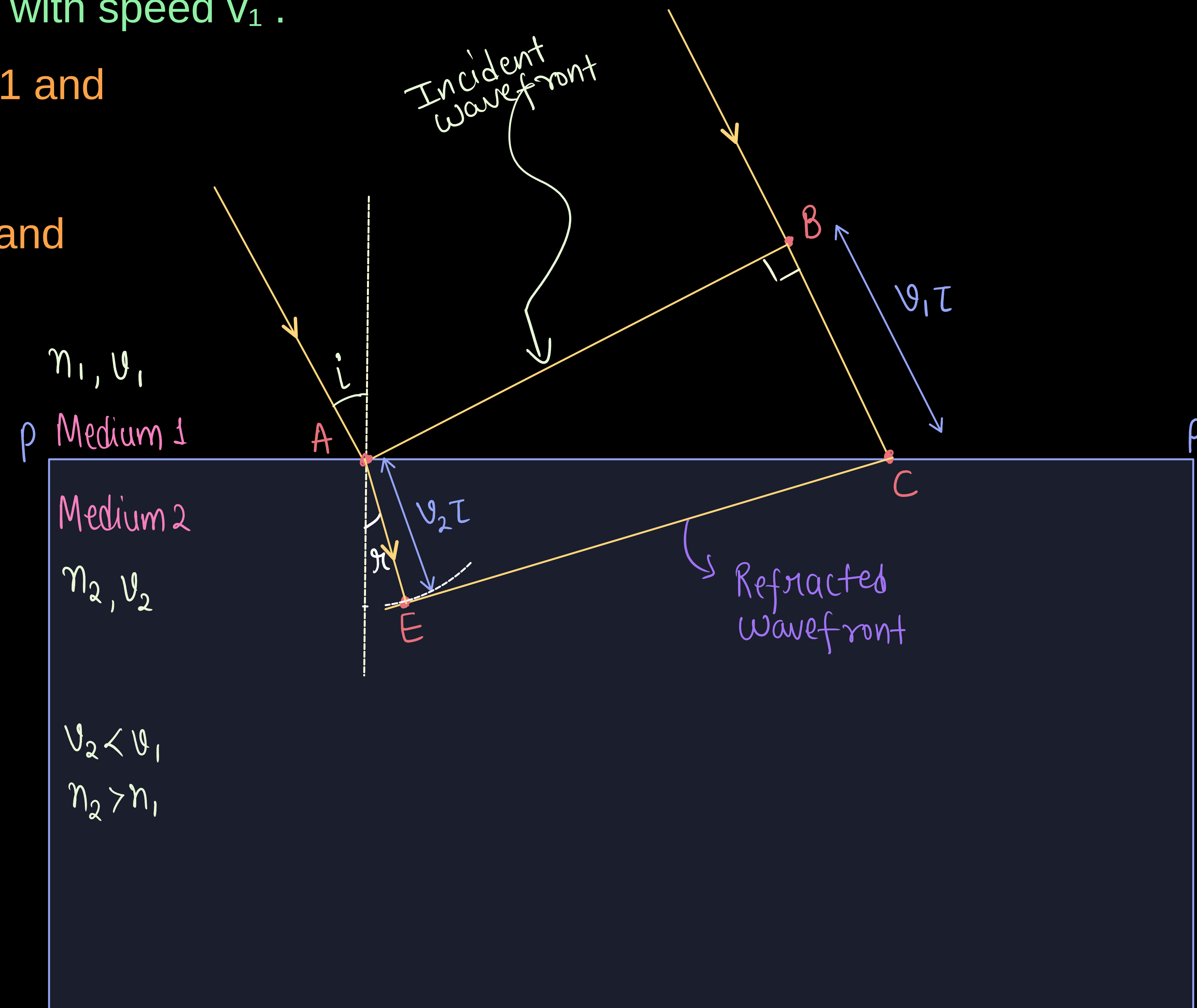
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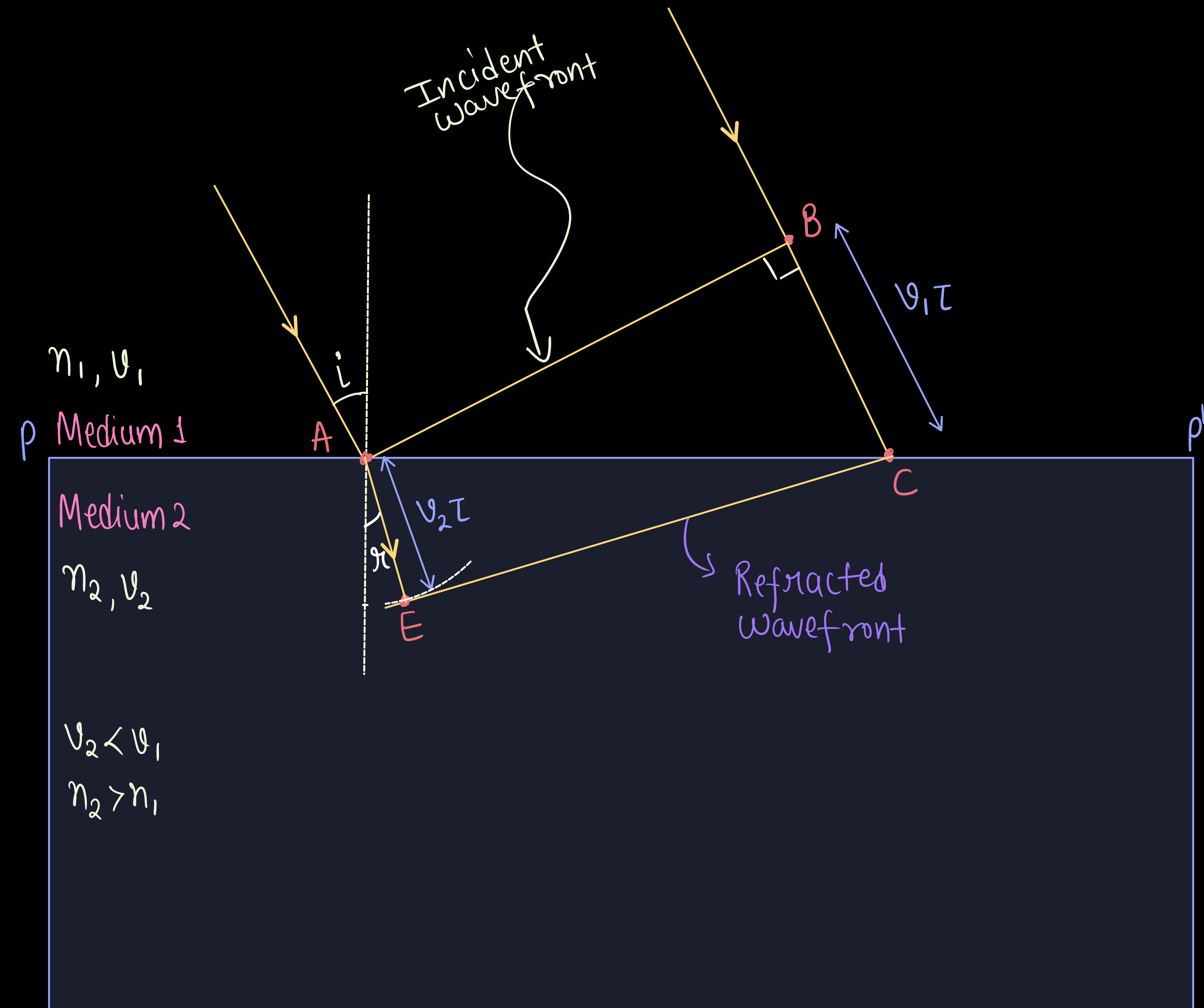
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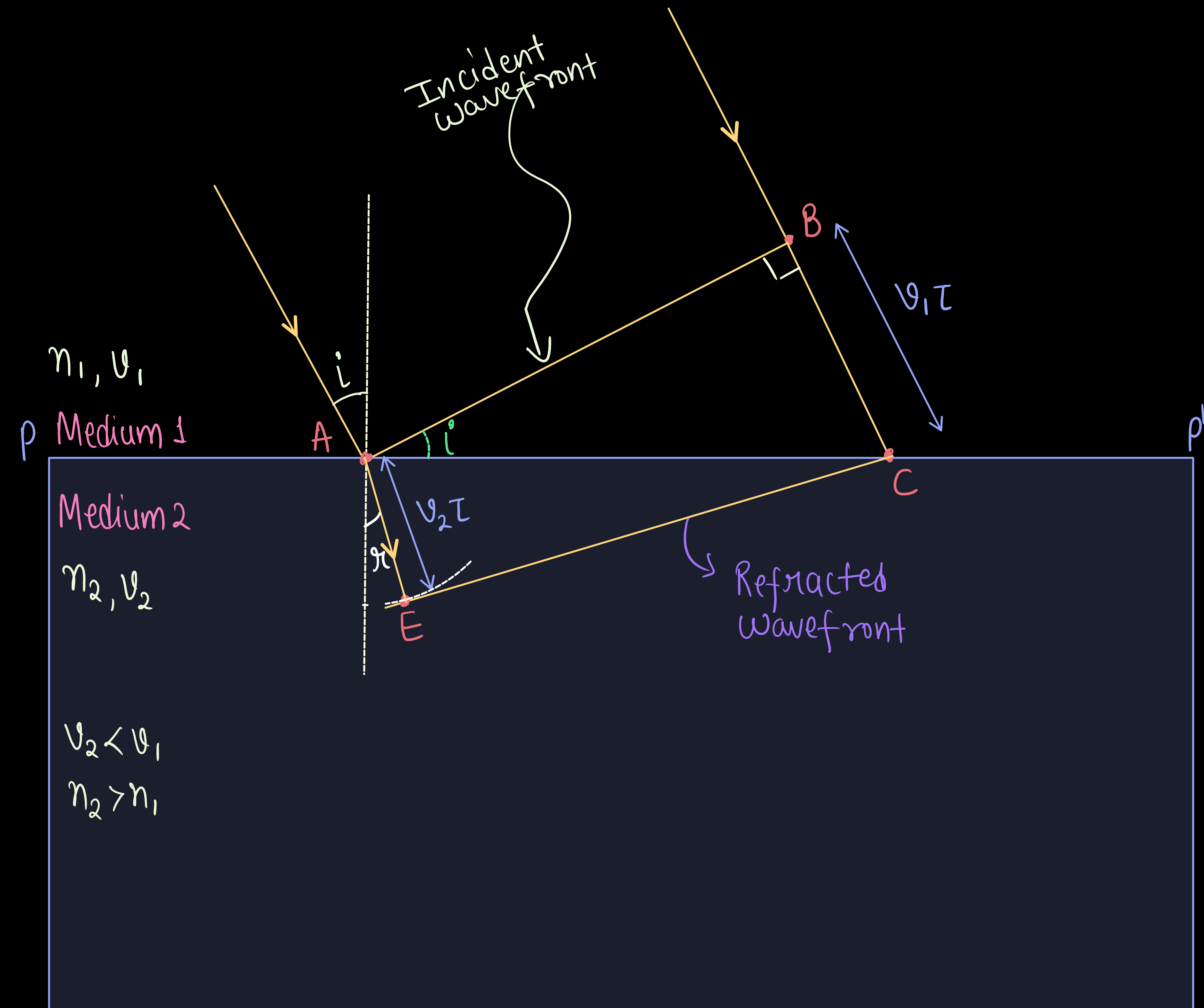
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Derivation of Snell's Law -



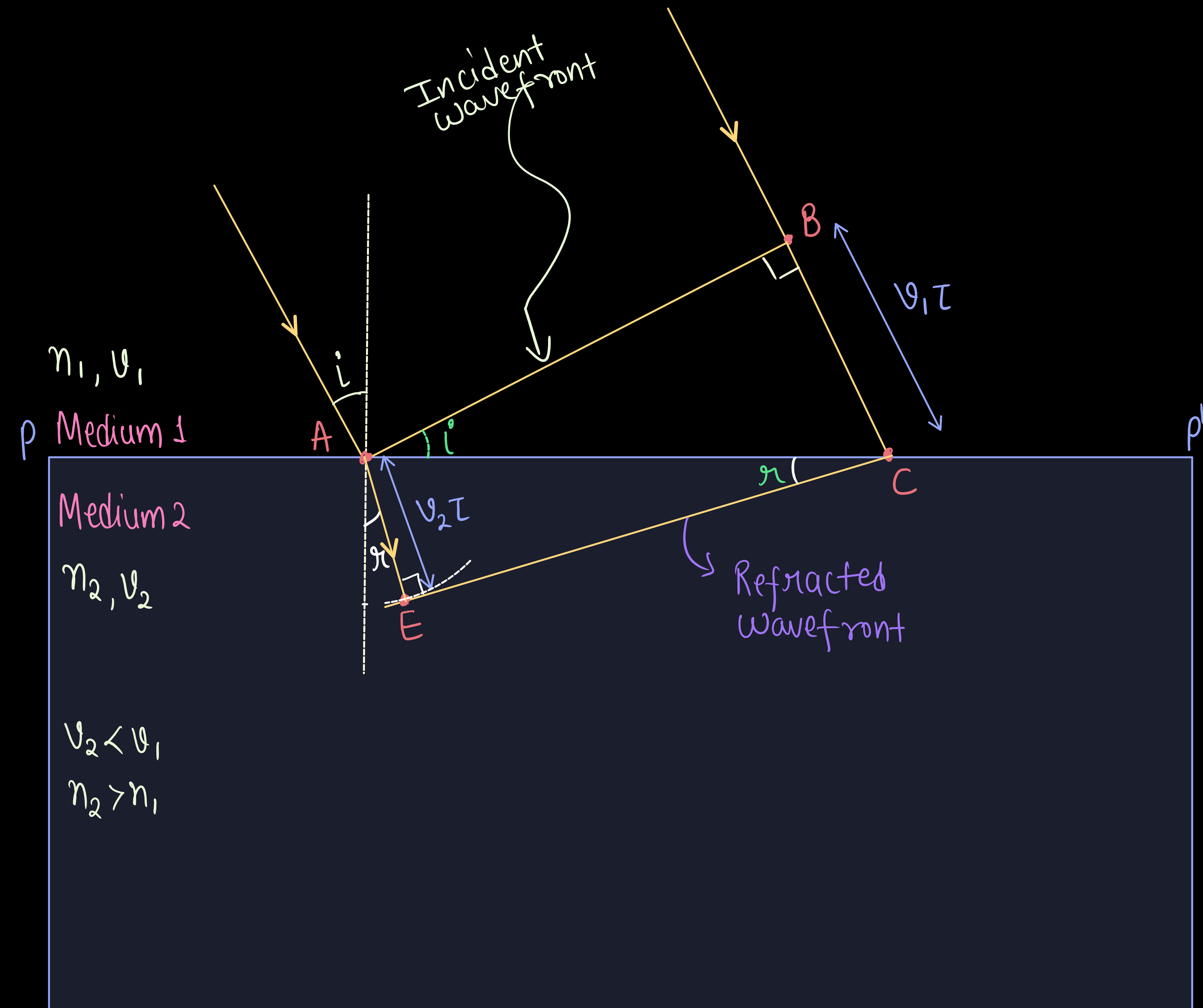
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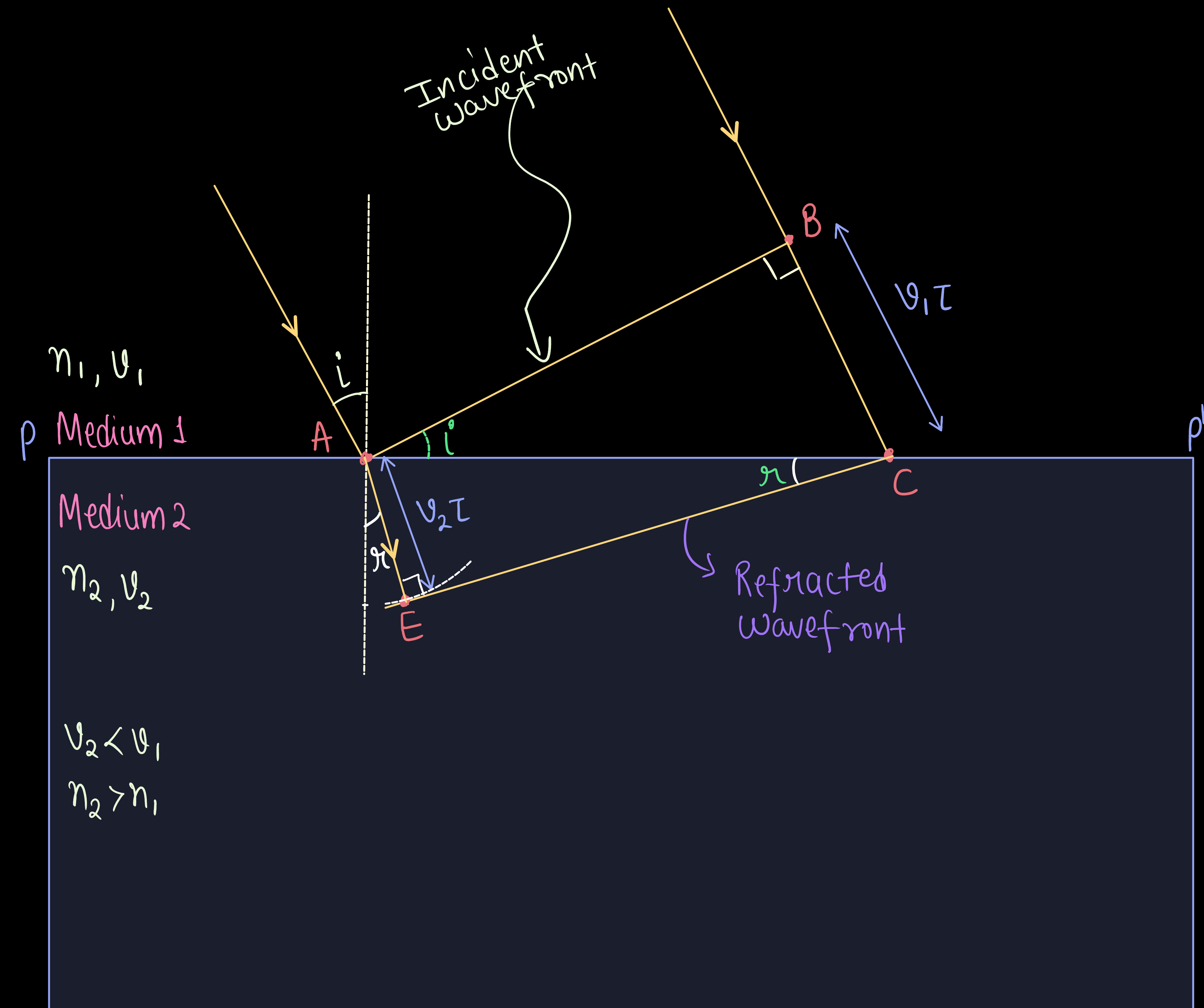


Reflection and Refraction of Plane Wave at a Plane Surface

Derivation of Snell's Law -

In $\triangle ABC$

$$\sin i = \frac{BC}{AC} = \frac{v_1 T}{AC} \quad \text{--- (1)}$$



Reflection and Refraction of Plane Wave at a Plane Surface

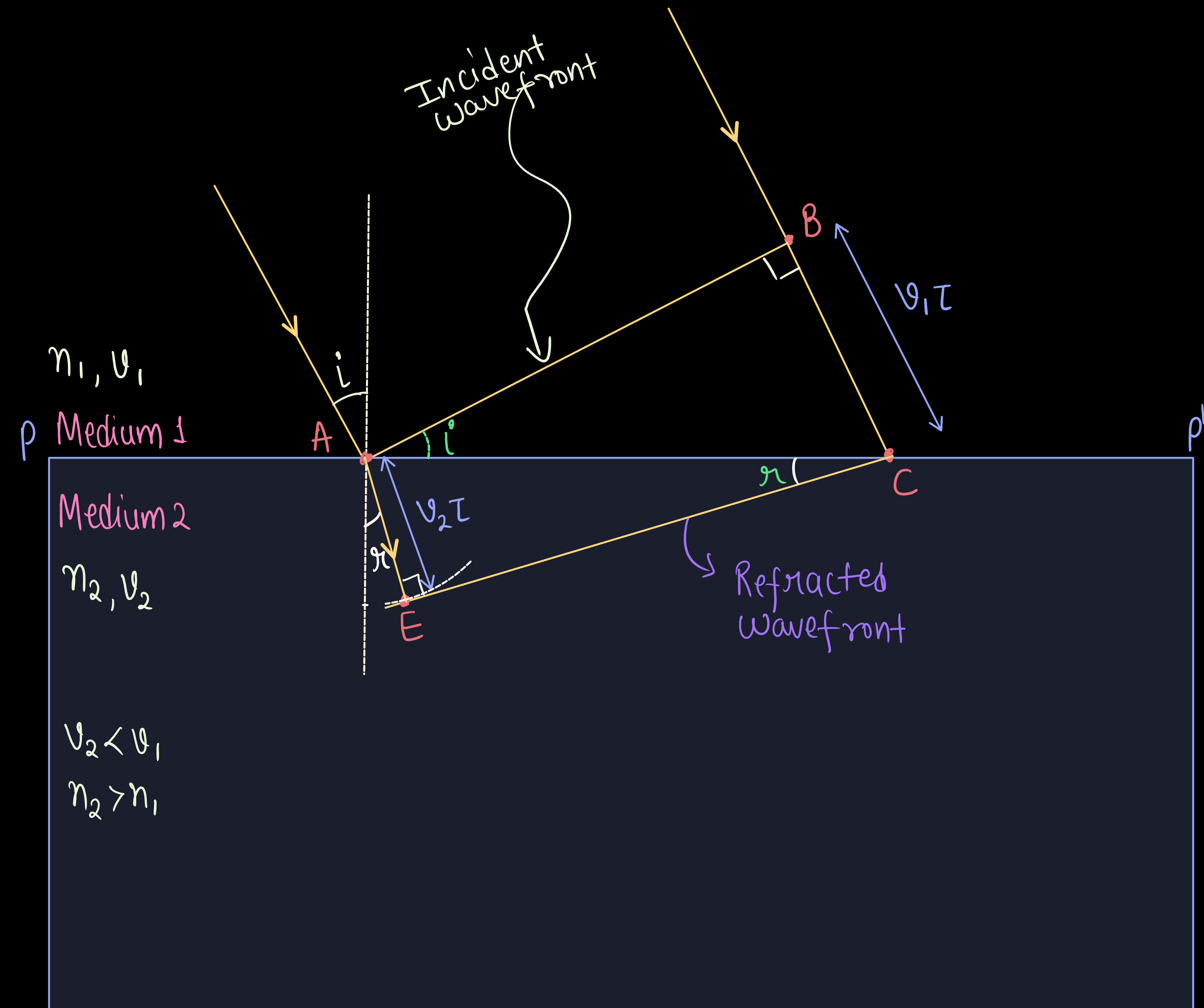
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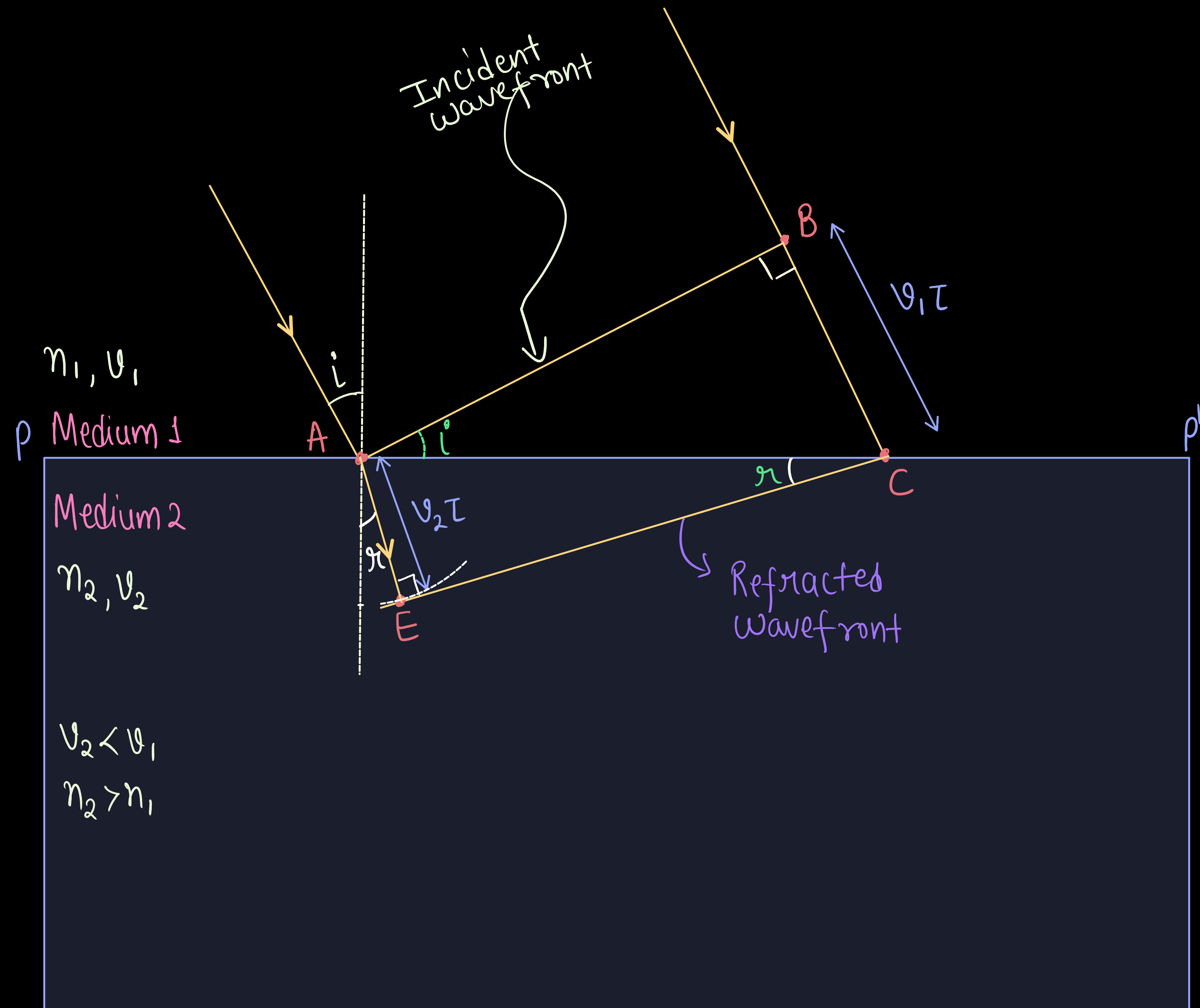
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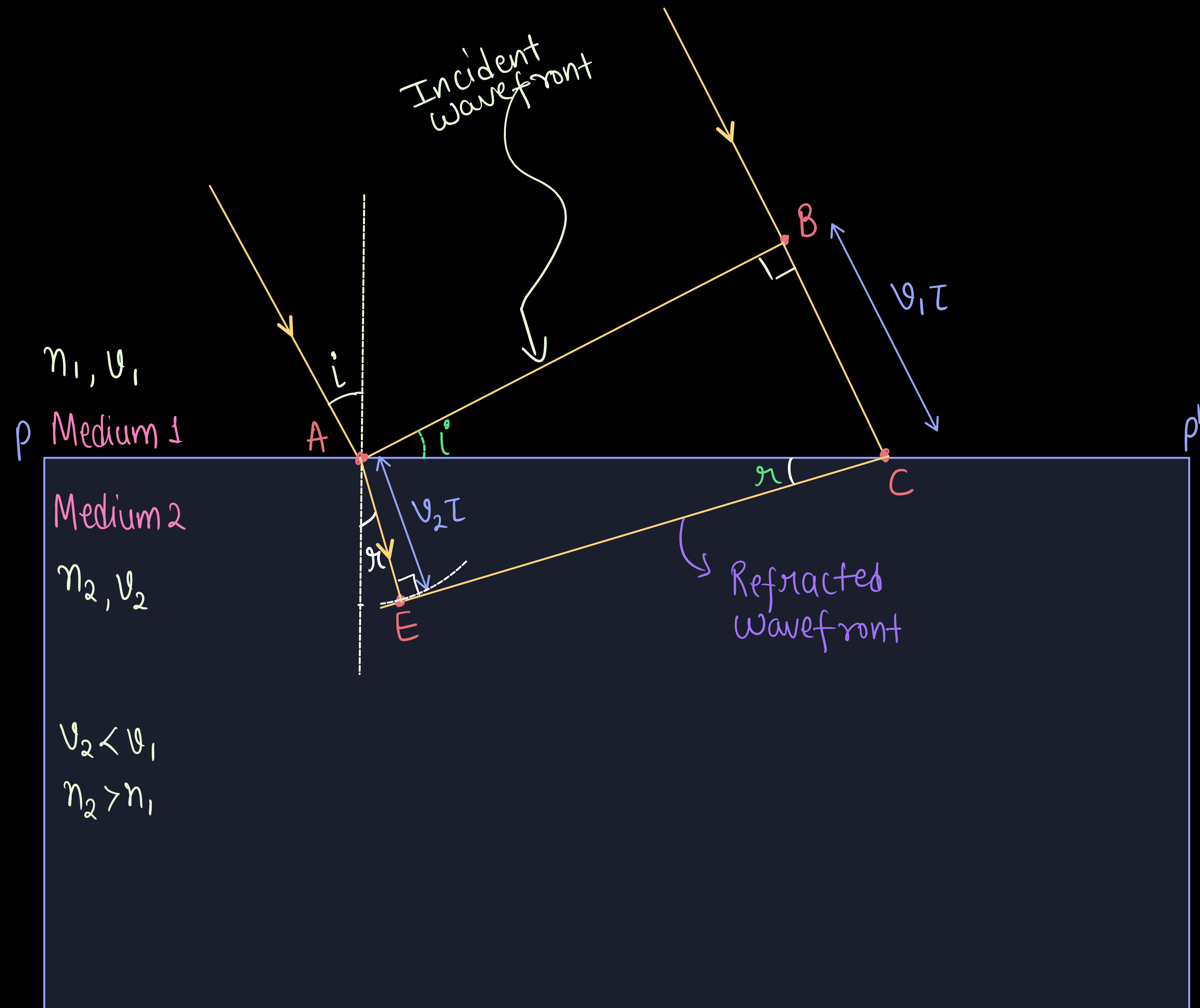
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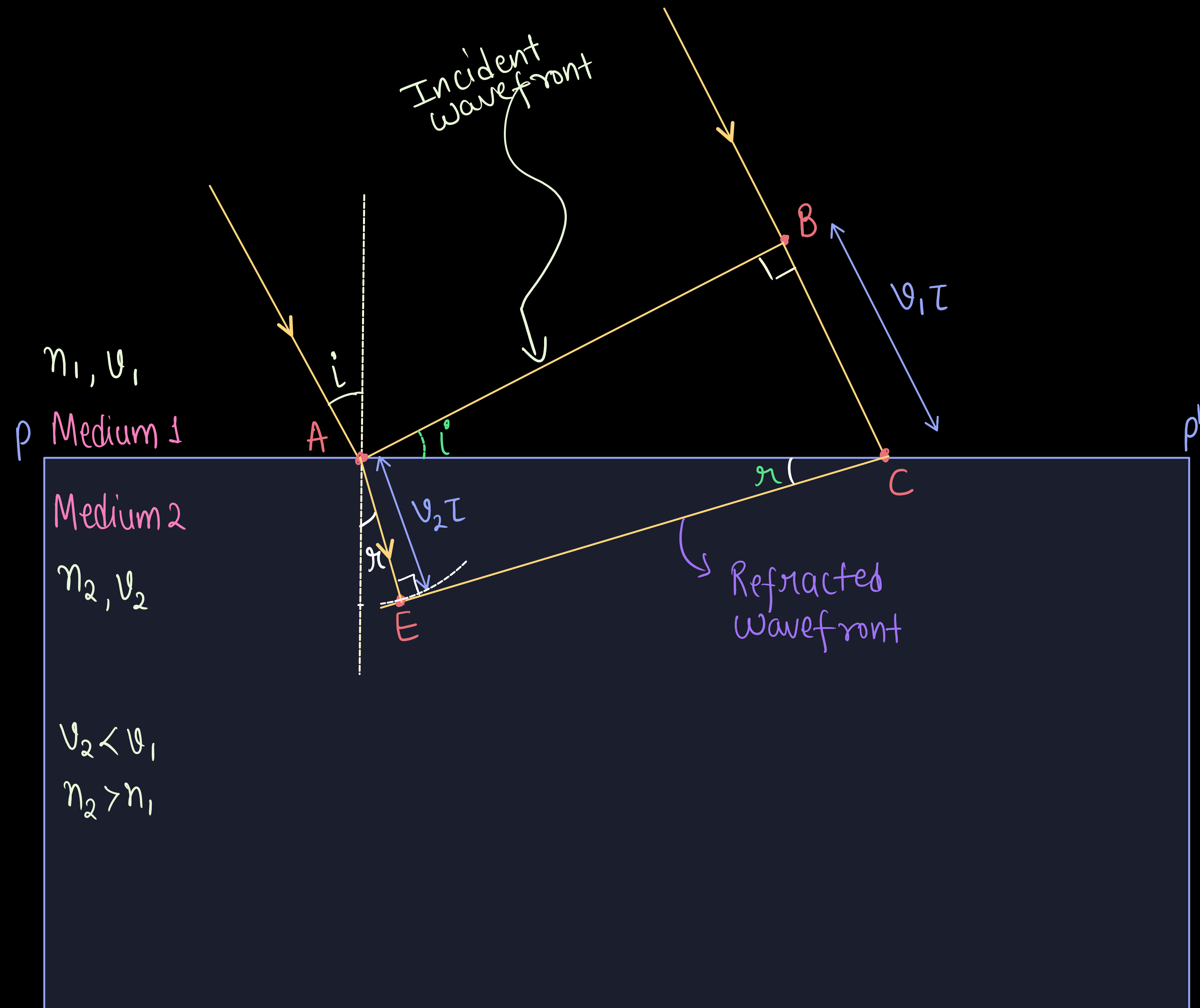
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Also, $n_1 = \frac{c}{v_1}$ & $n_2 = \frac{c}{v_2} \Rightarrow \frac{n_2}{n_1} = \frac{\frac{c}{v_2}}{\frac{c}{v_1}}$



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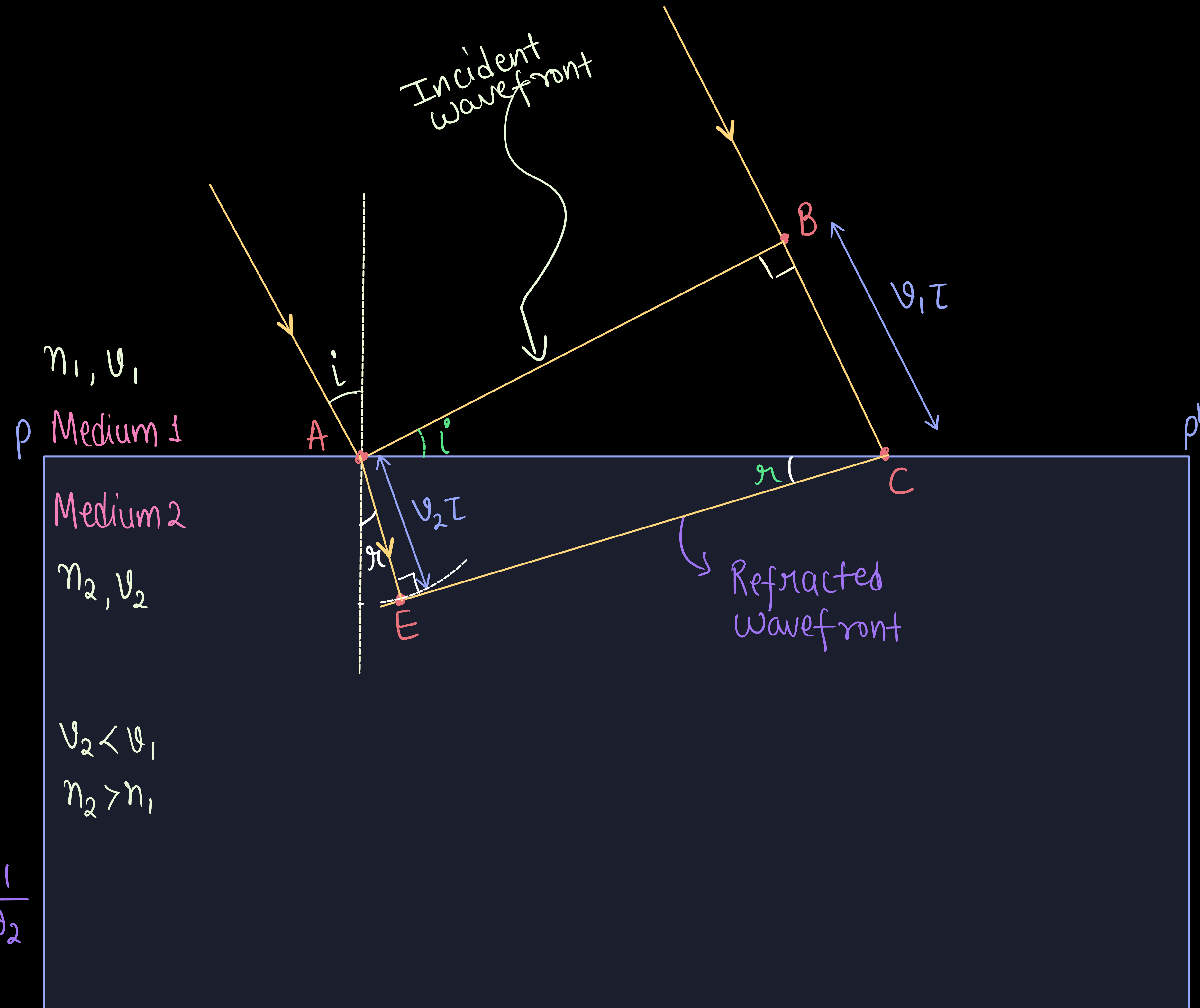
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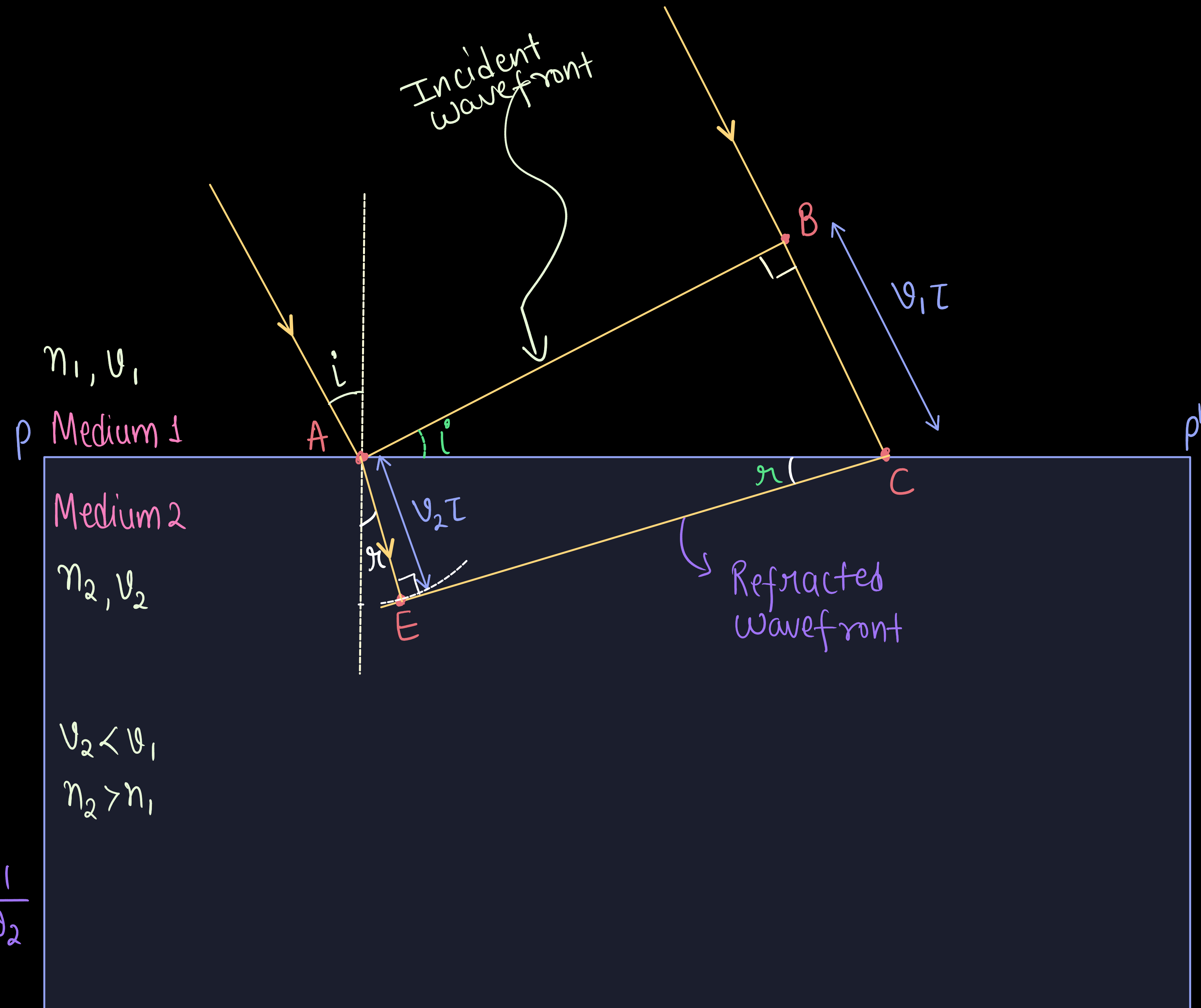
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$\therefore \boxed{\frac{\sin i}{\sin r} = \frac{n_2}{n_1}} \Rightarrow \text{Snell's Law}$

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Reflection and Refraction of Plane Wave at a Plane Surface

(2) Refraction at Rarer Medium -

Reflection and Refraction of Plane Wave at a Plane Surface

(2) Refraction at Rarer Medium -

n_1, v_1

Medium(1)

Medium(2)

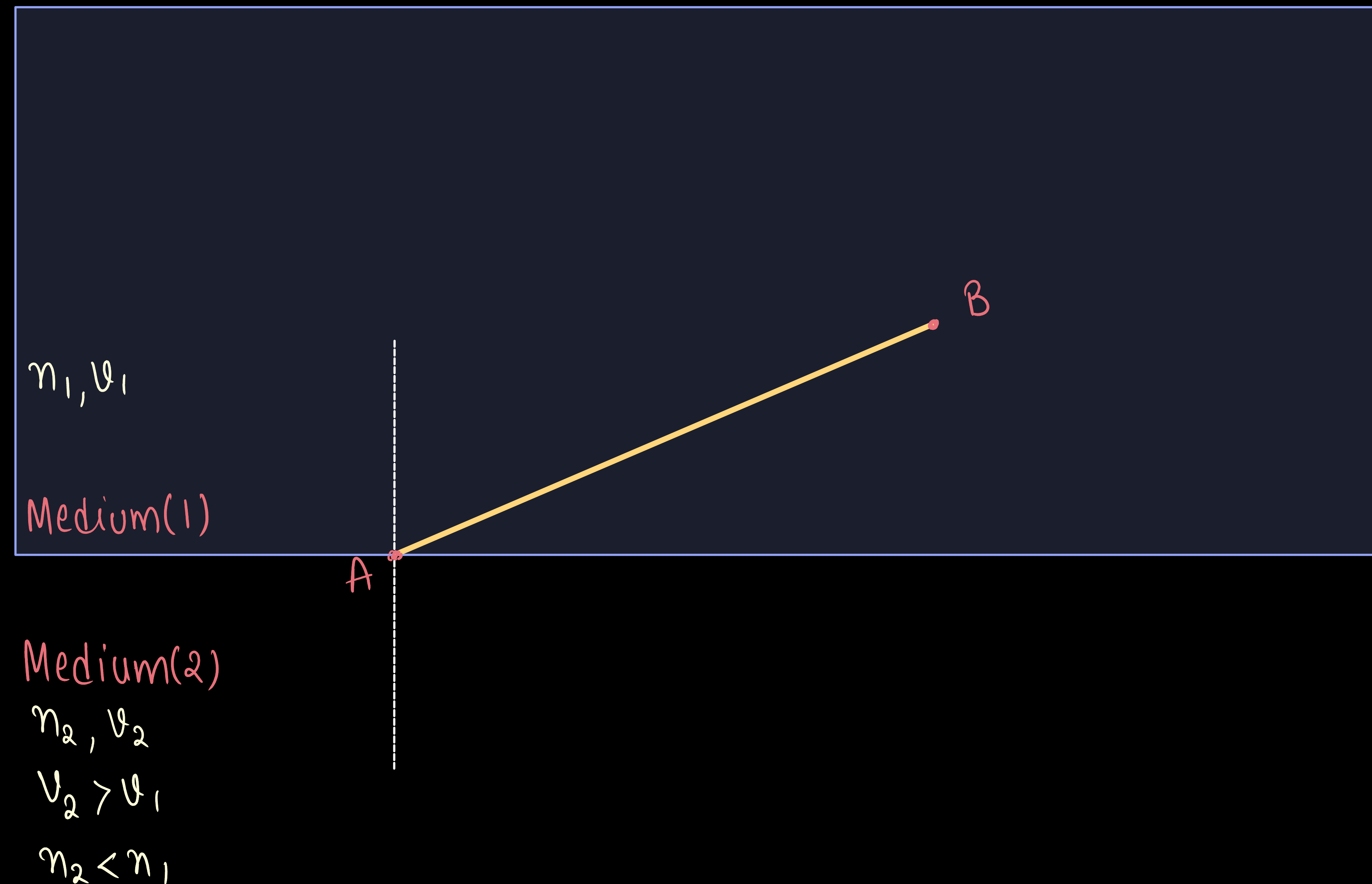
n_2, v_2

$v_2 > v_1$

$n_2 < n_1$

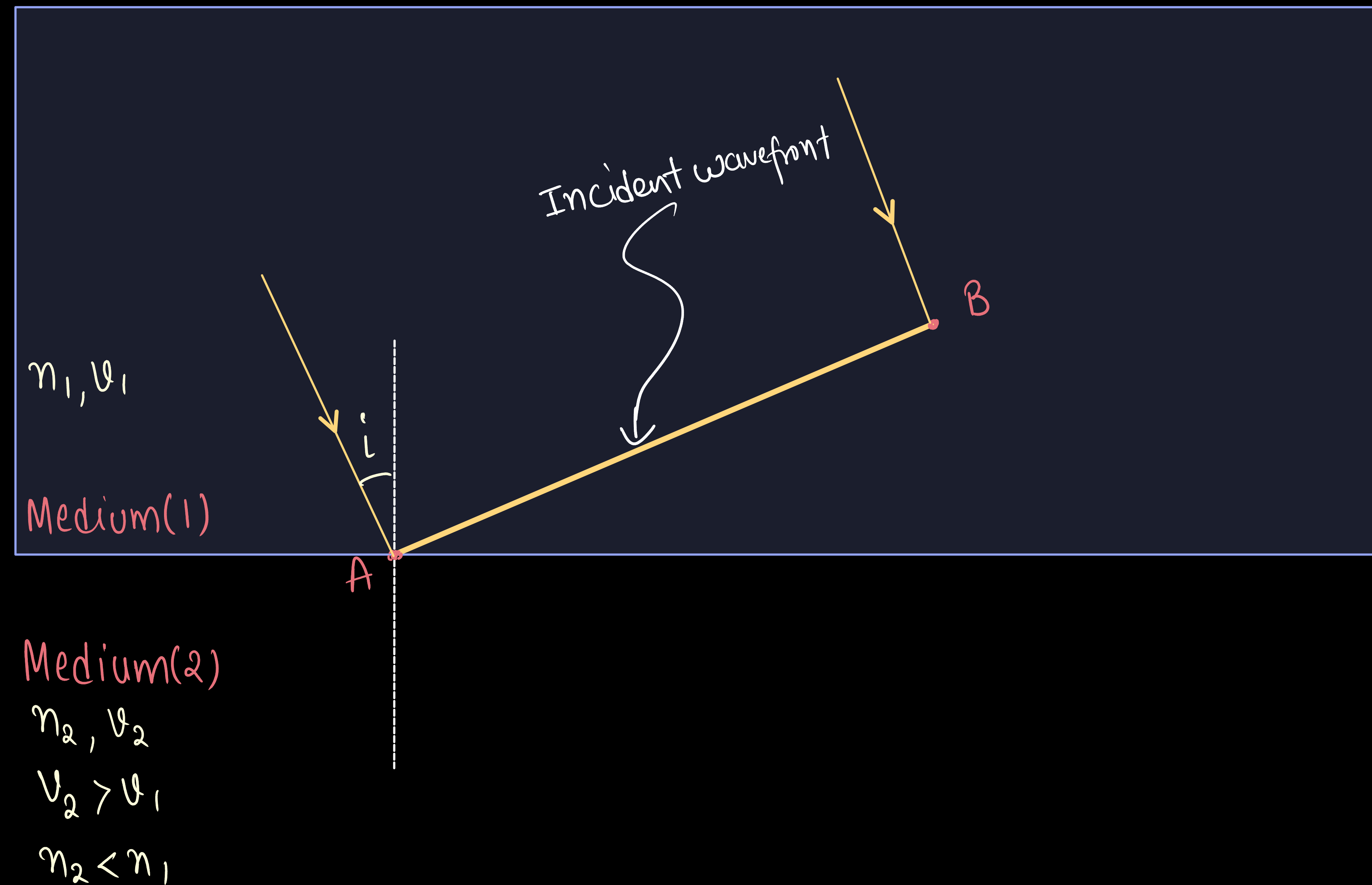
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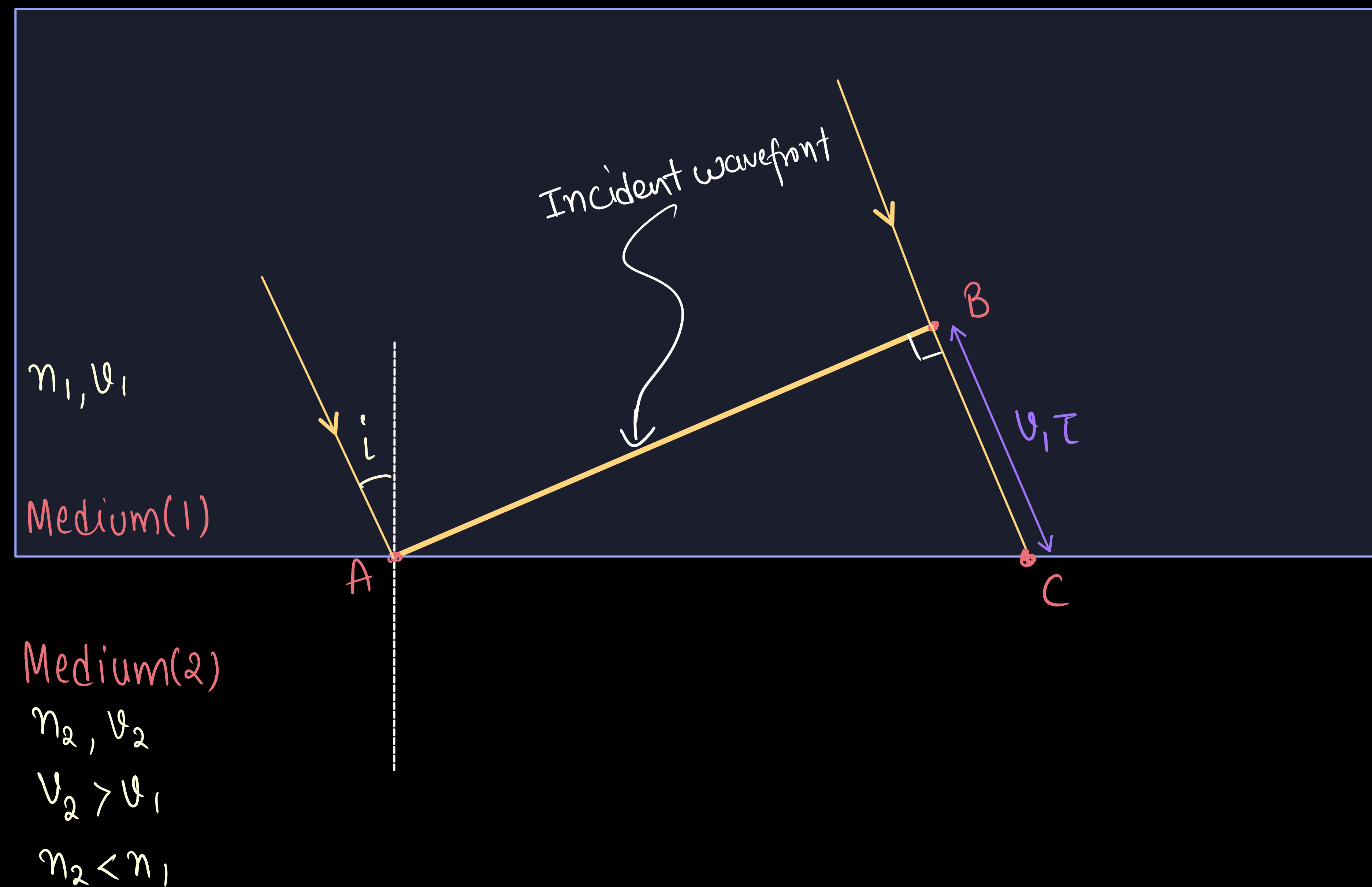
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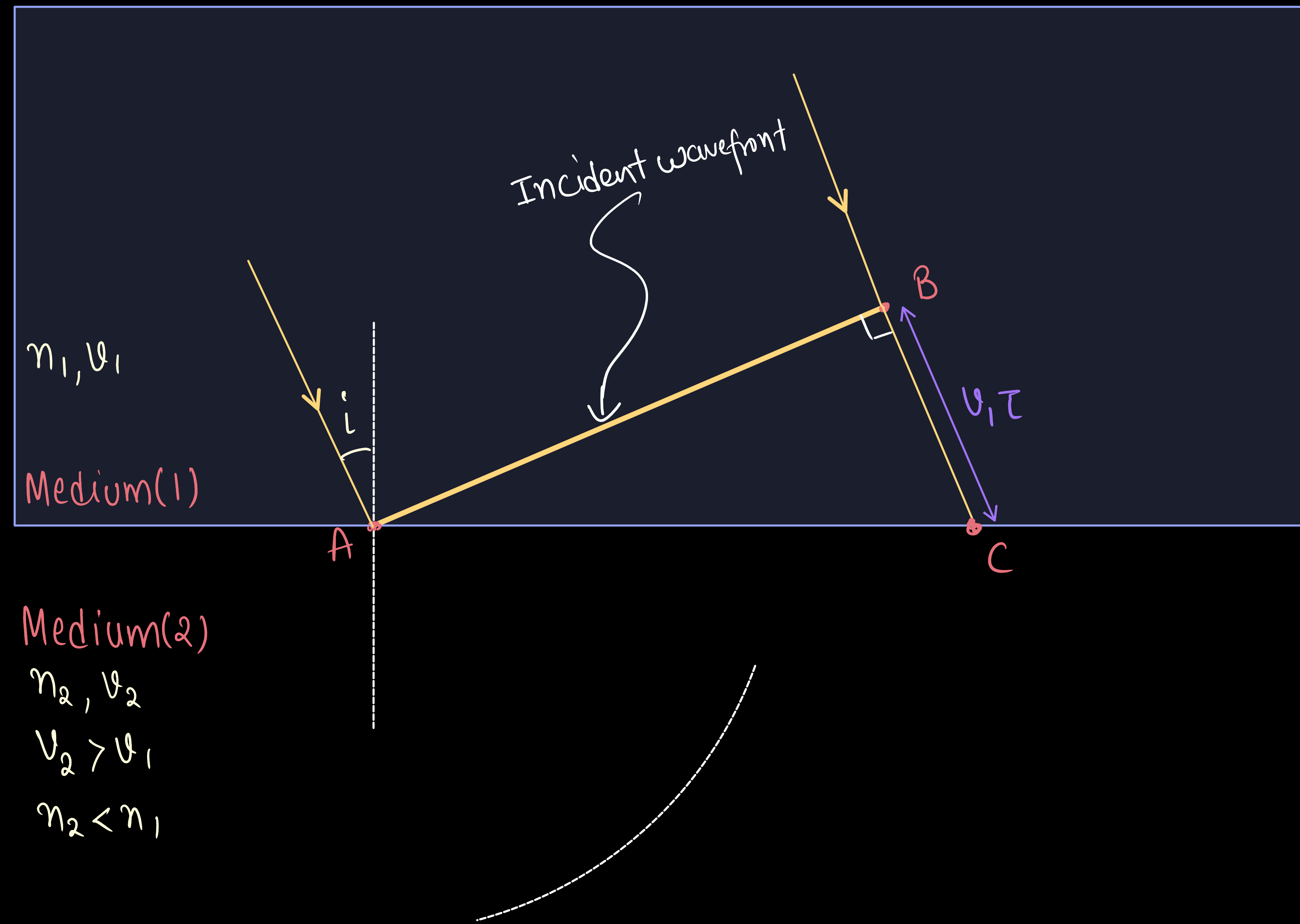
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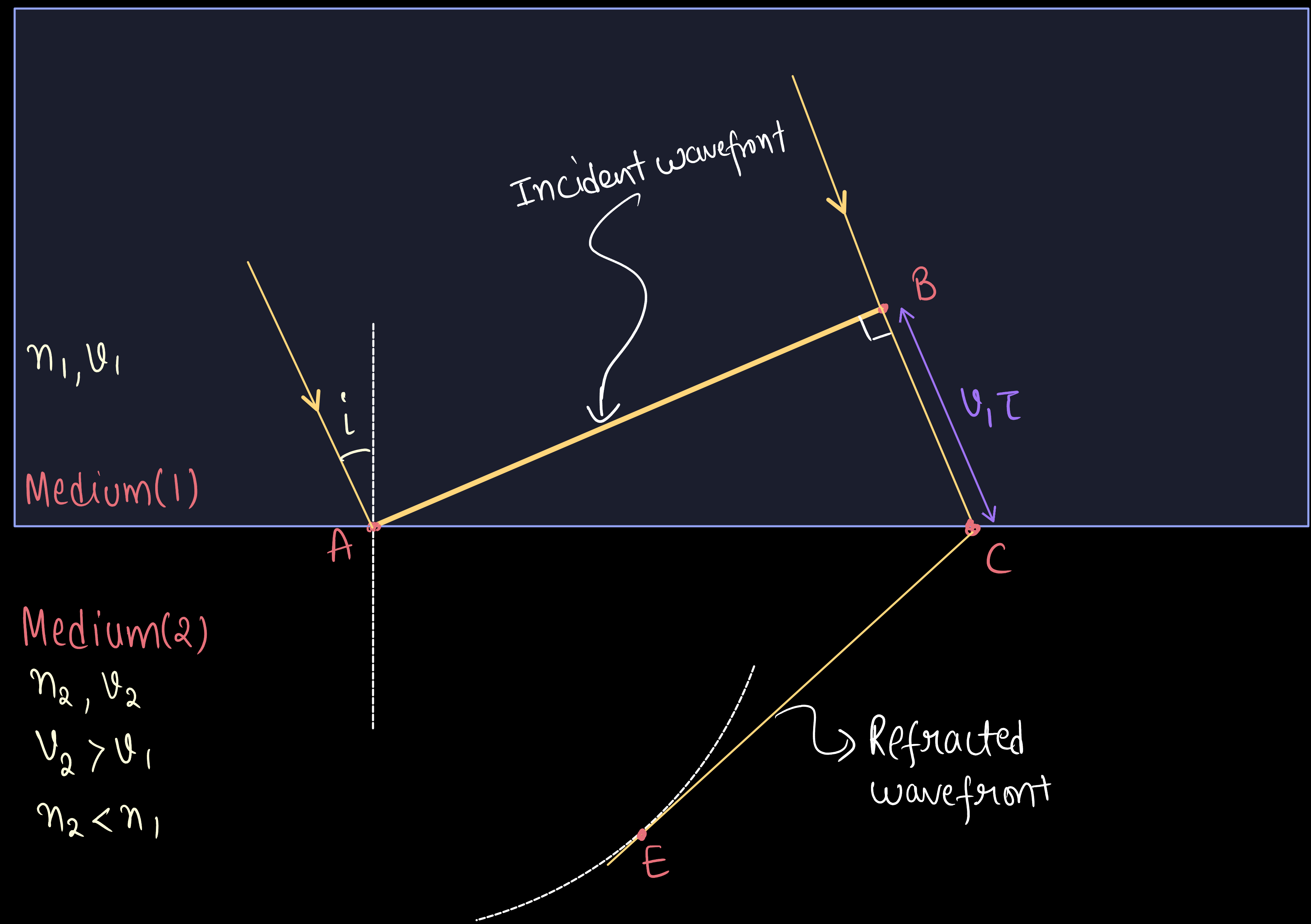
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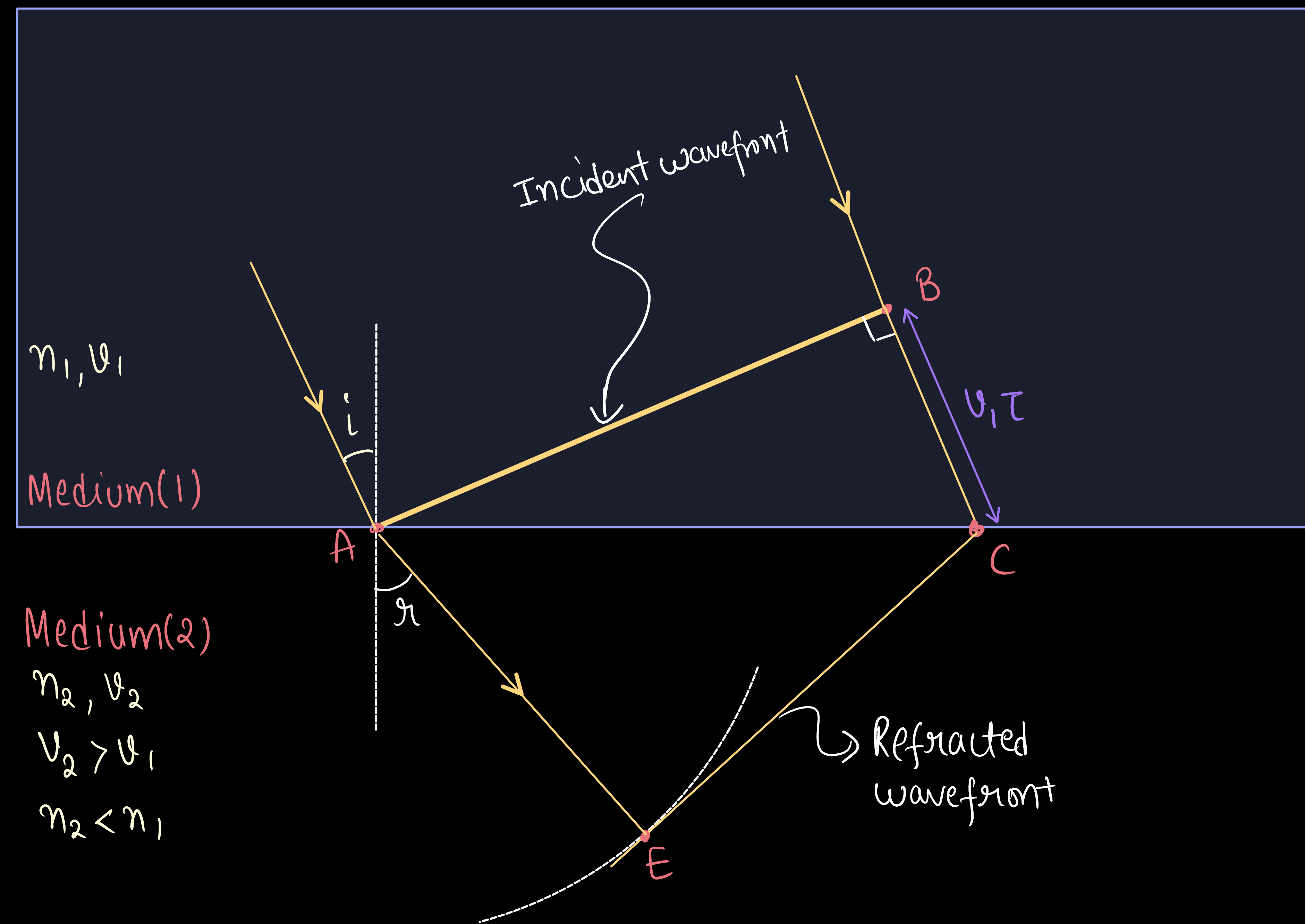
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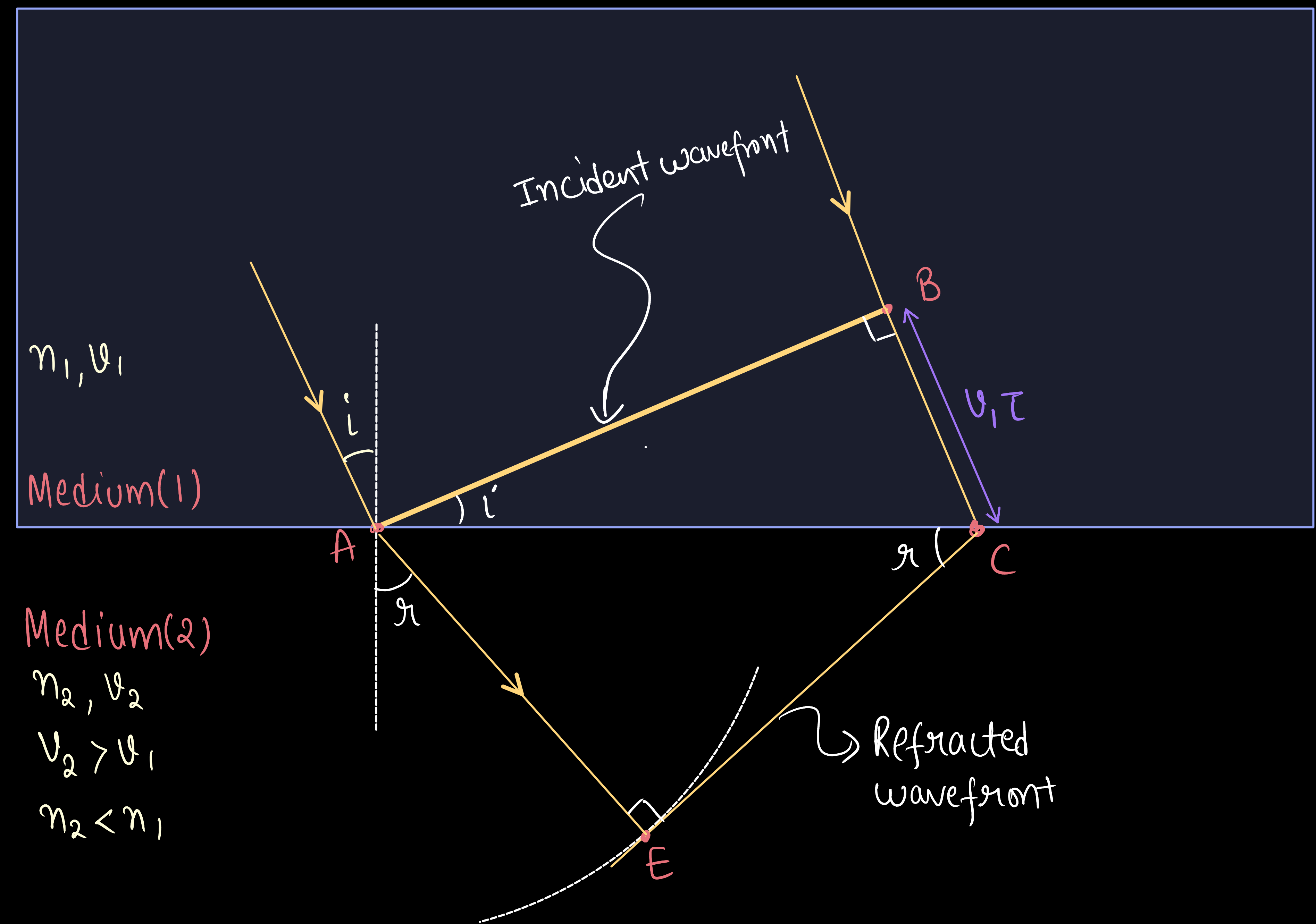
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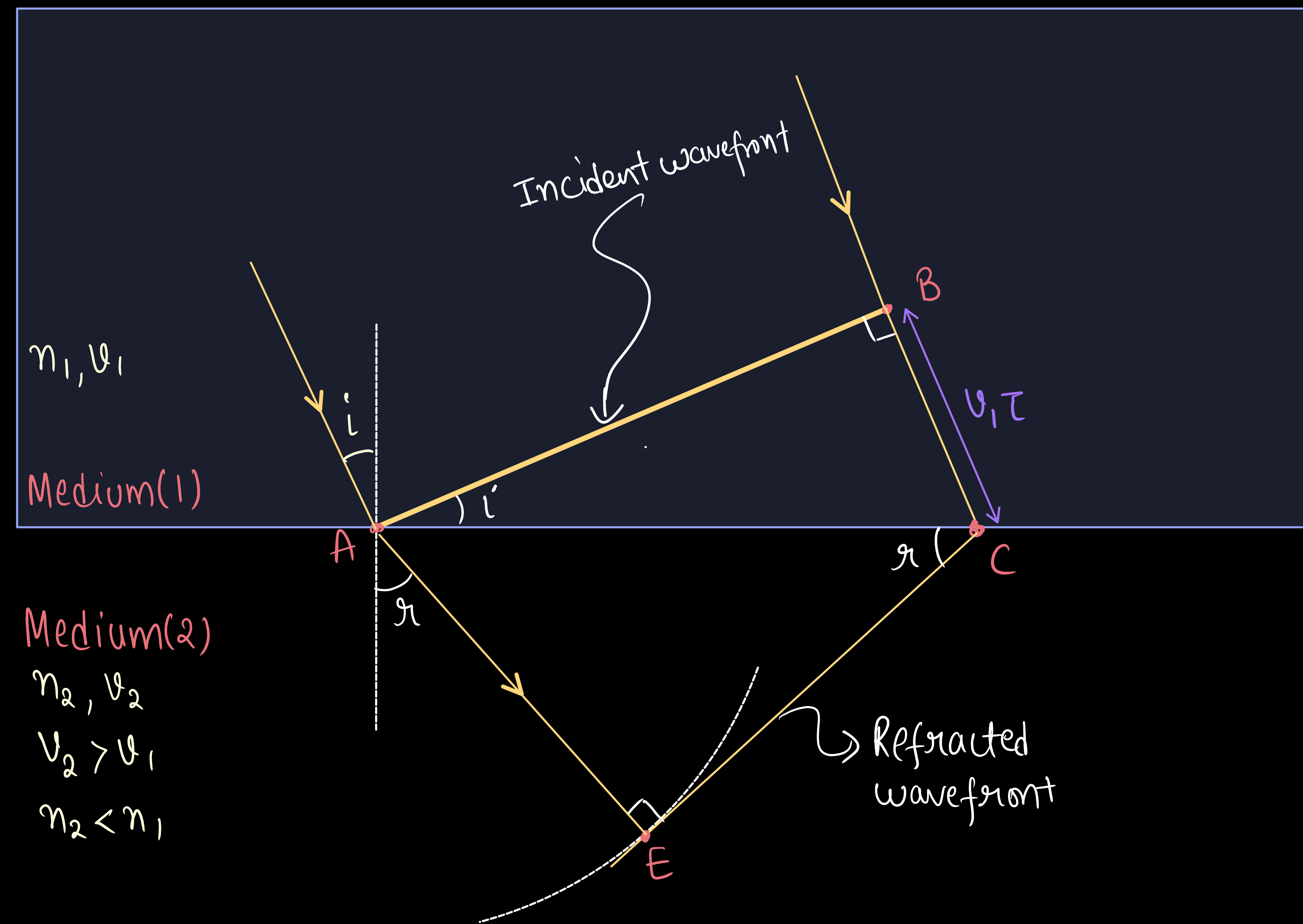
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$$n_1 \sin i = n_2 \sin r$$

$$\text{at } i = i_c, \quad r = 90^\circ$$

$$n_1 \sin i_c = n_2 \sin 90^\circ$$

$$\sin i_c = \frac{n_2}{n_1}$$



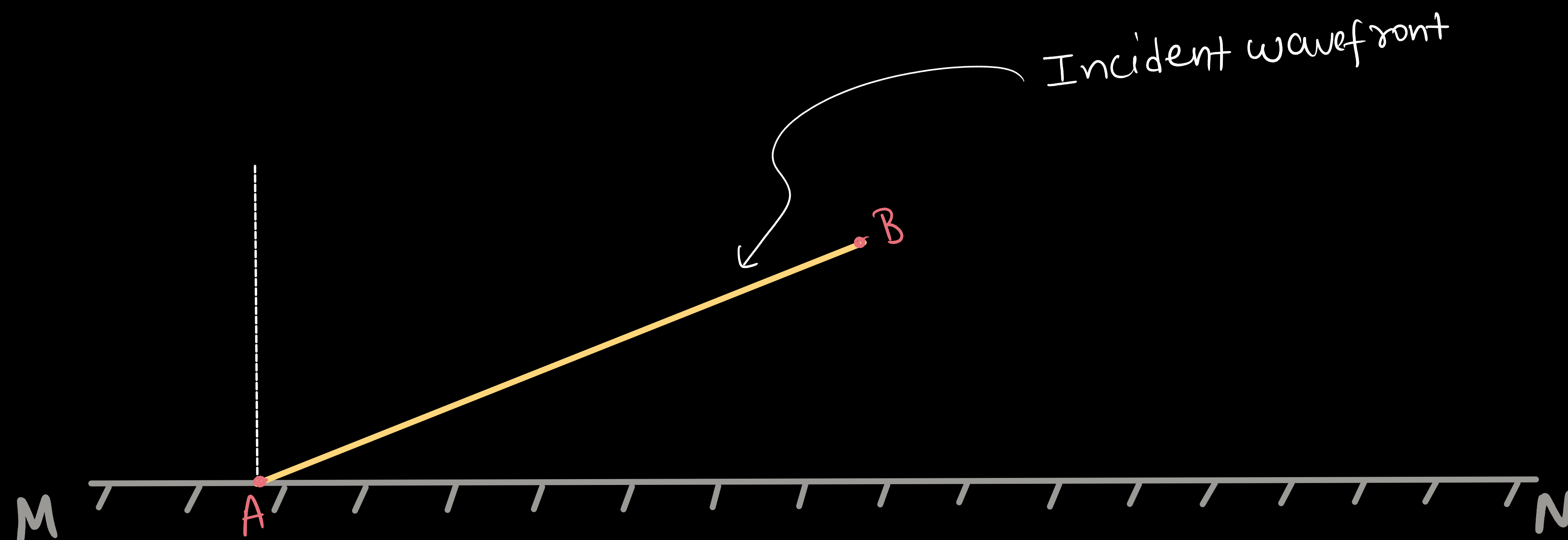
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* Reflection of a plane wave by a plane surface

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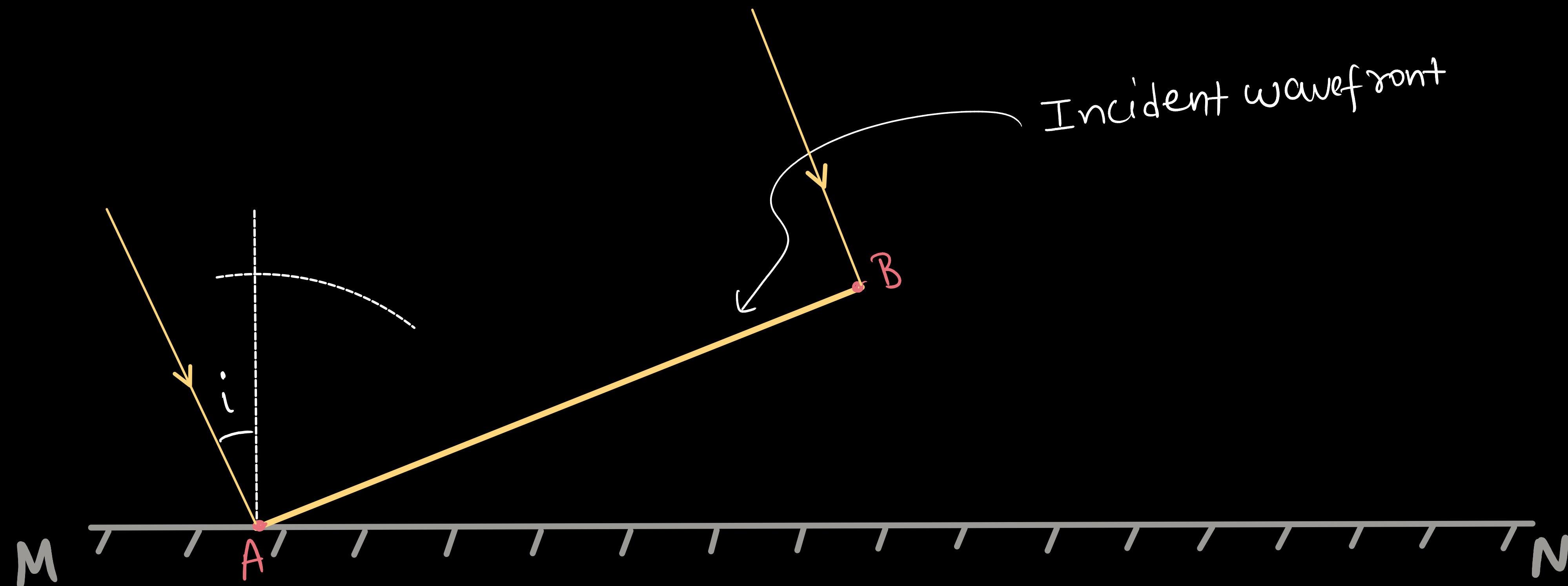
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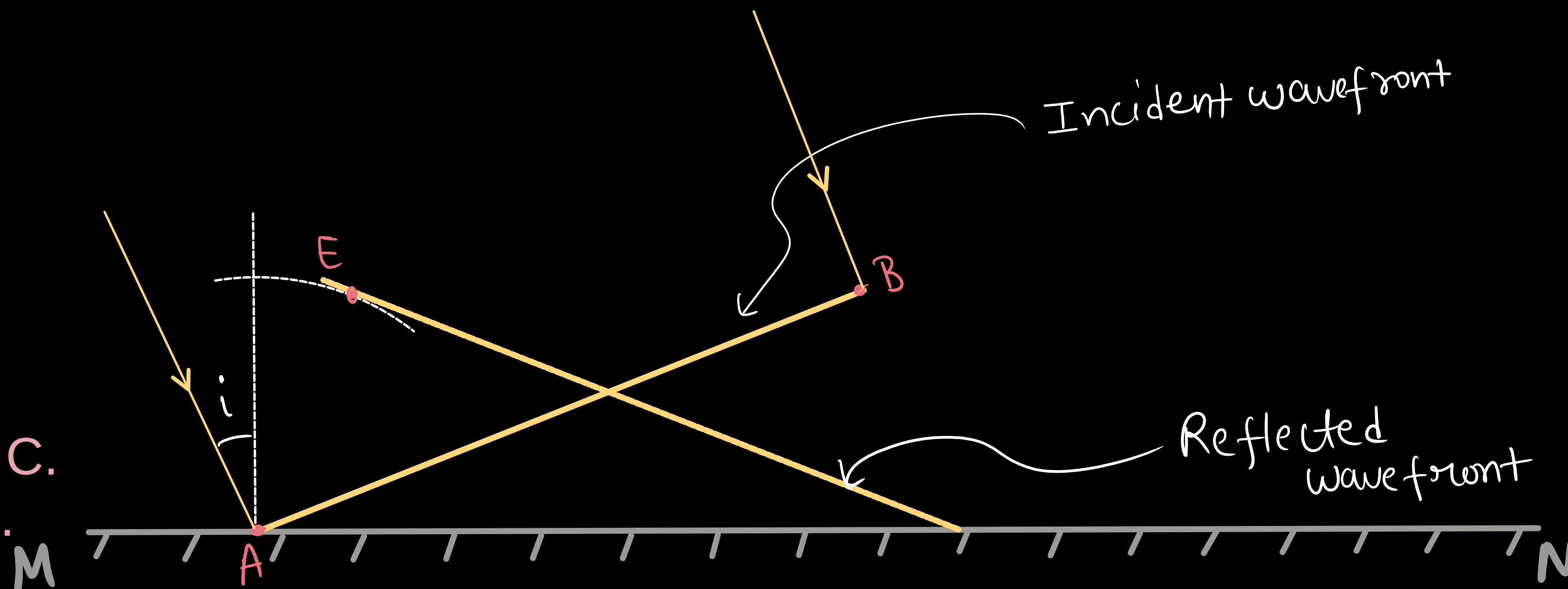
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Draw a tangent CE to this sphere from point C.
Plane CE represents the reflected wavefront.



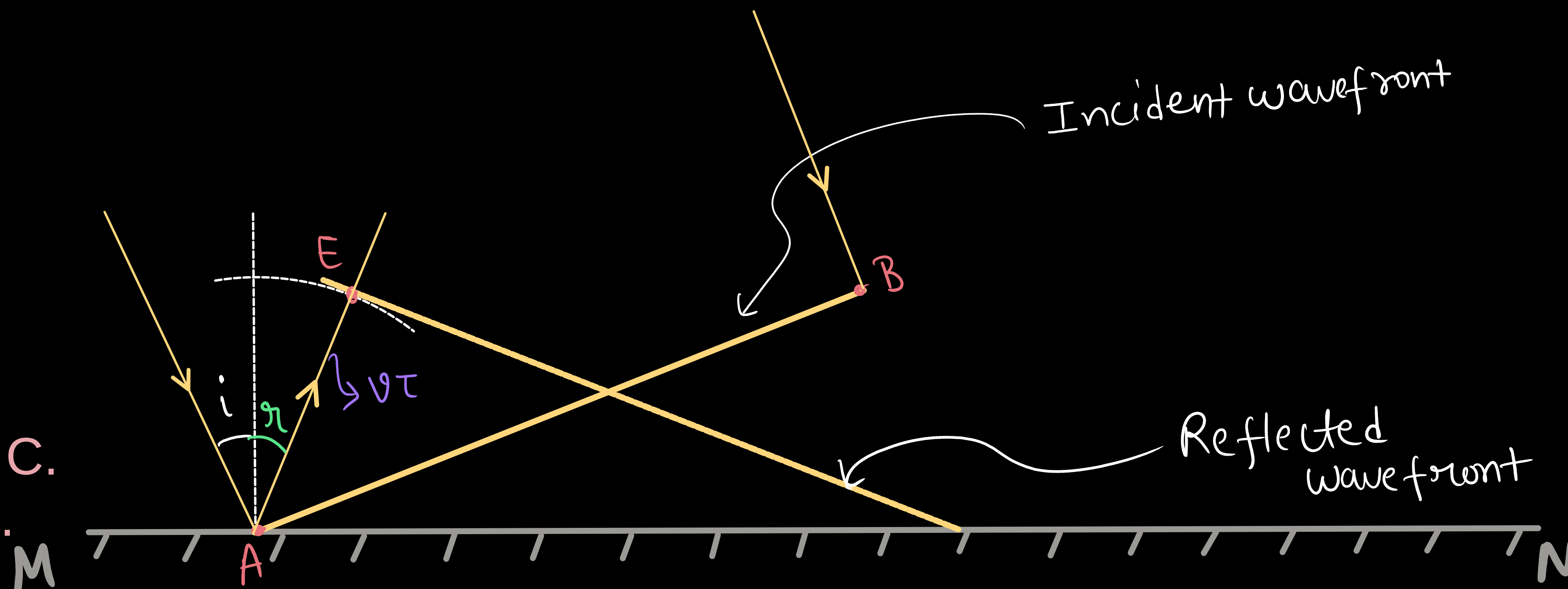
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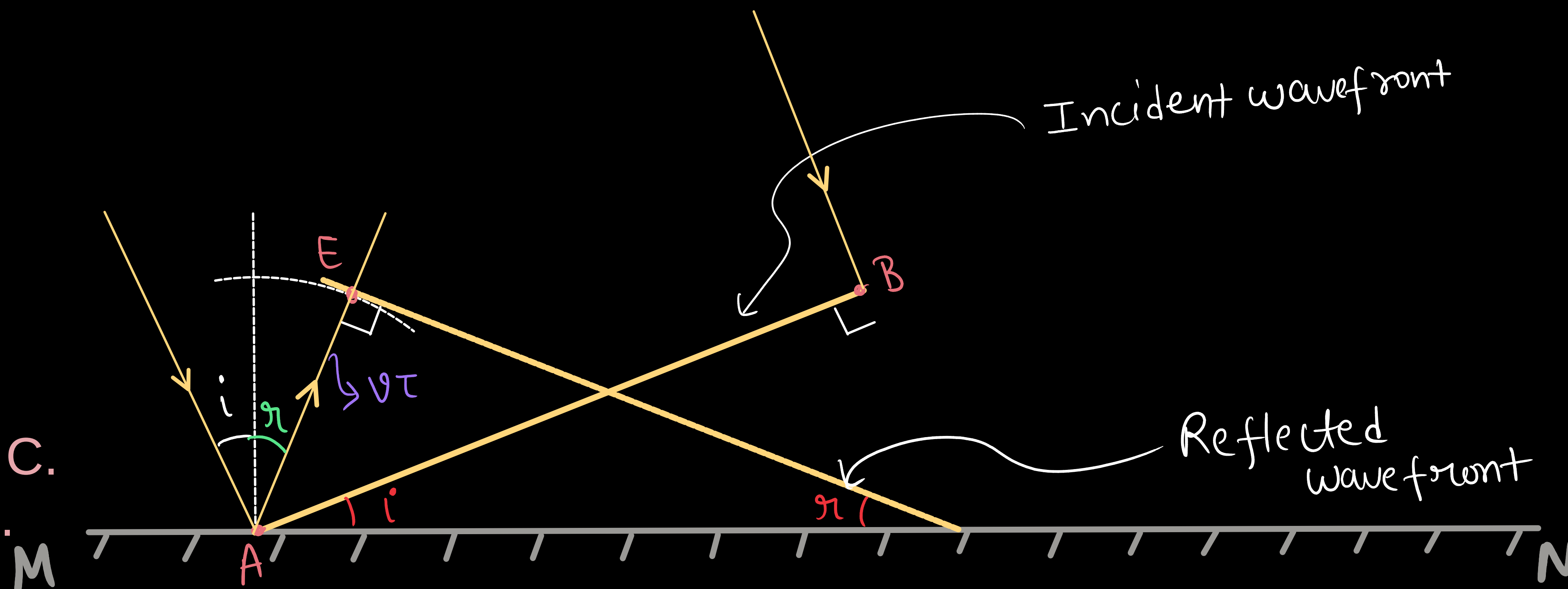
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In $\triangle ABC$ and $\triangle CEA$

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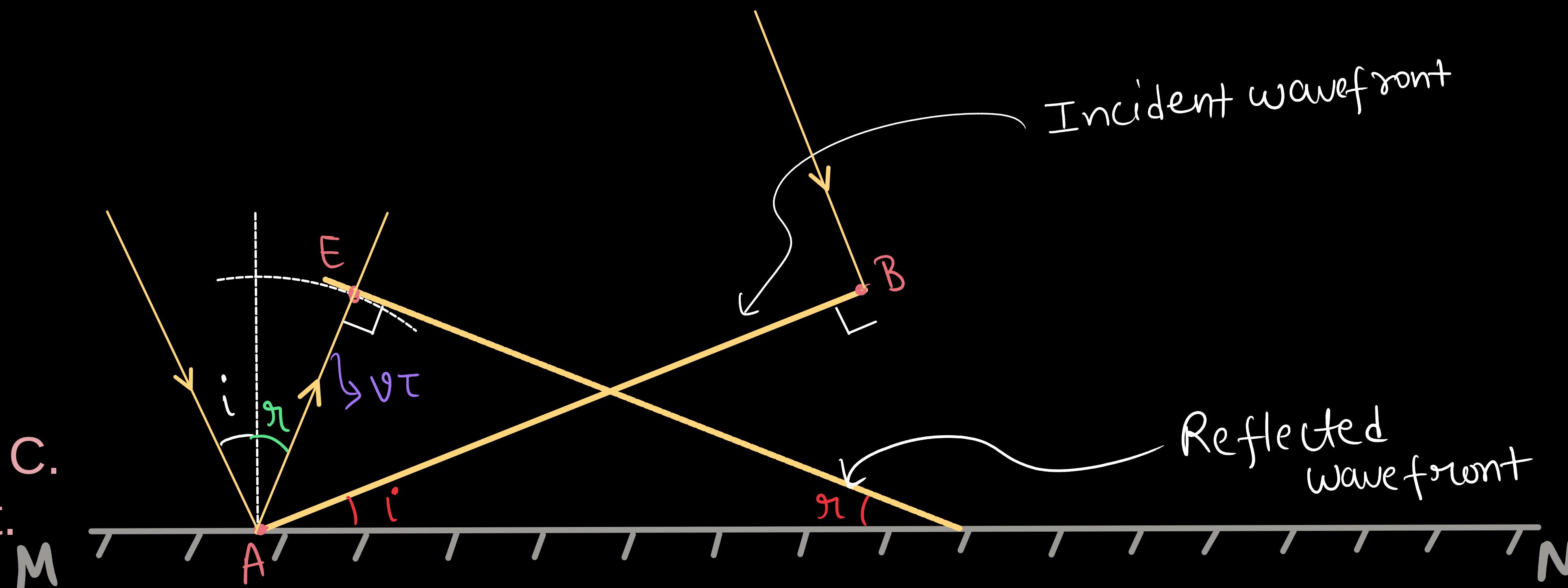
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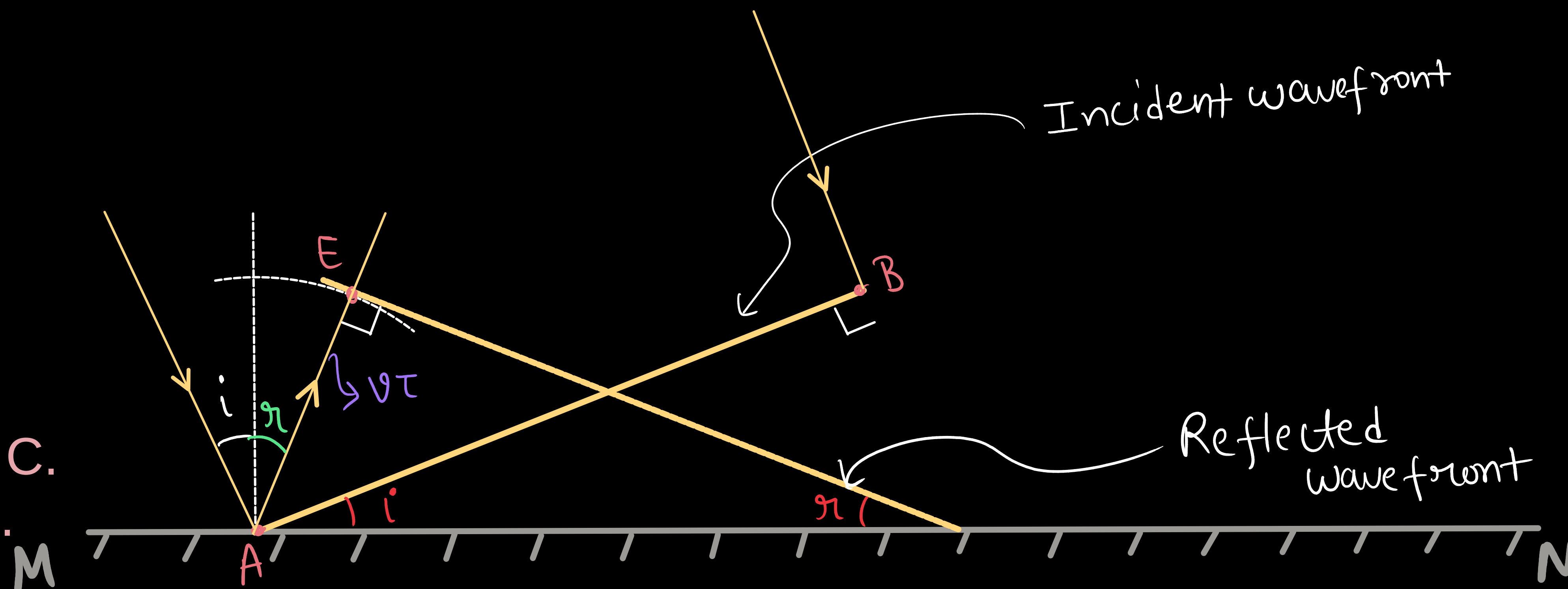
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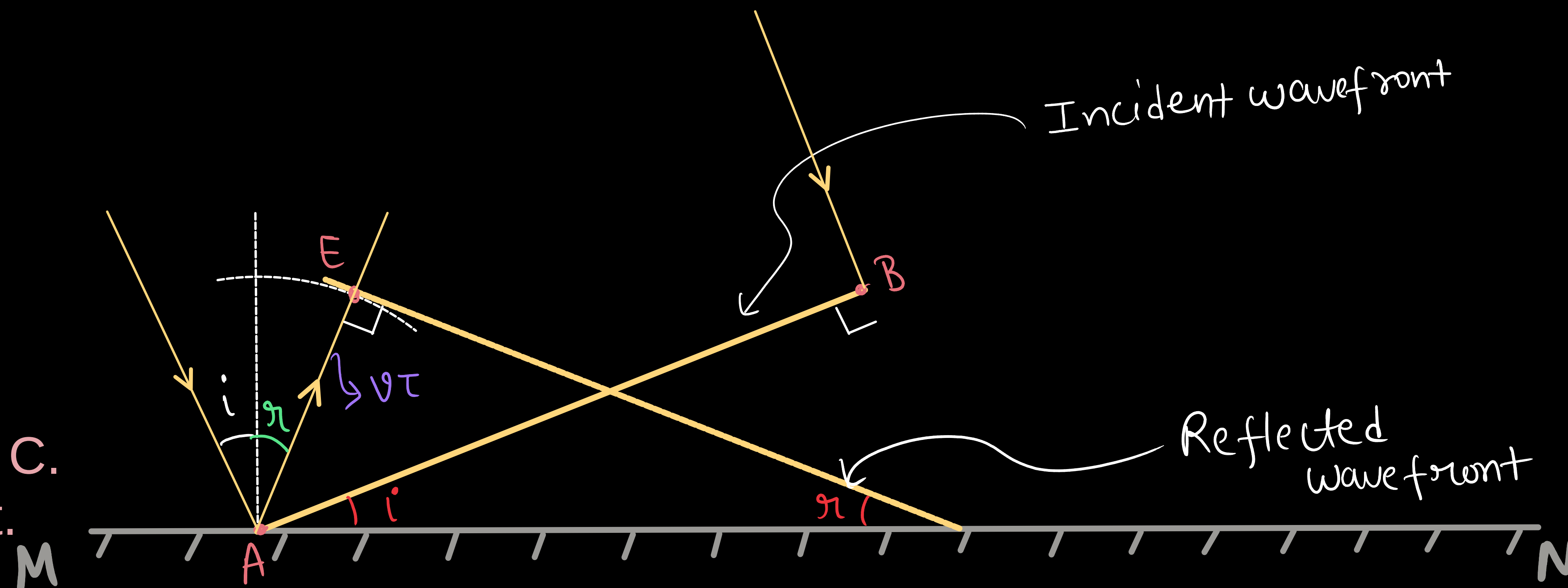
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$$\angle E = \angle B = 90^\circ$$

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In $\triangle ABC$ and $\triangle CEA$

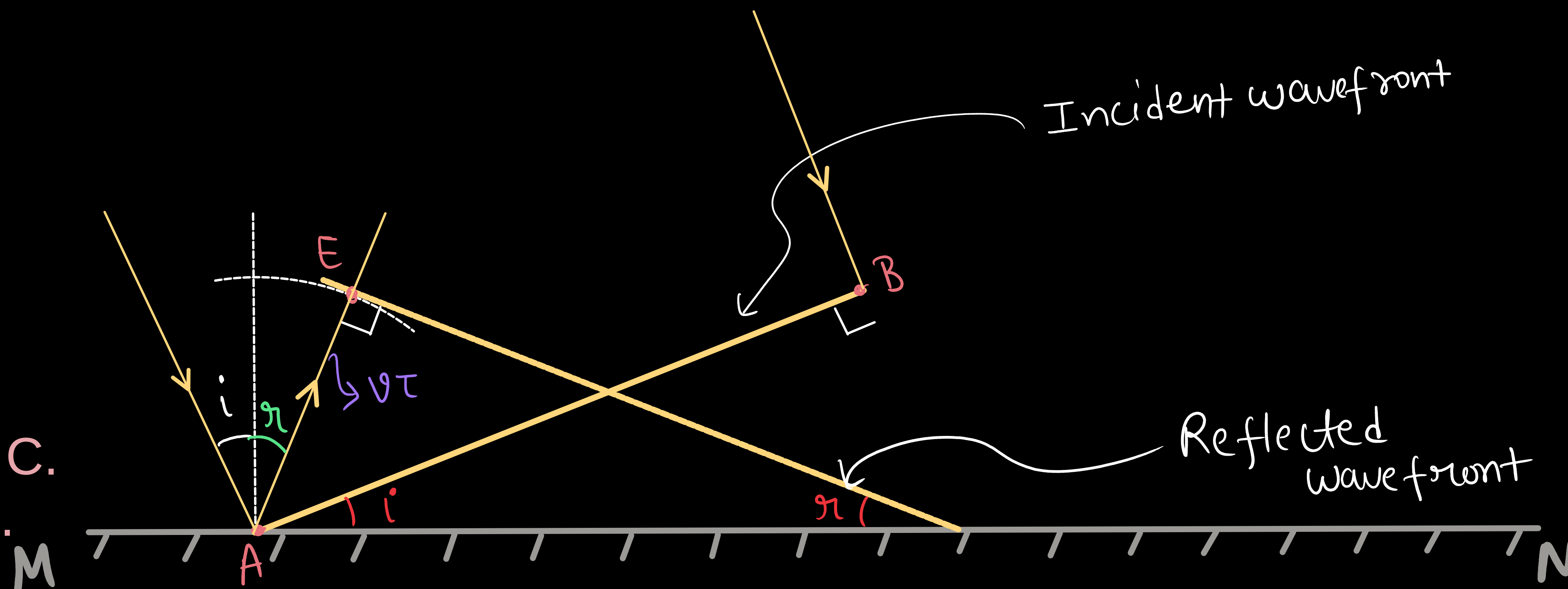
$$BC = EA = v\tau \quad (S)$$

$$AC = AC \text{ (Common)} \quad (H)$$

$$\angle E = \angle B = 90^\circ \quad (R)$$

(4) Reflected Wavefront:

Draw a tangent CE to this sphere from point C.
Plane CE represents the reflected wavefront.



Reflection and Refraction of Plane Wave at a Plane Surface

* Reflection of a plane wave by a plane surface

- * Consider a plane wavefront AB incident on a plane reflecting surface MN.
- * The angle of incidence is i .
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* Huygens' Construction

- (1) AB represents the incident wavefront.
- (2) Point A touches the reflecting surface MN first.

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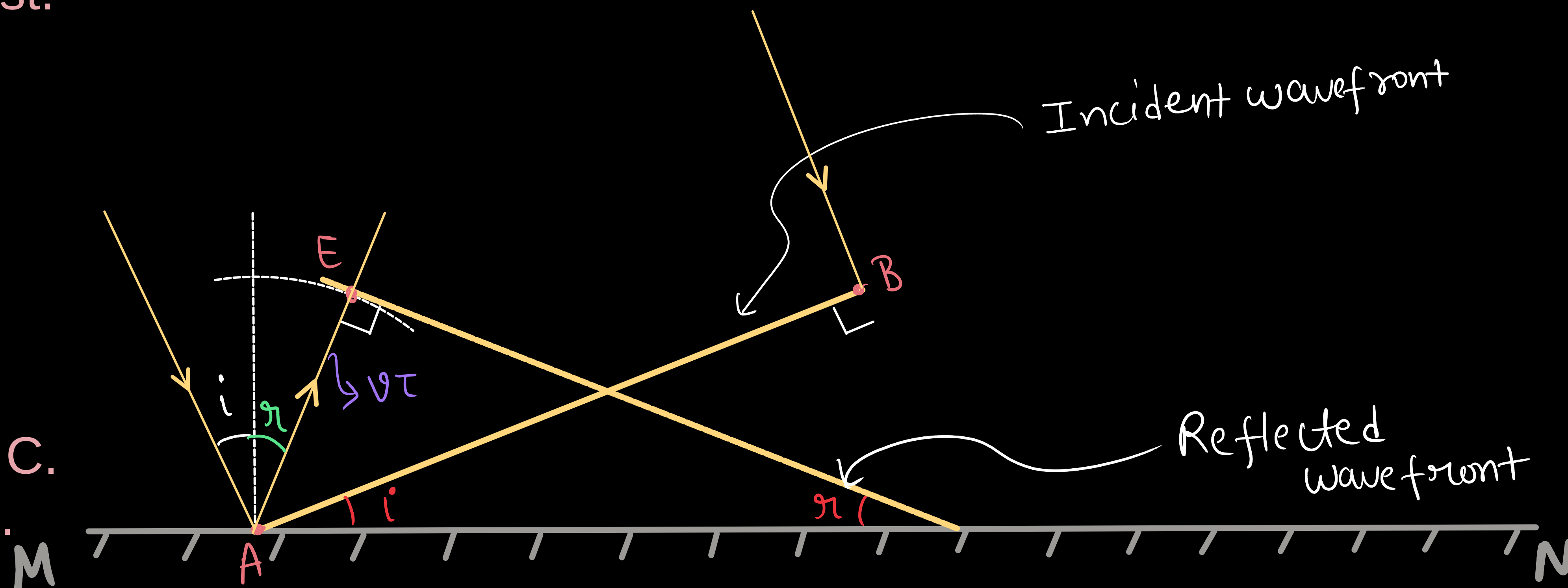
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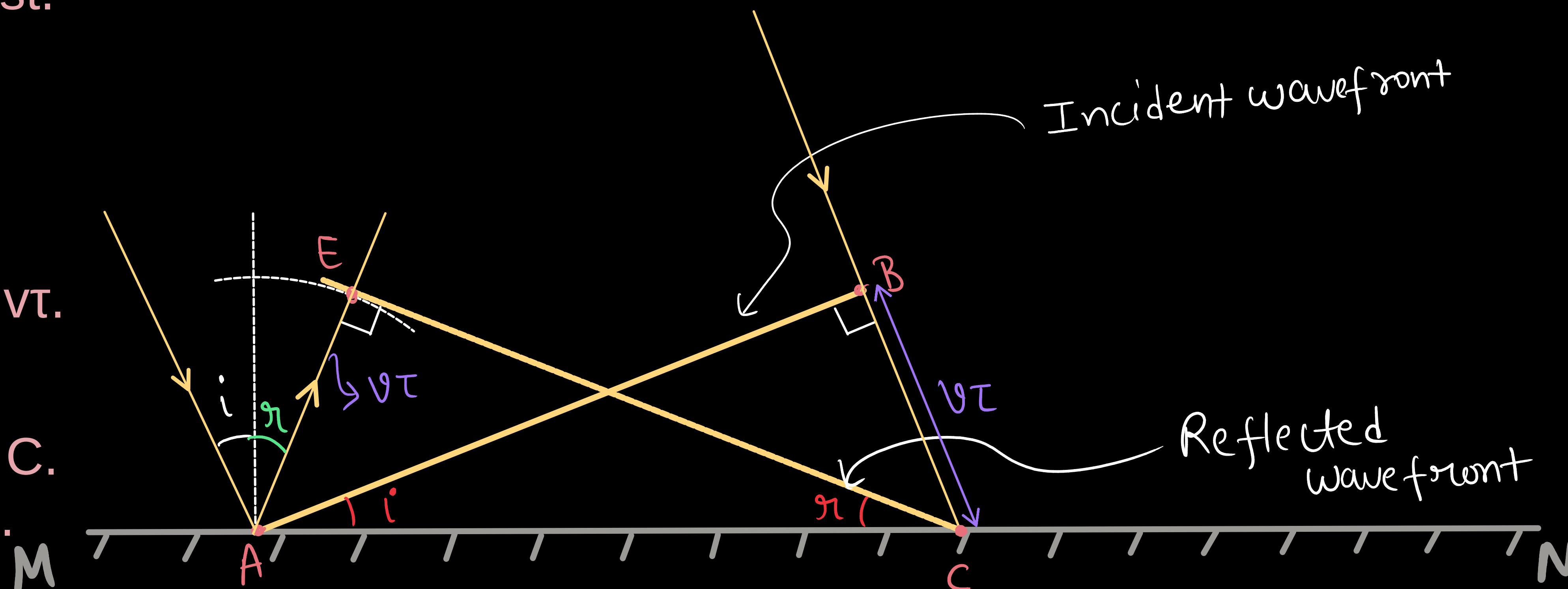
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- (1) AB represents the incident wavefront.
- (2) Point A touches the reflecting surface MN first.
- (3) After Time τ : Point B moves forward to C such that $BC = v\tau$

Meanwhile, from point A (which lies on the reflecting surface), draw a sphere of radius $v\tau$.

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Draw a tangent CE to this sphere from point C. Plane CE represents the reflected wavefront.



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$$\triangle ABC \cong \triangle CEA$$

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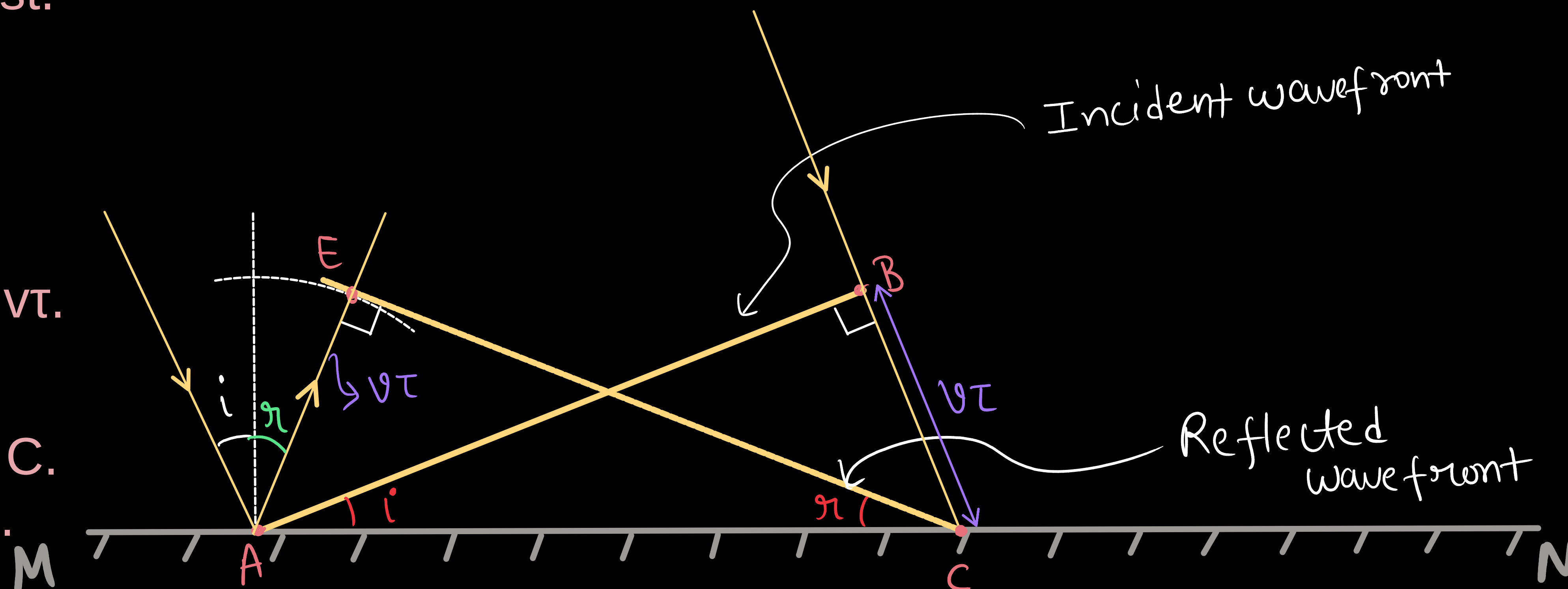
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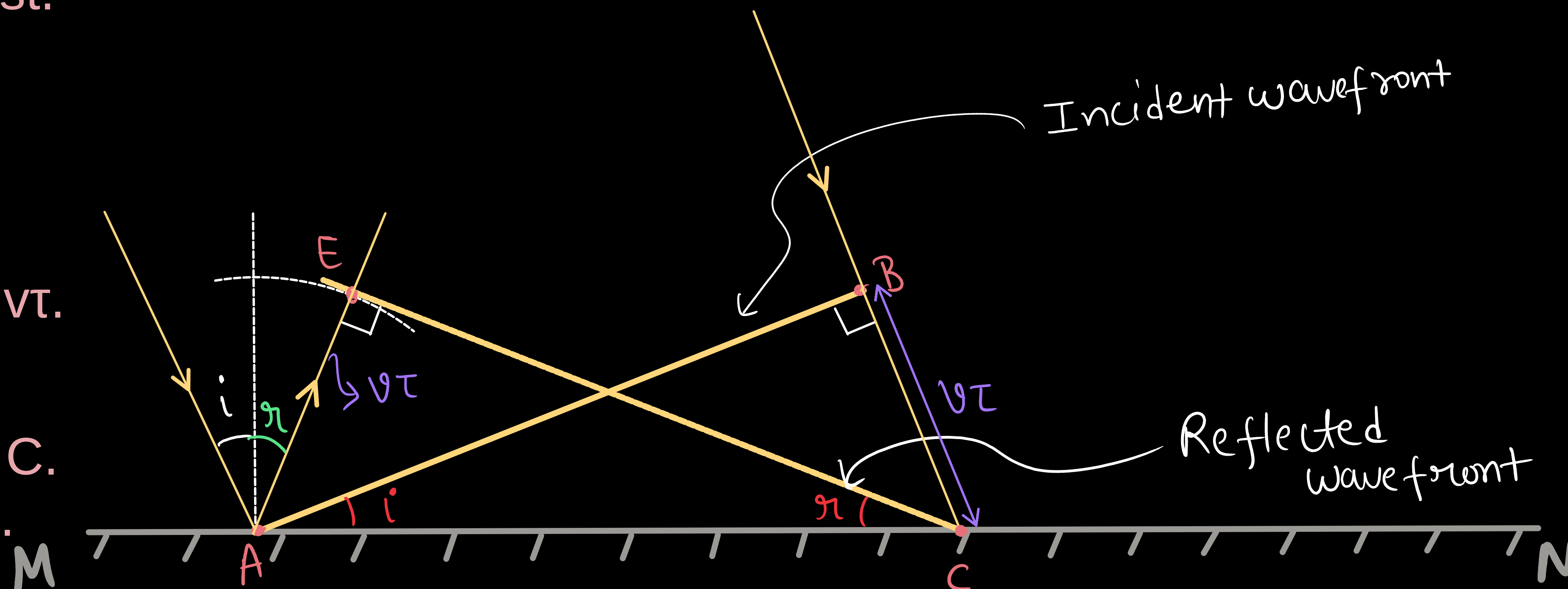
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$$\boxed{i = r}$$